Observers Are All You Need

Records, Quantum Identities, Thermodynamics and Closure Equations for Continuing Observers

Bernhard Mueller[1*] Mario Poneder Alexander Osika[2] Kai Xue Ben Cassie
Peter Nguyen Daniel Ari Friedman[3,4]

October 28, 2026

Paper release: r2055 **Released:** October 28, 2026

Author affiliations: [1]Pragma Research Inc.; [2]EtherWorks; [3]Active Inference Institute; [4]FractiAI.

Abstract

This paper develops the observer layer of Observer Patch Holography (OPH). Its hypothesis is that observers are continuing native processes of the executable carrier: each has an accessible algebra and state, retains records in a commutative public record algebra, reads them back, acts on them through record-controlled instruments and continues through checkpoints. At one finite regulator a machine-checked receipt inhabits all seven observer clauses, and one event packet fixes an event's support region, public record observable and chart coordinate. On finite matrix algebras, every nonnegative normalized valuation additive on coexisting effects is the Born weight of a unique density operator, every pointwise-continuous automorphism flow of a full private block is Hamiltonian, and on the regional-net interface no full matrix factor sits above a scalar overlap region. Given a registered repair law, a declared clock and energy calibration and a refinement-uniform gap family, the four laws of thermodynamics, fluctuation relations and the Landauer bound form one conditional theorem, and the mean entropy production of one repair step equals the relative entropy from its input to the repaired image. The derived action of a positive source chain is fixed up to an additive constant and a multiplier. The local closure $P = \varphi + \sqrt{\pi}/A_T(P)$ has interval-certified unique roots for each declared map, and the global closure $N = \log M_0(\mathfrak{U}_N)$ reads exact public-record capacity as the independence number of a compound confusability graph; a bounded counterfamily shows that its finite base structure does not fix a unique zero. A supplied charged-scalar/Maxwell action gives first-order convergent real continuum trajectories and a self-adjoint interacting Hamiltonian on its fixed mesh, read out by exact classical record feedback.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and

*Corresponding authors: Bernhard Mueller, bernhard@floatingpragma.ai; Alexander Osika, alex@snrgystudios.com.

intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap*

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning. The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results supply finite components for the larger construction. The finite campaign itself does not freeze a carrier/refinement class; that boundary belongs to the separate executable program contract. Its observer services must be constructed or counted on the declared source, and any physical use needs the actual accessible algebra and implemented clock. A defect sector or controlled lifetime also needs its own specification outside the excluded commuting-charge class. The finite campaign adds no independent physical comparison or whole-program exclusion.

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

Same-action finite evidence. A supplied three-dimensional Weyl/contact CAR action has one isolated two-fermion bound band whose wavepacket moves, carries the conserved CAR number charge and has complete contact/scattering spectral records. Its positive contact section makes the band topologically trivial. An exact rational curvature enclosure disagrees with the rest/kinetic mass relation using the same speed parameter; these are coordinate energies and masses, not identified particle species. One invariant pair and dark-reference subspace is attached by a supplied isometry to a native two-level corner. Its record-derived modular clock and complete herald, continuation and energy records are executed with a finite unconditional error bound. This does not encode the whole spatial field.

On the same local action, signed CAR regional restrictions, exact local energy conservation and finite quantum Gibbs/entropy identities have explicit witnesses. They retain all local number sectors, references and detector outcomes. Regional bare-energy modular identification fails, and the contact interaction leaks from any proper Weyl momentum pocket into its complement. Finite thermal positivity and KMS identities do not remove that coupling or establish a chiral continuum. These results reduce specific finite obligations in the existing premise map; physical geometry, Lorentz stress, common normalization and integrated acceptance are not established. The current four construction targets are native defect protection, common energy/kinetics, the interacting chiral field and finite-theorem integration.

A further complete-band execution uses the native record-derived rate with counted local field controls. Its unconditional error bound is below 0.038565 on $|s| \leq 1$, conditional on a supplied global field Gibbs reference. Finite independent copies of the original native reference cannot furnish that same field clock: their integer modular-gap ratios conflict with an exactly bounded noninteger pair gap. Directly on the native carrier, a faithful coherence has an exactly reference-matched exterior through every raw history and a finite contrast lifetime, but zero modular-energy excess and a fixed coherent core. The exact-code exterior and clock restrictions prevent combining these separate properties into an established moving particle.

The fair and endpoint-biased actively corrected memories have separate exact core laws. Positive coin rescaling transfers the sharp-code classification to the biased law, extending its optimal asymptotic reset bound to arbitrary adaptive exact code choices. A separately checked variant removes identity recoveries while preserving the admitted channel. Contact-only field protocols also admit a seven-state cyclic reduction in the declared side-four zero-momentum fiber. These simplifications reduce finite service or evaluation costs. A clock on another factor composes with the memory, but supplies no physical mass or propagation relation.

This paper's hypothesis and place in the tower

This paper is layer 3 of the tower, the executable observer, and it carries the two closure equations that layer 7 tests. Its hypothesis is the following. *Observers are continuing native processes of the executable carrier with accessible algebras, retained records and record-controlled feedback. The operational quantum identities (Born weights, Lüders conditioning, no-cloning, the Tsirelson bound and Hamiltonian form of continuous symmetry flows) and the thermodynamic identities (second law, Clausius inequality, fluctuation relations, Landauer bound and finite-regulator third law) follow from that observer structure once one repair law and one shared reference are supplied on the same carrier.*

The unconditional content is a set of finite theorems: the inhabited observerhood receipt and common-origin theorem, the character-block obstruction, the finite Stone converse, the finite effect-valuation theorem, the event-algebra classifications, the finite history-weight and derived-action

theorems, the public-capacity theorems and the bounded capacity counterfamily. The four-law package, the closure readings of P and N and every physical reading of a modular parameter are conditional implications whose premises are displayed where they enter.

The executable model must realize several open joins before these conclusions apply to it: one continuing native observer with every service and resource accounted; a source-produced, complex-tomographically complete effect and instrument web together with an operational derivation of noncontextual additivity; one source reference shared by the state and transition optimizers; the physical clock and energy dictionary; a complete capacity source antecedent with one physical zero of the finite-size slack; and the same-quantity bridge and target-independent map selection for P.

The hypothesis or a branch of it is refuted by a counterexample to a load-bearing finite theorem under its stated premises; by a native law whose records admit no common state and transition reference (the audited direct obstructions below are such results for one artifact); by observables that change with hidden carrier presentation after the visible interface is held fixed; by a source-realized complete effect web on which a non-Born valuation is additive; by a nonunique or target-leaking closure solve; or by the qualifying comparisons stated in the companion papers. Gauge structure, matter, spacetime and gravity are developed in the companion papers [18, 21, 17]; this paper points to them rather than re-deriving them. Appendix A records the finite field histories, on a supplied charged-scalar/Maxwell action, that observer records read and restore.

Keywords: observer patch holography; quantum foundations; observers and records; finite thermodynamics; Born rule; closure equations

Matter operators and gravitational response

The finite response has an explicit conditional matter and interaction readout. A declared spin lift and trace redistribution act on $\Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$. Charge and Casimir projectors recover the five internal multiplets of one Standard Model generation. With an independently declared physical Weyl algebra, one scalar doublet admits the three charged-fermion interaction contractions. The optional neutrino interaction is dimension five. These exact current identities select no Yukawa coefficients, Higgs vacuum or observed masses. Identifying both family factors with the same unbroken irreducible source triplet instead forces scalar intertwiners and degenerate family masses.

The conditional Lorentzian scalar action also admits a common local tetrad dictionary for scalar, spinor and Maxwell principal symbols. A finite CAR construction shares its ideal spatial operator and incident-count clock. Completing a separate tetrahedral cochain control removes edge-cycle zero modes introduced by truncation and preserves its scalar action block. These statements keep their distinct global domains, boundary conditions and reference states; they establish no complete physical chiral field net.

For coefficient density $C = \mu\,\mathrm{diag}(-1, 1, a)$, the induced screen area is $dA = \mu\, d^2x$, while proper spatial volume is $dV = \mu N\, dr\, d^2x$, with $N^2 = \mu\sqrt{\det a}$. Counting stationarity consequently requires a further identification with fixed-volume geometric entropy stationarity. The static radial-product metric has vanishing radial null Einstein contractions. Its allowed scalar excitation can have positive relative null stress, so a fixed background, with the reference and other sources fixed, cannot satisfy both Einstein balances. A responsive geometry must supply the corresponding area focusing. This is a scoped restriction of that field/metric construction, independent of the validity of the conditional Einstein theorem.

The exact finite tensors and complete observation controls are given in `model/observer_particles/`. Their representation, field-operator and geometric witnesses instantiate finite hypotheses on their stated branches. They do not remove the separate source-selection, physical-quantity or controlled-refinement hypotheses.

Carrier signature, screens and observers

The tower panel above orders the layers. Three conventions are shared by every layer that uses the carrier.

Three meanings of screen

The word "screen" is used for three related objects that are identified only through an explicit map.

1. The *local carrier boundary* is the twelve-port oriented interface of one icosahedral carrier. Its incidence has $(V, E, F) = (12, 30, 20)$.
2. The *federation screen* is the routed system of interfaces, records, repairs and checkpoints of many carriers at finite cutoff.
3. The *support screen* is the observer-facing geometric chart. On the spherical branch it is the refined conformal S^2 used for caps, collars, modular flow and Lorentz reconstruction.

Local icosahedral incidence does not determine the topology of the federation nerve. Identical local carriers can be routed as a path, a cycle, a higher-genus complex or a spherical complex. The map from routed carriers to a support-visible spherical nerve is a physical bridge with its own premises.

Presentation invariance and architecture sensitivity

OPH is invariant under changes of presentation that preserve the complete observer-visible carrier signature. On the icosahedral branch that signature is

$$\mathcal{C}_{i,r} = \big(\mathcal{A}_{i,r}, \rho_{i,r}, P_{i,r}, I^{\mathrm{or}}_{i,r}, \mathcal{R}_{i,r}, \mathcal{U}_{i,r}, \mathsf{Chk}_{i,r}, \mathsf{Resp}_{i,r}, c_{sr}\big),$$

with accessible algebra, state, port set, oriented incidence, record algebra, repair or feedback interface, checkpoints, visible response law and refinement lineage. Hidden coordinates, port names, worker partitions, materials and wiring presentations are silent when an isomorphism preserves this tuple and its error model. A change of port number, incidence, orientation, accessible algebra, response, repair law, clock or refinement lineage can change the admitted physics. A cube and an icosahedron are different carrier contracts even when built from the same material.

When a carrier is an observer

A carrier body realizes an observer when it supplies bounded access, self-readback, durable records, record-conditioned feedback, boundary prediction against controls and checkpoint continuation. One carrier may pass that test, or a connected subfederation may pass it instead; no theorem fixes the primitive observer size by counting carrier bodies. Equal-frequency phase locking can synchronize an interface; it makes the interface an observer only together with these record and feedback clauses.

How the layers compose

Consensus, geometry and gravity are composable branches of one carrier tower: accepted repair produces the quotient-visible normal form, the exact port response supplies the rank-three slow-response carrier, and the support screen feeds the modular and Lorentz constructions of the space-time layer. The gauge layer uses the same carrier architecture, where the complete twelve-port

response and endogenous holonomy force the abstract local Lie algebra $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. The matrix current and matter action are conditional realizations; a common source construction joining them to the geometric tower is the matter layer's open join. Matching dimensions or symmetry labels does not supply that join.

Contents

1 Observers, Records and Clocks **9**

2 Observerhood and the Common-Origin Bridge **15**

3 Records and Quantum Identities **18**
- 3.1 Finite event algebra . . . 18
- 3.2 The Born weight: effect-valuation theorem and frame boundary . . . 18
- 3.3 Record algebra, measurement, Bell correlations and checkpoints . . . 20
- 3.4 Finite regional observer nets . . . 21
- 3.5 A noncommutative join of the retained observer carriers . . . 22

4 Common-World Joins on the Twelve-Port Carrier **23**

5 Histories, Actions and Mechanics **29**

6 Thermodynamics from Conditional Repair **39**
- 6.1 Protected records with decaying repair memory . . . 42
- 6.2 Elementary corollaries of the record and screen structure . . . 46

7 The Closure Equations **49**
- 7.1 Local closure for the pixel ratio . . . 49
- 7.2 The unified-coupling proof record and the clock hierarchy . . . 51
- 7.3 Global closure for total capacity . . . 53
- 7.4 Comparison coordinates that are not outputs . . . 58

8 Comparison with other unification approaches **59**

9 Conclusion **61**

A Finite Field Histories Read by Observers **62**

B State-and-Law Habitat **99**

C Observer Registry and Clock-Naturality Certificate **102**

1 Observers, Records and Clocks

An abstract observer patch is the algebraic object

$$\mathsf{O}_i = (\mathcal{A}_i, \rho_i, \mathcal{R}_i, \{(\mathcal{I}_e, \pi_{i,e})\}_{e \ni i}, \mathcal{U}_i, \mathrm{Chk}_i)\,.$$

Here $\mathcal{A}_i$ is the finite or regulated accessible algebra, ρ_i the accessible state, $\mathcal{R}_i$ the exact or approximate record algebra, $\mathcal{I}_e$ the overlap-visible interface algebra, $\pi_{i,e}$ the visible restriction, $\mathcal{U}_i$ the allowed update and repair instruments, and Chk_i the checkpoint data used for continuation.

A support patch, such as a cap $P_i \subset S^2$ or a causal diamond, is a geometric chart of the abstract patch on a geometric branch. A carrier patch is a physical or digital implementation that realizes the same visible interface and record statistics within a declared error. Echosahedral hardware is therefore a reference carrier architecture for a fixed-cutoff implementation surface. The physical claim lives at the quotient of visible records, interface statistics, repair maps, and checkpoint continuation.

The quotient removes presentation data such as coordinates, labels, worker layout, and hidden ancillas. Quotient-visible port number, oriented incidence, topology, and response maps are physical branch data when the observer can recover them. Implementation invariance therefore does not imply carrier neutrality.

The observer patch is therefore the formal access structure used by the proof. A human observer, a horizon cap, and a hardware module can realize or chart it, but none of those examples is the definition. Physical predictions attach to quotient data: visible interface statistics, record readout, repair maps, and checkpoint continuation. A carrier implementation is admissible only when those data are invariant within its declared error model.

Definition 1.1 (Semantic observer history)**.** *For an observer token O, the observer-readable history is a labeled informational poset*

$$\mathsf{H}_O = (E_O, \preceq_O, \ell_O).$$

The elements of E_O are semantic record events, $\preceq_O$ is informational dependence, and ℓ_O is the observer-visible event label. The strict part of the order is the transitive closure of certified read-from parenthood, and $\preceq_O$ is its reflexive closure. The strict part is machine-checked as the least strict extension of the direct parent edges. A direct edge requires the child to certify the resource, the parent to write it, the child's pre-commit snapshot to name that parent as writer, and the pre-commit value to equal the parent's post-commit value; a raw writer label alone creates no edge. One adjacent independent swap preserves the authenticated direct-parent relation and its closure only under the fresh-ID and duplicate-free hypotheses on both executions; the general swap-chain theorem preserves only the raw writer-citation relation. Executor stutters are removed before the commit list is formed. Event identity is defined from the canonical semantic payload, observer token, visible footprint, and semantic dependency parents. It is not derived from worker ID, repair iteration, queue position, retry counter, timestamp, or packet latency metadata. This definition is not a theorem that arbitrary schedule rewrites preserve labels or payloads. Latency is execution gauge only when it does not alter informational dependence, delivered data, timeout-sensitive operations, or operational clock records.

The strict order of Definition 1.1 is causal-set-like in a limited sense: Lean proves that reflexive authenticated ancestry is a partial order whose intervals are finite at each cutoff, so the order-theoretic causal-set axioms hold [1, 2]. The relation is informational read-from dependence. Its canonical source height, zero at roots and one plus the maximum direct-parent height otherwise, is exactly the longest authenticated parent-chain length, and every finite strict partial order compiles exactly into an abstract authenticated log. An OPH commit log is not a Poisson sprinkling and its

event count is not a spacetime volume; physical embedding, a volume law, a four-dimensional manifold limit and Lorentz kinematics are developed, with their premises, in the spacetime paper [17]. There a rank-three source Gram quotient and a real axis give a four-dimensional target carrier of inertia $(1, 3)$, source placements become exact finite causal-order embeddings under stated speed and spacelikeness conditions, and conservative record populations with a declared model clock have flat $1 + 3$ order and count-volume limits. Thirteen finite covariance and isometry table identities of the rank-three port-record completion use Lean's `native_decide`, so those subreceipts trust the native compiler and runtime in addition to the kernel and are not kernel-only proofs.

Definition 1.2 (Observer registry groupoid and namespaces)**.** *Observer identities live in a registry groupoid, not in a flat UUID table. Objects are observer tokens*

$$\mathrm{RID}(O) = (\mathrm{kind}(O), [\mathrm{birth}(O)]_{\mathsf{H}}, \mathrm{lineage}(O)),$$

where $\mathrm{kind}(O)$ *lies in the disjoint namespace*

$$\{\textit{patch observer},\ \textit{cap observer},\ \textit{future observer}\}.$$

The birth event is a semantic event class in H*, and lineage is carried by explicit continuation, split, and merge arrows. Local registries on overlapping presentations must preserve observer kind, birth event, lineage arrows, visible observer structure, and checkpoint cuts. The global registry is the descent groupoid or colimit of those local registries when the cocycle and trivial-monodromy checks pass. A split creates child tokens with parent links; it does not silently assign one identity to multiple descendants.*

Definition 1.3 (Operational observer clock)**.** *The modular parameter of an observer-facing cap pair is not, by itself, an arbitrary operational clock reading. A clock claim supplies an observer-accessible instrument*

$$\Theta_O : \mathrm{Chk}_O(D) \to \mathrm{Prob}(\mathbb{R})$$

and a declared affine calibration law. For a corresponding observer O'*, the required finite or exact certificate has the form*

$$\Theta_{O'}(\Phi_{O*}\rho) \approx (A_{a,b})_*\Theta_O(\rho), \qquad A_{a,b}(\tau) = a\tau + b,$$

with an explicit residual bound from state mismatch, calibration error, and record-readout error. Execution counters are provenance; observer record order, modular parameter, and clock uncertainty are separate semantic or operational fields.

Remark 1.4 (Observer-clock naturality boundary)**.** *This paper proves observer-first quotient and readback claims only where the cited theorem surface supplies the required data. Scheduler-independent observer history and clock naturality require a history-augmented quotient relation with history-coherent diamonds, registry descent, state-preserving observer-algebra extraction, support-cap chart naturality, and the operational clock instrument above. A public evidence bundle can falsify the claim by showing counter-dependent semantic event keys, duplicate or namespace-colliding registry identities, broken registry cocycles, non-isomorphic semantic history DAGs across executors, or clock residuals exceeding the declared bound. Implementation field names alone are not the proof.*

The registry, history and clock data above form one finite evidence contract. Theorem C.1 in Appendix C states the conditions under which executor metadata cannot change the observer-readable history class, registry class or clock law.

The formal time-and-order ledger enforces this boundary at the type level. Universe closure, repair execution order, observer record order, modular parameter, worldline realization, clock readout, proper time, and an optional global time function are eight separate structures. In the committed source environment, the complete matrix of 56 ordered transitive coercions between distinct ledger layers fails to elaborate. Interpretations pass through an explicitly named realization map without a function coercion. More strongly, composition with any strictly increasing real function preserves the bare clock interface. A three-record control sends readings (0,1,2) to (0,1,8) and proves that the resulting clock is not related to the original by any positive affine map. Record order therefore supplies order data and an order-compatible scalar readout, while fixing neither affine structure, origin, nor physical unit.

The bounded observer-time continuation keeps the missing receipt visible. If a supplied event atlas and record map satisfy an affine coordinate law along a supplied future-unit timelike direction, every precedence has future-timelike displacement in every overlapping chart. On this calibrated branch the positive clock increment along that same supplied affine history is additive, and its square equals the chart-invariant Lorentz quadratic interval. One shared public event cannot determine affine rate and origin. Two ordered shared events determine the unique positive-affine interpolation of their four readings. At a third shared event, affine consistency is equivalent to an exact cross-multiplication equation. Distinction of the event and both readings from the two anchors makes this a nondegenerate check with no new fit parameter. Calling it held out requires a separate predesignation and custody protocol. The affine and cubic controls synthesize separate atlases from their clocks; they pass and fail this algebraic check without constructing two physical clocks on one calibrated geometry. A separate literal three-record, two-chart example proves the geometric interface is inhabited. These are bounded conditional algebraic theorems with finite controls: the source atlas, affine unit-speed law, shared-event correspondences, refinement transport, physical instrument, common physical clock geometry, predesignation protocol, three-clock network, and SI unit are not constructed, and no global time function or modular-to-clock identity follows.

The companion formal tower interface packages the next structural layer in one object: finite observer and record fibres, observer-indexed record orders, private matrix algebras, commutative public subalgebras, selected states and generators, and functorial refinement maps with explicit compatibility laws. An exact constant adaptor packages one existing projective partition and certified state using a discrete order and zero generator. This is an inhabited packaging witness, not a nonconstant source-derived tower, repair endpoint, causal net, physical evolution, geometry, clock, or continuum limit.

The observer-patch definition does not fix the primitive observer size. A1–A3 require finite or regulated access at fixed cutoff and permit connected support regions with different finite or regulated algebras. A primitive observer may therefore be a variable finite federation of screen cells. The stronger claim that elementary carriers are isomorphic fixed-capacity cells would require a separate branch: homogeneous UV cellulation, isomorphic local cell algebras, equivariant overlap maps, refinement preservation of the cell type, and a proof identifying one cell or a fixed block of cells with the primitive carrier. One Echosahedron is therefore a candidate primitive carrier on the homogeneous branch. It is not the definition of an observer patch.

Calibration is an import. The local readout of a selected normal form supplies an internal record history, not a physical clock. Operational time requires an observer-readable transition process, event correspondence and affine calibration; it is not the external update parameter. In the Lean corpus the 2019 SI anchors, the cesium hyperfine frequency, the speed of light and the Planck constant, enter as exact rational literals with a unique inhabitant, so the dictionary itself

carries no measurement uncertainty, and the entire empirical content of the import is one declared tick duration. Under a declared tick the internal step count, angular rate and energy gap convert exactly to laboratory seconds, hertz and joules, the conversions compose affinely with the committed clock algebra, and two distinct tick declarations give distinct laboratory frequencies for the same internal rate, so the calibration is an import and never a derivation.

The premises A1–A3.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0 ,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \arg\min_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

The closure hypothesis. The theorem premises A1–A3 state the library's hypotheses on the carrier and its response, on internal transport and on local state selection. The closure hypothesis states that the Universe is a timeless, self-referential structure whose observers recover the specification of their own world from their records and realize it, so that the description and the

described system are one object. Every quantity that has both a construction-side reading and a readback-side reading must take the same value once a typed bridge proves that the two readings denote one invariant. The equality is forced by self-identity. Constructing that bridge and the return map is part of the physics. Existence, uniqueness, and stability are separate determinacy tests on the resulting equation. The name given to such an equation carries no mathematical weight: after the typed identification, unequal readings would describe two systems rather than the single self-referential universe.

The closed object is the complete inhabited history, whose internal events and responses to interventions are its proposed intrinsic computation. This reading requires no external computer or playback time. It is distinct from fixedness of a repair state, a public record, or a probability law, and does not itself construct the history or prove a final cosmic endpoint.

The local screen-grain proposal

$$P = \varphi + \sqrt{\pi}/A_T(P),$$

with $A_T(P)$ the inverse coupling returned by the declared source map, has one interval-certified fixed point per declared map on its analytic domain. Interpreting that root as the laboratory coupling additionally requires a target-independent choice of the physical map, a typed bridge identifying the two sides as readings of one quantity, and same-scheme hadronic spectral transport to the Thomson endpoint. The displayed map's certified fixed point sits at $P = 1.6309720959$ and returns $A_T = 136.9948352$; the certified gauge-width variant's fixed point sits at $P = 1.6309682414$ and returns 137.0356601. Through the same outer equation the measured 2022 CODATA coupling $\alpha^{-1} = 137.035999177(21)$ reads back the grain value $P = 1.6309682094$. The comparisons stand at 3.0×10^{-4} relative for the displayed map and at 2.5×10^{-6} for the gauge-width variant, as diagnostics of the declared source maps. The latter absolute residual is 3.3904×10^{-4}, compared with experimental standard uncertainty 2.1×10^{-8}; this is not agreement at measured precision. The gauge width supplies 99.2 percent of the gap between the bare root and the measured value; the residual 3.3904×10^{-4} is addressed to same-scheme hadronic transport to the Thomson endpoint.

The direct global proposal is

$$N = \log M_0(\mathfrak{U}_N).$$

The bounded completion class defined by base agreement, positivity, and the carrier bound does not select a unique cosmic value. Universal all-rung membership in a complete A1–A3 capacity-source contract and the corresponding executable-to-Lean bridge are absent. The incomplete antecedent determines no direct N, and no theorem extends the bounded-class conclusion to the complete source class. A positive result must complete the source antecedent and prove one physical zero, either within the A1–A3 source contract or through a separately named stronger source law. A separate common-load branch starts from

$$N_0 = \pi \exp\!\left(\frac{6\pi}{P\,\alpha_U(P)}\right).$$

On the finite collar branch, the declared total reserve expectation $P/4$ and six-class equidistribution give presence probability $P/24$ for each declared class. If one class is physically selected as the blocked event, its scalar-weighted presence receipt is discharged, and that normalized collar-survival factor is proved to act on the global capacity, the corresponding candidate is

$$N_{\mathrm{pres}} = N_0\left(1 - \frac{P}{24}\right), \qquad \ln\frac{N_{\mathrm{pres}}}{\pi} = \frac{6\pi}{P\,\alpha_U(P)} + \ln\left(1 - \frac{P}{24}\right).$$

The exponential alternative $N_{\mathrm{Pois}} = N_0 e^{-P/24}$ requires a separate mean-count or continuum carrier. Exact neutral, one-class, and six-class-total actions share the declared local datum, obey positive composition and finite cut-count regrouping, and disagree globally. The finite source therefore selects no action or blocked-event semantics. The source class determines no named-law value, and its horizon branch has no capacity object. A stronger source-derived action would require the screen/electroweak bridge, physical common-load bridge, physical $\mathbb{Z}_6$ seam action, scalar-weighted receipt, and horizon-record identification. On the source-forward numerical branch,

$$N_0 = 3.5321 \times 10^{122},$$

so the finite-presence candidate is 3.2921×10^{122} and the exponential candidate is 3.3001×10^{122}. The full weighted Planck base-ΛCDM chain gives the comparison coordinate 3.3129×10^{122}. The respective residuals are -0.63 percent and -0.39 percent. Both comparisons were exposed before the branch choice and carry no predictive weight.

The descent repair law turns compatible local corrections into a schedule-independent public record once termination and the local diamond hold; the consensus paper proves this finite normal-form theorem and its countermodels [19]. Counting and trace give exact measures on the carrier algebra, and agreement and repair decide whether those measures survive on the public quotient. A physical readout must preserve the resulting measure before it can be identified with a laboratory or spacetime quantity. The local gauge Lie type that complete reversible response and endogenous overlap transport force, and the conditional matter image, are proved in the gauge paper [18]; the particle paper carries the matter and mass calculations [21].

2 Observerhood and the Common-Origin Bridge

The observer named in the title has a theorem-level bounded receipt. Over any instance of the companion formal tower interface, its base structure packages clauses 1 through 6 and the own-observer half of clause 7 as typed data: a bounded accessible star subalgebra of the private algebra containing the observer's own public record algebra; a self-readback map sending accessible elements into that public algebra and fixing each of its elements; durability, with every record element annihilated by the observer's declared generator; record-conditioned control maps preserving the accessible interface; a prediction map whose output never strictly precedes its input in the observer's own record order; label and prediction families commuting with the declared refinement maps; and a declared readout equal to the trace pairing of each record element against the observer's own state, with every record element fixed by every declared control. Readback idempotence and refinement stability of the readout follow. An explicit witness inhabits the receipt away from the constant adaptor: private algebra $M_2(\mathbb{C})$, a nonzero generator given by the scaled commutator with $\mathrm{diag}(1,2)$, which annihilates every diagonal record element while moving the raising matrix unit, control by conjugation with the involution $\mathrm{diag}(1,-1)$, three chain-ordered records, and a nonconstant non-identity prediction map. Four negative controls (an erasing readback, an identity generator, a resetting prediction, and a constant misreading) each violate one attained field on the same data. The base structure is operational and finite: it selects no unique observer over a given tower, attaches no physical instrument, makes no consciousness claim, and supplies no source realization of the tower data. A separate fixed-regulator mixin closes the cross-observer half on an access cut. It requires two distinct labels, exact agreement of both accessible interfaces with the cut, equality of every committed record after the two typed restrictions to the owner-region meet, equality of own readouts, and a nonzero common restriction. The common section is thereby accessible to both observers. Its proper-meet witness uses two distinct diagonal owner algebras,

a scalar bottom algebra, and corner-character restrictions: observer-indexed records differ before restriction and agree nontrivially only after restriction. A one-corner mutation makes the overlap receipt false. The operational overlap-event consumer takes two operational observers owning two charts in which one event is visible and proves equality of the two owner-region restrictions of its packet record together with two-sided self-readback. Thus all seven operational clauses are inhabited at one finite cut. Cross-regulator naturality, triple-overlap coherence, and source realization are not supplied [29].

An access cut on the finite regional observer net of Section 3.4 assigns each observer one declared support region and one accessible star subalgebra squeezed between the observer's public record algebra and the regional algebra of that region, stable under the declared restrictions below it; the observer-local algebra at a region is the meet of the regional algebra with the accessible algebra, and the net receipts transport to this meet. A finite event world over the supplied Lorentz soldering binds one packet p_e, with packet carrier fixed independently of the quotient state, to each event e, with declared chart ownership. The common-origin theorem states that for every active visible chart/event pair (i, e),

$$\operatorname{supp}(\mathsf{p}_e) \preceq R_{O(i)}, \qquad \operatorname{rec}(\mathsf{p}_e) \in \mathcal{L}_{O(i)}\big(\operatorname{supp}(\mathsf{p}_e)\big) \cap \mathcal{P}_{O(i)}, \qquad x_i(e) = \operatorname{geo}_i(\mathsf{p}_e),$$

where $O(i)$ is the declared owner of chart i, $R_{O(i)}$ its observer region, $\mathcal{L}_{O(i)}$ its observer-local algebra, and $\mathcal{P}_{O(i)}$ its public record algebra: one event packet determines the event support region below the owner's region, the public record observable in the meet of the observer-local algebra at that support with the owner's public algebra, and the Lorentz chart coordinate as a readout of the same packet. A selected coarse-to-fine refinement transports all three conjuncts. The witness chain is strict, with the public algebra strictly inside the accessible algebra strictly inside the private algebra, and the witness records separate packets. Charts and observers are distinct types, ownership is declared data, and no physical region, coverage, instrument, spacetime, causal cone, event actualization, or continuum object is claimed [29].

Coexistence of regional operator algebras on this net interface has an exact boundary.

Theorem 2.1 (Scalar-character obstruction and character-block coexistence)**.** *For every $n \geq 2$ there is no unital complex-algebra homomorphism from the full matrix algebra $M_n(\mathbb{C})$ into the scalars, and no unital complex-algebra homomorphism into the scalars accepts an anticommuting pair of units. On this net interface, at any regulator of nonzero dimension, a region whose local algebra consists of scalars admits above it no region whose local algebra contains a unital copy of $M_n(\mathbb{C})$ with $n \geq 2$: the declared star-homomorphic restriction retraction would compose with the copy to a scalar-valued unital homomorphism on a full matrix algebra. At any such regulator, a region lying above a region with commutative local algebra carries no anticommuting pair of units: the declared restriction would send the pair to commuting units whose product equals its own negation. No full matrix factor of dimension at least two sits above a scalar overlap region, and the standard tensor-factor local-algebra picture with trivial overlap is impossible on this interface. The boundary is inhabited: over a dimension-four private algebra, the block algebras of matrices* $\operatorname{diag}(a, a, A)$ *and* $\operatorname{diag}(B, b, b)$ *with arbitrary two-by-two blocks A and B are each noncommutative, commute elementwise, neither contains the other, and each retracts onto the scalar overlap algebra through the character evaluating its one-dimensional corner block.*

Proof. The matrix ring $M_n(\mathbb{C})$ is simple, so the kernel of a unital homomorphism into the scalars is trivial and the homomorphism is injective, while the off-diagonal matrix unit is a nonzero element whose square is zero and whose scalar image would be a nonzero complex number whose square is zero. For the anticommuting form, a unital homomorphism into the scalars sends units to nonzero

scalars, and applying it to the anticommutation relation forces the product of the two image scalars to vanish. The interface statements compose these arguments with the declared restriction into the scalar or commutative region. The inhabitant is finite linear algebra over one declared regulator with identity repair maps; the general matrix obstruction, the interface compositions, the interface laws, both noncommutativity witnesses, the mutual noncontainment, and the corner-character retractions are checked in Lean [29]. □

The declared finite C^*-algebra/retraction interface is therefore compatible with noncommutative regional coexistence through the displayed character-block witness. The obstruction excludes the full-matrix and anticommuting presentations stated in the theorem; it does not classify every possible coexistence presentation. The declared overlap algebra is a proper subalgebra of the set intersection of the two block algebras, and genuine coverage semantics or regional-factor receipts for a source-produced net are not supplied.

Dynamics on the private block carries a matching rigidity statement.

Theorem 2.2 (Finite Stone converse on one full private block)**.** *Let α_t, $t \in \mathbb{R}$, be star automorphisms of one full finite matrix block $M_n(\mathbb{C})$ with $\alpha_0 = \mathrm{id}$, $\alpha_{s+t} = \alpha_s \circ \alpha_t$, and every orbit map $t \mapsto \alpha_t(x)$ continuous. Then there is a self-adjoint $H \in M_n(\mathbb{C})$ with*

$$\alpha_t(x) = e^{-itH}\, x\, e^{itH} \qquad \text{for all real } t \text{ and all } x \in M_n(\mathbb{C});$$

the conjugating family $U_t = e^{-itH}$ is unitary with $U_0 = 1$ and $U_{s+t} = U_s U_t$, and H is unique up to one additive real scalar multiple of the identity.

Proof. Continuity upgrades to differentiability at parameter zero by interval averaging: the group acts on the matrix-unit basis as a norm-continuous matrix family whose short-interval average is invertible, and the fundamental theorem of calculus recovers the family from its primitive. The derivative at zero is a star-compatible Leibniz endomorphism. Every complex-linear Leibniz endomorphism of the block is an inner derivation, with an explicit matrix-unit witness proved from scratch; star compatibility symmetrizes the witness into one self-adjoint generator; two generators with one commutator action differ by a real scalar multiple of the identity; and the group differential equation identifies the flow with conjugation by the coherent unitary family. Every step is checked in Lean [29]. □

On the finite private block, a pointwise-continuous symmetry flow of the observables therefore has no choice except Hamiltonian form; the additive energy offset is the only freedom. The theorem extends to the general finite private algebra: every unital star subalgebra of a finite complex matrix algebra, containing the ambient identity, is star-isomorphic to a finite direct sum of full matrix blocks, and on that block model a pointwise-continuous star-automorphism group fixes every block projection and is blockwise conjugation by the unitary family of one time-independent self-adjoint Hamiltonian per block. Block fixing is derived rather than assumed; a single automorphism may swap isomorphic blocks, and the swap is realized on the square two-block algebra. The parameter is a real number; no physical clock, source-selected Hamiltonian, or physical time identification is claimed.

The measurement and dynamics surfaces share one object. A composed frame-duality theorem (`schroedinger_frame_duality`) consumes the registered record representation, operational effect additivity, and one supplied pointwise-continuous flow through a single typed bundle and proves that the unique Busch–Gleason state of the Heisenberg-shifted registered frame is exactly the propagator conjugation of the unique state, at every flow parameter: the Schrödinger picture of the Born frame is forced by the Heisenberg action of the rigid generator. The flow stays a supplied

premise and its parameter is not physical time; what the theorem removes is any freedom to represent the measurement statistics and the dynamics on different state objects.

3 Records and Quantum Identities

3.1 Finite event algebra

The span of a projective record partition is a commutative star subalgebra exactly star-isomorphic to complex functions on its nonzero projector labels. A common linear isometry can copy two distinct sharp states from one normalized blank only when they are orthogonal; the full mixed-state no-broadcasting implication enters through an explicit adapter. Partition pinching and partition averaging have explicit normalized Kraus families and preserve trace, and a separate module proves the averaging, pinching and nonnegative-parameter relaxation channels completely positive and trace preserving. The two-scale public relaxation and Poissonized repair families obey exact semigroup laws, and on bounded endomorphisms of a real Banach space the latter is exactly $\exp[t\gamma(E-I)] = E + e^{-\gamma t}(I-E)$; this operator exponential supplies no source rate or physical clock.

Positive unital maps of the active-record function algebra are exactly real row-stochastic kernels. Every star automorphism of the public function algebra is uniquely pullback by a label permutation, so every pointwise-continuous real-parameter group of public star automorphisms is trivial. On one finite full private matrix block, every star automorphism is unitarily inner, and a fixed self-adjoint Hamiltonian gives a unitary real-parameter flow obeying the von Neumann equation; the converse is Theorem 2.2. For nonselective projective pinching, $S(\mathcal{E}\rho) - S(\rho) = D(\rho\|\mathcal{E}\rho) \geq 0$; this analytic matrix identity is separate from a normalized selective Lüders branch. Arbitrary partition pinching is exactly the uniform random-unitary average over all 2^k independently signed block reflections. A complementary support boundary is exact: the totalized matrix logarithm sends every projection to zero, so its raw relative-entropy formula assigns zero to an orthogonal pure-state pair whose support inclusion fails. The projective partition and input state are finite source data, and the public automorphism classification is complete for the exact finite active-record function algebra and its matrix realization.

3.2 The Born weight: effect-valuation theorem and frame boundary

Theorem 3.1 (Finite effect-valuation representation)**.** *Let $\mathcal{A}$ be a finite matrix algebra. Every valuation on the effects of $\mathcal{A}$ that is nonnegative, normalized and additive on coexisting effects equals $E \mapsto \mathrm{Tr}(\rho E)$ for a unique density operator ρ. This holds in every finite dimension including two, with no continuity axiom.*

The theorem is checked in Lean [29]. Its premise, noncontextual additivity on all coexisting effects, is the load-bearing clause, and the finite receipts locate it rather than derive it. The twelve declared central port atoms form one classical context with an eleven-dimensional weight simplex. A distinct qubit spinor adapter has six disjoint antipodal binary contexts whose additive weights have affine dimension six; the trace-one Hermitian/Born slice is three-dimensional and obeys three exact golden-ratio relations. Representation there is unique when it exists, but exact weights show that neither existence nor positivity follows. On the full celestial or Bloch sphere, the continuous weight $F(n) = (1+n_z^3)/2$ is normalized on antipodes yet non-affine. Sharp binary webs admit the cube valuation as a non-Born extension, and one unsharp three-effect context excludes that particular extension. The same finite battery admits the planar cubic response $F_y(n) = (1+n_y^3)/2$,

which agrees with the maximally mixed Born weight on every displayed axis while remaining non-affine away from them. The projector webs alone therefore underdetermine the weight, and one finite unsharp battery does not force a global valuation.

The real S_3 algebraic web attached to source-realized gauge labels is blind to the Pauli-Y direction: two distinct complex states give equal weights on every declared projector candidate. The missing algebraic direction is exact. If P is the declared diagonal projector and Q its noncommuting image under the declared two-dimensional representation of a source-realized gauge label, then

$$\frac{I}{2} - \frac{2\sqrt{3}}{3} i(QP - PQ) = \rho_{Y+}.$$

Together with P and Q, this phase lift separates every fixed-trace 2×2 matrix. Conversely, every effect generated by real linear coarse graining, complements and real Kraus pullbacks remains real symmetric and gives the same weight to ρ_{Y+} and ρ_{Y-}. Entrywise conjugation fixes the complexified real web pointwise and exchanges ρ_{Y+} with ρ_{Y-}, so the ambiguity between the two candidate completions is one conjugation orbit, and any conjugation-fixed frame conflates each state with its conjugate. The phase lift lies in the complex operator algebra generated by the source-attached pair; it is a declared effect, not an operation or instrument.

A post-hoc statistic on one bounded run, executed under an unrelated locally hash-pinned common-reference contract, is reversal-odd on raw counts: a designated pair counts 1343 against 0, and a designated three-cycle has count products 1239691068 against 0, machine-checked as the unique maximizing orbits of the post-hoc designation rule. Transposing the counted order flips the selection bit, and a synthetic symmetric count table fails the strict-inequality applicability condition. Under a declared transport convention the selected completion coincides with the phase lift, either candidate completes state tomography, and Theorem 3.1 pins the represented state by the three completed-frame weights. The opposite pairing is equally admissible, neither the statistic nor its designation rule was preregistered, and the counted orientation carries no arrow-of-time claim.

On the committed static effect and count fixture this boundary is exact. Every committed effect has trace one, so a sum of two committed effects is an effect precisely when it resolves the identity, and any assignment sending the sure effect to one and normalized in each context is additive on every coexistent sum formed within the committed set. A cubic deformation of the run Born functional, shaped to vanish on the committed set, reproduces every generated expected frequency, stays in $[0, 1]$ on all effects, obeys the complement rule and is additive on every committed coexistent sum, yet it fails additivity at an explicit effect pair with values $35/64$ against $143/256$ and is the Born functional of no state. Conversely, full cross-context additivity together with the static fit values forces the Born functional of the declared matrix uniquely. Between the committed clauses and full cross-context additivity lies an injective affine family through the run Born member with additivity residue exactly $-(3/256)\,t$ at the witness pair. On the convex interval $0 \le t \le 1$ every member is probability-valued on every effect; outside that interval only the algebraic identities are asserted. Full additivity selects the Born endpoint from the probability-valued continuum.

A declared Lüders instrument on the committed effect table supplies completely positive outcome maps, a trace-preserving summed channel per context and induced effects equal to the table; the table does not determine the instrument. The placeholder binding schema splits static compatibility fields from the evidence a source-bound run would have to supply. Its determined matrix and declared expected-frequency table are inhabited by every compatible declared instrument; they are not a source-produced preparation, public outcome or readback. The record's natural-number digest and inhabited proposition fields are freely stipulable, so the schema authenticates no producer or custody artifact. Positive closure of the Born attachment therefore requires a source-realized complex-tomographically complete effect and instrument web, an operational coarse-graining or

equivalence theorem deriving noncontextual additivity, and only then preregistered frequencies as validation; frequencies alone do not prove the universal valuation law.

3.3 Record algebra, measurement, Bell correlations and checkpoints

The fixed-cutoff record, measurement, Bell and checkpoint packages are proved in the screen-microphysics paper [20] and the consensus paper [19]. They are stated here in the form the observer layer uses.

Public normal forms. On its declared fixed-cutoff branch, the consensus construction gives a finite constraint code whose codewords are the globally consistent states $C = \Phi^{-1}(0)$; it is not automatically a quantum error-correcting or topological code, and its graph min-cut does not determine code distance. For the declared accepted repair law, the local-fit contract makes Φ a Lyapunov functional, the union-collar gluing package supplies the local diamond on the physical quotient, and that confluence condition together with repair completeness yields a unique schedule-independent normal form from a fixed initial quotient state. On the abelian branch, global consistency holds exactly when cycle holonomy vanishes; on the noncentral branch the crossed-module gluing orbit $q_\Sigma \in \check{H}^2(N_\Sigma, H_\Sigma \to G_\Sigma)$ labels the fixed-cutoff gluing orbit. The repair law descends to the overlap-invariant quotient, and the induced terminal state on the declared physical observable algebras is unique there even when microscopic representatives differ by gauge or sector relabelings.

Record algebra and the Born–Lüders package. For each completed compare/write/verify slice of the regulated microphysics, the declared pointer and overlap-sector projectors generate a finite commutative central record algebra $\mathcal{Z}_{\mathrm{rec}}$. Given the supplied density-state valuation and declared Lüders instrument, every observer-accessible event E in that algebra has probability $\mathbb{P}(E) = \mathrm{Tr}(\rho P_E)$, and the selected conditioning on E gives

$$\rho\,|_E = \frac{P_E \rho P_E}{\mathrm{Tr}(\rho P_E)}.$$

The effect projectors do not determine this instrument; same-effect non-Lüders instruments remain possible. If a practical readout uses projectors Q_a with commuting central reference projectors $\widehat{Q}_a$ on the same slots and $\delta_{\mathrm{rec}} = \max_a \|Q_a - \widehat{Q}_a\|$, then $\|[Q_a, Q_b]\| \le 4\,\delta_{\mathrm{rec}}$, and any accessible-state perturbation $\|\widetilde{\rho} - \rho\|_1 \le \varepsilon$ changes each declared elementary record-event probability by at most $\varepsilon + \delta_{\mathrm{rec}}$. Records carried by central projectors are shareable without violating no-cloning for generic quantum states.

Bell correlations. On the declared fixed-cutoff observable algebra, fix commuting left and right wing subalgebras, central binary setting registers, binary projective readouts on each wing and a source state on the joint wing algebra. Then

$$p(a, b \mid x, y) = \mathrm{Tr}\big(\rho_{LR}\, P^{(L)}_{a|x} P^{(R)}_{b|y}\big), \qquad E(x, y) = \mathrm{Tr}\big(\rho_{LR}\, A_x B_y\big),$$

the local marginals are independent of the remote setting, and $\big|E(0,0)+E(0,1)+E(1,0)-E(1,1)\big| \le 2\sqrt{2}$. An explicit two-qubit branch with the stated Pauli readouts saturates $2\sqrt{2}$; Bell-pair preparation and two-qubit factor structure are branch conditions. Reachable states on the committed pair carriers are separable across the declared slot split and obey the classical bound 2 against every slot-local unit-interval readout, jointly diagonal or not; a diagonality-preserving map cannot reach the declared Bell-state witness, and a product-diagonal state paired with cross-commuting settings

outside both slots reaches $2\sqrt{2}$. Slot membership relative to the split therefore carries the classical bound.

Collar factorization, checkpoints and restoration. For a collar tripartition A-B-D, the exact Markov normal form is used only when $I(A:D|B)=0$ holds literally, or along a controlled fixed-collar family whose exact-Markov replacement modulus tends to zero. Then

$$\rho_{ABD} = \bigoplus_{\alpha} p_\alpha \, \rho^{(\alpha)}_{Ab^\alpha_L} \otimes \rho^{(\alpha)}_{b^\alpha_R D},$$

with classical sector label α in the center. A checkpoint is the tuple $\mathcal{C} = \big(R, \alpha, \rho^{(\alpha)}_{\mathrm{int}}\big)$, and for a compatible target environment state the restored state is $\rho^{(\alpha)}_{\mathrm{int}} \otimes \sigma^{(\alpha)}_{\mathrm{env}}$. For approximate Markov collars, with ε the conditional mutual information in nats, a Fawzi–Renner recovery channel gives

$$\big\|\rho_{ABD} - (\mathrm{id}_A \otimes \mathcal{R}_{B\to BD})(\rho_{AB})\big\|_1 \le 2\sqrt{1-e^{-\varepsilon}} \le 2\sqrt{\varepsilon}.$$

The fixed-cutoff checkpoint theorem is stronger than this recoverability estimate: for an observer patch whose checkpoint consists of the observer-facing record algebra, the accessible local state, the future update schedule and the visible overlap interface data, exact restoration reproduces the full future law of observer-accessible events, and an ε-accurate restoration changes that law by at most ε in total variation [20]. The package is same-interface restoration; substrate transfer, continuation across redesigned environments and uniqueness of an observer-supporting sector are outside it.

3.4 Finite regional observer nets

A proof-carrying enrichment of one finite consensus tower packages a finite region order, isotony, exact commutation on declared-disjoint regions, covariant refinement, compatible regional expectations and idempotent local repair. A declared repair that fixes remote observables gives exact Heisenberg nondisturbance for every matrix state. Compatible local sections have a unique glue only on nonempty finite subregion families carrying the explicit restriction and uniqueness receipts; the interface name `FiniteCover` includes no joint-coverage law. A separately declared basis split and Kraus completeness give a generic reindexed marginal identity, and no theorem identifies the regional algebras with its matrix factors. Contravariant restriction maps and their retraction of isotony are strong extra premises, not consequences of an ordinary isotonic net.

A rich-fibre packet gives a stronger conditional construction on one preregistered fresh-seed bounded run: four target-blind greedy-disjoint observers realize split-fibre counts $3, 3, 4, 4$ on pairwise-disjoint twenty-node windows, and the mirrored net carries a noncommutative block algebra with an embedded two-by-two designated-fibre matrix corner at every observer region, kernel-backed window disjointness, elementwise commutation across distinct windows, the character-and-block restriction laws, the computed four-window coverage of the top algebra, and a drop-one control showing that every window is load-bearing. The enrichment rule, region lattice, restriction system, base-point state and identity repair are declared readings, and the matrix corner is not a regional tensor factor.

For each of four retained observers, counted transition operators together with diagonal field projectors generate the full matrix algebra on the realized alphabet. The fully support-disjoint retained pair 86/247 defines a positive rational correlated state with exact counted marginals, recovered by a conditional-expectation diamond whose regional pair erases the correlation, so coverage cannot imply unique reconstruction. A typed linear join embeds the 182-dimensional 86/88

and 169-dimensional 86/247 diagonal layers into a 338-dimensional pushout over their shared 13-dimensional observer-86 record space; the pushout is a linear record construction whose quotient has no induced multiplication. On the same three-observer label carrier, identity-spectator embeddings of the full pair matrix algebras intersect in exactly the full M_{13} hinge and generate M_{2366} (Section 3.5). A separately declared slot exchange gives an alternating regional family whose evolved admitted data generate the next regional algebra. The path-read permutations and the slot exchange are supplied adapters; physical regions, quantum preparations and outcomes, a physical clock and a Cauchy-surface interpretation are outside these finite statements.

Theorem 2.1 bounds every such net: no region containing a unital full matrix algebra of dimension at least two sits above a scalar region, while the character-block witness realizes noncommutative coexistence over a scalar overlap.

3.5 A noncommutative join of the retained observer carriers

The retained observers 86, 88 and 247 have realized label spaces of dimensions 13, 14 and 13. Their 32 aligned source rows define one triple path $q_t = (a_t, b_t, c_t)$, $0 \le t \le 31$, with the two pair paths as its coordinate projections. Use the declared tensor reading $H = (\mathbb{C}^{13} \otimes \mathbb{C}^{14}) \otimes \mathbb{C}^{13}$, and write $M_n = \mathrm{End}(\mathbb{C}^n)$. The operator join uses this 2366-dimensional carrier; it assigns no multiplication to the 338-dimensional linear pushout of diagonal records.

Theorem 3.2 (Full operator join on the common carrier)**.** *The maps $\iota_{88} : M_{182} \to M_{2366}$ and $\iota_{247} : M_{169} \to M_{2366}$, defined by*

$$\iota_{88}(X) = X \otimes I_{13}, \tag{1}$$

$$\iota_{247}(Y)_{(a,b,c),(a',b',c')} = Y_{(a,c),(a',c')}\delta_{bb'}, \tag{2}$$

are injective unital star homomorphisms. Their images intersect in exactly $M_{13} \otimes I_{14} \otimes I_{13}$ and generate all of M_{2366}. The exclusive observer-88 and observer-247 factors commute. The two overlapping pair algebras do not commute.

Proof. Identity-spectator inclusion and coordinate reindexing preserve products, adjoints and the identity. Evaluating a fixed spectator diagonal entry proves injectivity. If $\iota_{88}(X) = \iota_{247}(Y)$, evaluation at $c = c' = 0$ gives

$$X_{(a,b),(a',b')} = Z_{a,a'}\delta_{bb'}, \qquad Z_{a,a'} = Y_{(a,0),(a',0)}.$$

Thus $X = Z \otimes I_{14}$, and injectivity forces $Y = Z \otimes I_{13}$. Conversely these two matrices have the same triple image. This proves the full M_{13} intersection, whose complex vector dimension is 169.

For any triple indices $q = (a, b, c)$, $r = (a', b', c')$, and any hinge label s, the pair matrix units obey

$$\iota_{88}(E_{(a,b),(s,b')})\,\iota_{247}(E_{(s,c),(a',c')}) = E_{q,r}.$$

Every triple matrix unit therefore belongs to the generated algebra. Operators on the exclusive factors commute by the tensor-product rule. The common hinge contains E_{01} and E_{10}, whose two products are the distinct E_{00} and E_{11}, so the pair algebras cannot commute elementwise. □

Counted states and observable retractions. The empirical state and its pair marginals are

$$\rho_{ABC} = \frac{1}{32}\sum_{t=0}^{31} |q_t\rangle\langle q_t|, \qquad \rho_{AB} = \mathrm{Tr}_C\,\rho_{ABC}, \qquad \rho_{AC} = \mathrm{Tr}_B\,\rho_{ABC}.$$

For arbitrary pair observables, these give

$$\mathrm{Tr}(\rho_{ABC}\iota_{88}(X)) = \mathrm{Tr}(\rho_{AB}X), \qquad \mathrm{Tr}(\rho_{ABC}\iota_{247}(Y)) = \mathrm{Tr}(\rho_{AC}Y).$$

Observable retractions instead use normalized partial traces:

$$\frac{1}{13}\,\mathrm{Tr}_C\,\iota_{88}(X) = X, \qquad \frac{1}{14}\,\mathrm{Tr}_B\,\iota_{247}(Y) = Y.$$

These reverse maps are linear, not multiplicative. The density-state marginals carry no spectator normalization. The empirical states are distinct from the independently supplied uniform tower states; no state-preserving tower/refinement morphism follows from these equalities.

Source generators and finite transport. Each observer's counted transition operator and diagonal field projectors generate the full matrix algebra on its realized alphabet. Their pair lifts, followed by ι_{88} and ι_{247}, therefore generate the full triple algebra. This is a family of operators constructed from source data, not a history of source-executed quantum gates. A singleton path projector does not replace this generating family.

For each $0 \le t < 31$, transpose the two adjacent path labels separately in each factor, and let U_t be the tensor product of the resulting permutation matrices. The triple conjugation α_t and its pair conjugations satisfy

$$\alpha_t\iota_{88} = \iota_{88}\alpha_t^{88}, \qquad \alpha_t\iota_{247} = \iota_{247}\alpha_t^{247}.$$

These identities hold for arbitrary matrices, including off-diagonal observables. The evolved source generators generate the full joint algebra because α_t is an automorphism. Set $P_0 = I$, $P_t = U_{t-1}\cdots U_0$, and $U_{s,t} = P_tP_s^{-1}$. Then $U_{r,t}U_{s,r} = U_{s,t}$ for $s \le r \le t$, with exact pair projections and path transport. The final-to-initial edge is absent from all three transition-count tables and is excluded. The empirical state is not invariant under these row-dependent permutations.

The general operator identities, counted-state readouts and evolved source generation are kernel checked in `TripleCarrierOperatorJoin`. Independent source and permutation replay in the public repository [29] checks all 31 adjacent transitions and 496 nonempty ordered row intervals; its sparse matrix-unit controls are finite examples, separate from the general proofs.

The label extraction is post hoc; the tensor assembly and path-read transpositions are supplied adapters. They provide no preregistered physical validation. Observer 86/88 source supports overlap, so their tensor slots do not prove physical support disjointness. This construction supplies no universal algebraic pushout, physical region assignment, quantum preparation or outcome record, physical clock, source-selected Cauchy geometry or interacting continuum limit. It does not identify this carrier with the separate 64-mode scalar instrument.

4 Common-World Joins on the Twelve-Port Carrier

This section places one observer's mass, clock, worldline and a declared Maxwell evolution on one typed record of the twelve-port carrier. Every join below is a declared regression surface: it shows which identities the shared record forces and which choices remain free, and none is a source-selected physical action. Six receipts of the formal stack share one typed record. A common-world architecture record carries one declared positive mass, one state frame, one flow frame, one projective partition with state, one declared Maxwell evolution bundle on the twelve-port local carrier, and one declared electroweak premise bundle whose base instantiates the Standard-Model structure premises. For every inhabitant, the kinematic mass-shell, internal-clock, free-evolution,

and causal-order receipts hold of the same mass, frame, flow frame, partition, and state on one Hermitian Lorentz module, with the shared fields identified definitionally, and four join theorems place the four-momentum, the clock worldline, and every nonnegative flow displacement inside the cone order that carries the finite causal nets, with the four-momentum future-causal from the causal net's bottom vertex. Those nets and their Lorentz-module cone order are declared kinematic data, not the source-derived informational read-from order. On the screen side, the temporal Maxwell receipt and the electroweak-breaking receipt hold of the same record, and the pinned port-dual measure of the record's structure premises evaluates the Maxwell Gauss load on the one twelve-port index type, with every weight equal to $1/12$. An explicit instance inhabits the record (`CommonWorldKinematicsWitness`). On the base record the two islands are joined only by conjunction: no typed dictionary sends ports, seams, or faces to points or cones of the module, the identification of the Maxwell step index with the kinematic proper time is a declared dictionary carried by the instrumented extension below, and no common variational principle spans both carriers; clock calibration and carrier identification are not supplied. Gravitation, thermodynamics, matter dynamics, quantization, and observer readout appear in no field of the record, so the record does not establish the common-world target.

One typed dictionary joins the two islands beyond conjunction. The rank-three source Gram quotient of the six-axis port adapter, read from the registered twelve-port incidence table, maps onto the rest fiber of the record's declared frame by a linear equivalence composed from committed isometries, and the pullback of the rest metric through that dictionary is exactly the screen quotient Gram at every declared frame, with the standard-frame case equal to the committed candidate readback and with every integer record control carried to the committed screen chain up to one committed normalization radius. An extended record carries the dictionary and the metric clause as fields; an explicit inhabitant extends the committed instance, and a composed receipt proves the eight island packets, the bridge clause with its oriented-chart commuting square, and the transported inverse-square strength read on the screen Gram, of one record simultaneously (`CommonWorldIslandBridge`). Dropping the metric clause readmits a mismatched pair: the same base witness extended by the dictionary scaled by two is well typed and violates the clause at an exhibited unit class. The dictionary is a candidate local readout between two mathematical objects and identifies nothing physical; the shared clock, the common action, common dynamics, the map from ports, seams, and faces to the module, and the physical spacetime attachment are not supplied.

Four further typed joins extend the bridged record. A step-scaled Maxwell bundle sits beside the declared unit-step bundle on the same twelve-port carrier and shares its initial seam data and seam current; the unit-step law is the unit instance of the scaled law, every certified Courant constant is at least the committed eigenvalue 5, so every certified step lies below one, and the receipt carries the explicit unbounded unit-step mode beside the uniform bound on the scaled fields. The phase instrument joins on the two-dimensional record algebra: its diagonal-context channel is exactly the committed pinching of the diagonal partition, which refines the record's one-block partition as every partition does, and the swap-twisted instrument violates that identity at the run state. A declared positive step duration δ identifies evolution step n with the proper-time parameter $n\delta$ on the record's clock worldline: the clock at step n is $e^{-imn\delta}$, it repeats after N steps exactly when $mN\delta \in 2\pi\mathbb{Z}$, successive step events are causally ordered, and a zero or negative step collapses the steps or breaks the causal clause. A repair law whose reference is the diagonal of the instrument's preparation carries a horizon record, and the Einstein-branch register at that law states the first-law clause on the simplex tangent space (`CommonWorldInstrumentJoin`). A formal same-index product readout packages the two islands: a single function of the step index produces the pair of the scaled staggered-form value of the record's certified bundle and the clock

worldline event at proper time $n\delta$, with the index recovered from the joined object, successive joined events causally ordered, the record's step below the sharp certificate, and, for zero seam current, the joined value constant along every causally ordered index pair while the electric and magnetic energies stay within the committed multiples of the joined value at index zero. These identities are inherited independently from the event and field projections; no theorem relates the field value to the event beyond the shared index. On the committed inhabitant the joined value is constant while the clock does not repeat with period one step, and the product fixes no clock calibration (`CommonWorldMaxwellClockJoin`). A chosen quadratic clock functional can be added formally to the Maxwell functional: conditional on its declared endpoints, the worldline is the unique stationary path in the committed Lorentz form, whose exact expansion carries the action of the variation as the entire quadratic remainder, and the direct-sum action, the committed scaled Maxwell window action plus the worldline action on the shared window, is stationary in its Maxwell sector exactly at the scaled Ampère and Gauss clauses and in its worldline sector exactly at the record's clock line, with the committed configuration stationary in both sectors. The sum is proved a direct sum, partial variations independent in both directions, and because the form is Lorentzian a displaced spatial history strictly beats the stationary worldline, so stationarity is proved without any minimality claim (`CommonWorldJointAction`). This zero-coupling sum is a regression surface, not a source-selected physical common action; its displayed relative sector coefficient is likewise declared and cannot be fixed by the decoupled stationarity equations. The step dictionary is declared and identifies no physical time. For the carrier map itself a candidate exists: twelve exact golden-ratio-ring rays embed the ports into the spatial slice of the committed Lorentz module, reproduce the committed Gram table up to one squared-length normalization, respect the seam, face, and antipode incidence in both directions, carry the full sixty-element committed action through two exact rotation matrices and a sixty-word generation certificate, act compatibly with positive-factor barycentric refinement on nondegenerate mesh points, and land spacelike for the committed determinant form; the canonical assignment and its negation both satisfy every clause, so the selection is declared rather than forced, and the embedding carries no scale, causal content, or observer readout (`ScreenCarrierMapCandidate`). The candidate is dynamics compatible: a constrained transport structure, whose two generator inhabitants are kernel-verified row by row against the committed seam and face tables, intertwines the scaled Ampère update, the Gauss constraint with pulled-back charge, the staggered energy form, and the committed gauge transformation exactly and uniformly in the step, so no calibration is selected; through the committed word certificate every row of the sixty-element action is the port component of such a transport, the generators act geometrically through the two exact rotation matrices, and transport commutes with barycentric refinement along both routes. A declared interaction candidate couples the worldline increment to the seam electric field through the embedded seam directions: it is gauge invariant at every relative normalization, its zero value recovers the decoupled direct sum as the regression limit, and an explicit pulse configuration separates normalizations, so the coupling constant is declared rather than forced (`CarrierDynamicsCompatibility`). At a declared nonzero step h, varying the coupled action in both sectors shows what the candidate couples: the field equations are the committed Ampère and Gauss clauses at augmented sources, and the induced sources are polarization-type, a step difference of the worldline seam current and a neutral bound load, so the candidate carries no monopole charge and its force on the worldline is the coupling times the step difference of the embedded field, with the exact worldline equation $2m\,\Delta^2 x = \kappa\,\Delta E$. One declared identification of the induced Gauss-law load at a port with a declared value q fixes the normalization uniquely, $\kappa = -qh/(12 - 4\varphi)$ on a seam crossing, with the constant computed exactly from the golden-ratio seam data; the identification is step-local and increment-proportional, only κ/h is fixed, and the minimal-coupling shape pairing the potential itself with the increment is the live monopole alterna-

tive (`ChargeFixedInteraction`). The dynamical content of the mass–energy identity is located in the same setting as a conditional theorem: under a declared composite shape in which the rest parameter plus the modular energy of the cap first law multiplies the committed worldline functional, stationarity is exactly $(m + E)\,\Delta^2 x$ equal to the impulse of a declared potential, so internal energy enters the inertial coefficient with slope one, while a second declared shape entering the same ledger additively has coefficient m, agrees with the first at zero energy and on the committed worldline, and is separated from it by an explicit parabola history; the slope with which internal energy enters inertia is selected by the shape, both shapes are declared enrichments by the realized-history Legendre non-identifiability, and nothing derives the identity (`InternalEnergyInertia`).

The monopole route named there is built as a declared point charge hopping along the ports of the thirty-seam complex, minimally coupled to the seam and port potentials with the source weights of the committed window action. At a declared nonzero step h, its induced load has total q at every step and its induced current is $-(q/h)$ on the hopped seam, oriented so that the committed continuity equation holds identically; the coupled action equals the window action at the augmented sources with no endpoint term, so the field equations are the committed scaled Ampère update and Gauss constraint at the augmented sources. For a general source pair the gauge change of the source pairing is the gauge function paired against the continuity residual plus an endpoint term, and invariance under endpoint-vanishing gauge functions is equivalent to continuity on the interior steps. Along the Ampère evolution the staggered energy moves by $(q/2)(E_n(e) + E_{n+1}(e))$ on a forward hop across e and is constant at rest; no energy of the charge and no equation of motion for the path are defined. At the endpoints of a crossed seam the E-paired induced load is $-(\kappa/h)(12 - 4\varphi)$ times the step difference of the unit hopping load, and equals the hopping step difference exactly at $\kappa = -qh/(12-4\varphi)$; at the exhibited seam a port away from the endpoints carries a nonzero E-paired load while the hopping difference vanishes, so the polarization is the step difference of the monopole at the seam endpoints only. At nonzero h the charge is unconstrained by the field sector, and the path, charge, nonzero step, and coupling shape are declared (`PortChargeMinimalCoupling`).

The monopole route joins the clock action through a declared class of worldlines. A seam-step worldline is a declared start port and a declared step sequence, each step a rest or a signed seam of the carrier map candidate; it generates a Lorentz-module path with time coordinate τn for a declared unit τ and spatial coordinate the cumulative sum of the signed seam vectors, and a hopping port path. The seam vector is the ray difference of its endpoints in $\mathbb{Z}[\varphi]^3$, and at every step the ray of the occupied port minus the ray of the start equals the spatial coordinate exactly, so the injective ray map reads the port off the path. At every crossing step and a declared nonzero field step h, the E-paired induced load at the two endpoints of the stepped seam equals $-(\kappa/h)(12 - 4\varphi)$ times the step difference of the unit hopping load, and equals the hopping step difference of charge q at $\kappa = -qh/(12 - 4\varphi)$; the worldline seam current is the $\mathbb{Z}[\varphi]$ pairing of each seam vector with the stepped vector, ± 4 at the stepped seam and at its antipodal seam (the two carry one vector) and nonzero at a neighbouring seam, and the hopping current is $-(q/h)$ times its projection onto the stepped seam divided by four. At nonzero h, the transported action, the monopole coupled action plus the clock action of the generated path, has field-sector stationarity equal to the committed Ampère and Gauss equations at the transported sources; the clock action is $\sum(\tau^2 - 4)$ over crossings plus $\sum \tau^2$ over rests, timelike exactly when $\tau^2 > 4$. Distinct units give distinct paths with one hopping path and one source family, and at nonzero h the unit is unconstrained by the field sector (`WorldlineHopTransport`).

The transported charge has a discrete balance. Varying the transported action over the declared class of closed two-step variations, the replacement of one step pair by an admissible pair with the same endpoint, gives an exact action difference: the clock part is the original step norms minus the varied ones (rest 0, crossing 4, the unit cancels), the interaction part is q times the potential

pairing $h\varphi_n(u) - \langle \mathbf{1}_{u\to v}, A_{n+1}\rangle$ summed along the alternative route minus the original one, and the window action cancels. Exchanging a forward crossing of seam e with an adjacent rest moves the action by $q\,h\,E_{j+1}(e)$ at zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the exchange is stationary exactly when the scaled electric seam field vanishes at the delayed step; replacing two rests by a round trip across e moves it by $-8-q\,h\,E_{j+1}(e)$, so at zero field rest is neither stationary nor a local minimum in this class, the port form of stationarity without minimality. For static potentials the interaction difference is q times the port-potential difference at the intermediate port plus q times the discrete circulation of A around the four-leg loop, and a round trip on one seam has zero circulation. The class, the potentials, and the shapes are declared (`TransportedChargeForceLaw`).

The closed carrier also fixes the neutrality condition rather than hiding it in the field solver. A hopping load and current obey the discrete continuity equation identically; the sourced scaled Ampère and Gauss equations admit a history exactly when the initial load is neutral and continuity holds, so one isolated hopping charge is compatible only at zero charge. The viable neutral branch is explicit. Put charges q and $-q$ on two declared seam-step worldlines and form one action from the source-free Maxwell window, both minimal couplings, and both clock actions. Its field variations are equivalent, for $h \neq 0$ and endpoint-fixed seam-potential variations, to the interior Ampère equations at $m < N$ and Gauss equations at $n < N + 1$ for the pair's own sources. The action is invariant under gauge functions vanishing at the window endpoints. Replacing either path over the same closed two-step class, with both fields and the other path fixed, gives an exact off-shell clock-plus-interaction action difference, not by itself an equation of motion. At $q = 1$ and $h = 1/2$, crossings of seams 0 and 29 recover an explicit nonzero-current, Coulomb-started field-stationary inhabitant. It fails the declared closed-two-step exchange-stationarity condition in both path slots: for every $N \geq 1$, delaying either the positive seam-0 crossing or the negative seam-29 crossing while fixing the fields and the other path changes the same action by $5/12$ for every pair of clock-unit values within the fixed quadratic clock-action form. The two results are charge conjugates; applying both canonical exchanges simultaneously changes the action by $5/6$. These are only the displayed discrete exchanges, not continuum first-variation claims or a no-go for other paths, fields, or action shapes. The counterexample is specific to that Coulomb start. On the window $N = 1$, a second exact rational temporal-gauge history at the same $q = 1$, $h = 1/2$, and the same crossing pair satisfies the sourced Ampère and Gauss equations. The four possible intermediate ports for each closed two-step path replacement are exhausted exactly. Endpoint routes have zero action difference, and the two triangular detours have clock difference -4 cancelled by interaction difference $+4$. Thus one finite action is stationary in both field slots and unchanged under every in-window closed two-step replacement in either path slot. The seam currents are -2 and $+2$ on seams 0 and 29, and the initial loads are $+1$ and -1 at the corresponding ports. At the separately declared clock unit $\tau = 3$, both crossings are timelike in the carrier's Lorentz-module metric. The charge, paths, step, rational history, clock unit, coupling shape, and variation class remain declared; the theorem establishes existence without uniqueness or source selection. It proves no minimum, stability, or dynamic selection. No laboratory-time calibration or continuum limit is claimed (`SeamChargeContinuity`, `NeutralPairCoupledAction`, `NeutralPairJointStationaryWitness`).

Inside the rest-diluted class the law reads as a position law. A general one-step replacement with shifted later ports has the exact difference of the one-step clock term plus q times the pairing difference at the step plus q times the sum of the later pairing differences at the shifted ports, reducing to the one-step term under declared shift invariance and to the closed two-step difference when the route rejoins at the second step. Within a block of k rests and one crossing, moving the crossing by one position changes the action by $q\,h\,E(e)$ at the delayed index (minus that at the crossing index for an advance) with zero clock cost, so, for $q \neq 0$ and $h \neq 0$, the crossing position is stationary exactly when the seam field vanishes at the reachable indices, and under a

static nonzero field it is never stationary and drifts with the sign of qhE. The block momentum $2m((k+1)\tau, s_e)$ is unchanged by every in-block move, which swaps the per-step momenta. Along the respective scaled Ampère evolutions sourced by the original and delayed hopping currents, and with the original field static across the crossing step, the delay difference equals h times the hop's field-energy transfer $(q/2)(E_j + E_{j+1})(e)$ (`TimelikeClassForceLaw`).

The same class meets the field sector's stability window. For nonzero h, uniform boundedness of the electric seam energy for every zero-current datum holds exactly when $h^2(3+\sqrt{5}) < 4$; inside that strict window the magnetic face energy is also uniformly bounded, while equality and larger steps admit an explicit zero-current solution with unbounded electric seam energy. No bound on the gauge potential A follows. A crossing step is timelike exactly when the unit satisfies $\tau^2 > 4$. Under the declared identification of the worldline unit with the field step ($\tau = h$) no step lies in both, so inside the strict energy window every per-step crossing is spacelike and a timelike per-step crossing sits above the threshold with an unbounded electric-energy solution; the two thresholds differ by the factor $2\varphi^2 = 3 + \sqrt{5}$. A block of k rests and one crossing is timelike exactly when $2 < (k+1)\tau$, which inside the window forces $\sqrt{2}\,\varphi < k+1$, at least two rests per crossing, with the two-rest block timelike exactly for $2/3 < h$; the block speed $2/((k+1)\tau)$ is below one exactly on timelike blocks and its supremum over the window is one, approached and never attained. Without the identification the thresholds are independent ($\tau = 3$, $h = 1/2$ gives timelike crossings with uniformly bounded electric seam and magnetic face energies), and the corpus joined witnesses are not identified. The identification is declared and nothing selects it (`SeamStepSpeedLimit`).

Which rule the hosted internal process follows along such a worldline is compared exactly. Index accrual, one internal step per shared step index at the declared unit, is the per-step member of the internal-action family; proper-length accrual is the declared proper-time principle. Over a window with c crossings the two differ by exactly $E\,c\,(\tau - \sqrt{\tau^2 - 4})$. For $E \neq 0$ this difference is nonzero on every window with a crossing at every timelike unit and is zero exactly on resting windows; the midpoint refinement of a worldline with a crossing is the generated path of no seam-step worldline. For $E \neq 0$, index accrual fails refinement invariance for every nonzero unit while proper-length accrual passes, so the source chain advancing per shared index is the rule that refinement invariance excludes on every moving worldline with nonzero ledger. The uniform worldline with one crossing per $k+1$ steps has proper length $(k\tau + \sqrt{\tau^2 - 4})/((k+1)\tau)$ times that of a resting worldline over the same index count, below one for $\tau > 2$, tending to one in k, $\sqrt{5}/3$ at $\tau = 3$, $k = 0$; under the two declared readings of one internal unit the recurrence clock's mean return $61511/7155$ reads as that many shared steps on every worldline, or as the inverse factor times it on the uniform one. Neither rule is selected by the source or the join (`SourceClockRateAlongWorldlines`).

The length rule has a construction. Clock the hosted process by the accumulated proper length in units of a declared internal unit u: the clocked index $c(n)$ is monotone with each increment in $\{0, 1\}$ for $\tau \leq u$, equals the shared index on the resting worldline at $u = \tau$, and on the uniform worldline with k rests per crossing obeys $n\,d\tau/u - 1 < c(n) \leq n\,d\tau/u$ on full periods with d the dilation factor, so $c(n)/n \to d\tau/u$, exactly d at $u = \tau$. The clocked return count of the recurrence chain equals the chain's return count at $c(n)$, every per-chain-step quantity converts with the same bound, and the clocked index is invariant under midpoint refinement by construction. No constant step duration of the joined architecture reproduces the accumulated proper length on a moving uniform worldline with at least one rest between crossings; the all-crossing case is matched by the constant duration $\sqrt{\tau^2 - 4}$, so the clocked chain is a declared enrichment of the join, two internal units give two clocked indices, and nothing selects the unit from the source (`ProperLengthClockedChain`).

The slope with which internal energy enters inertia is then selected by two declared principles, each with an exact theorem. On the momentum map, let the composite four-momentum range over

the declared family $P_\lambda(m, E; u) = (m + \lambda E)\, u + (1 - \lambda) E\, e_0$, with u on the future unit hyperboloid and e_0 the rest direction; every member has scalar coordinate $m + E$ at the standard frame. Under an oriented Lorentz map L the family transports as $P_\lambda(Lu) = L\, P_\lambda(u) + (1 - \lambda) E\, (e_0 - L e_0)$, so P_λ is frame covariant exactly when $\lambda = 1$ or $E = 0$, the converse witnessed by an explicit boost with parameters $(5/4, 3/4)$; the Lorentz square is $(m + E)^2 + 2(1 - \lambda) E(m + \lambda E)(u^0 - 1)$, so the shell identity at the rest energy holds at every frame exactly when $(1 - \lambda) E(m + \lambda E) = 0$. The selection is kinematic, on the momentum map, and the slope-one member is the composite momentum of the inertia precursor with no factor (`CompositeMomentumCovariance`). On the action, the declared midpoint refinement of a worldline leaves the proper length invariant, doubles the step count, and halves the quadratic clock action; in the internal-action family $E(aL + bM)$ with L the proper length and M the step count, refinement invariance on timelike windows holds exactly when $b = 0$ for $E \neq 0$, which excludes the per-step additive ledger of the second composite shape at nonzero ledger. The proper-time principle, declared, sets the internal action to E per unit proper length; the rest phase of the declared internal clock advances by the mass parameter per unit proper time and its window increment equals that internal action, so the declared internal clock is the proper-length form, and within the family $E\lambda L$ agreement with the principle holds exactly when $\lambda = 1$ for $E \neq 0$. For the length action $(m + E)L$ plus a declared potential pairing, the derivative along every variation exists on timelike windows, and stationarity under endpoint-fixed variations is equivalent to $(m + E)$ times the unit-tangent difference equal to the impulse at every interior node, so the inertial coefficient of the refinement-invariant form is exactly $m + E$; on unit-increment windows the length and quadratic equations of motion coincide (`ProperTimeInternalAction`). The refinement, the proper-time principle, the momentum family, and the covariance requirement are declared, every shape is a declared enrichment by the realized-history Legendre non-identifiability, and nothing derives the identity; its dynamical content is located in these two principles.

The ledger itself composes exactly. Over a product reference the modular energy of any joint state, correlated or not, is the sum of the marginal ledgers; against a general joint reference the binding defect is the expectation of $\log(\tau_1 \tau_2 / \tau_{12})$, and against the Gibbs reference of a declared total energy $H_1 + H_2 + V$ it is $\beta \langle V \rangle_p + \log(Z_{12} / Z_1 Z_2)$, whose constant vanishes at $V = 0$. For a declared attractive $V \leq 0$ and $\beta \geq 0$ the state part is nonpositive while the constant is nonnegative, so the sign of the full defect is not fixed, and a two-point uniform state has the exact positive defect $-1/4 + \log((3 + e)/4)$; every normalized faithful joint reference is Gibbs for its own modular Hamiltonian, so the defect is always the expectation of a state-independent effective interaction. Under the declared composite shape the inertial coefficient of a composite is $m_1 + m_2 + E_1 + E_2$ plus the defect, a definitional composition whose physical reading is the calibration import (`ModularEnergyAdditivity`). Matter dynamics beyond these declared shapes and observer readout appear in no field, so the common-world target is unestablished. A physical common action, source-selected carrier identification, matter dynamics, and readout are four named residual categories, not an exhaustive theorem about all required attachments.

5 Histories, Actions and Mechanics

Two standard finite-horizon constructions clarify what the premises A1–A3 must supply before an action principle could be claimed.

Theorem 5.1 (Finite path projection and positive-gap tail)**.** *Let Ω be finite, let $\Gamma = \Omega^{\{0,\dots,N\}}$, and let P, τ be probability laws on Γ, with $\tau > 0$. Let $R > 0$ be a reference weight, let $F_a : \Gamma \to \mathbb{R}$ be*

finitely many declared observables, and suppose

$$\log \tau(\gamma) = \log R(\gamma) - \sum_a \lambda_a F_a(\gamma) - \log Z$$

and P and τ have the same F_a-moments. Then

$$D(P\|R) = D(P\|\tau) + D(\tau\|R), \qquad D(\tau\|R) \le D(P\|R).$$

With

$$A_{\text{eff}}(\gamma) = -\log R(\gamma) + \sum_a \lambda_a F_a(\gamma),$$

one has $\tau(\gamma) = \exp[-A_{\text{eff}}(\gamma) - \log Z]$, so every supplied modal history minimizes A_{eff} over Γ.

Separately, for any nonempty finite set Γ, function $S : \Gamma \to \mathbb{R}$, global minimizer γ_0, and

$$q_\beta(\gamma) = \frac{e^{-\beta S(\gamma)}}{\sum_\delta e^{-\beta S(\delta)}},$$

the mass at least $\Delta > 0$ above that minimum obeys

$$\sum_{S(\gamma) \ge S(\gamma_0)+\Delta} q_\beta(\gamma) \le |\Gamma| e^{-\beta\Delta}$$

for $\beta \ge 0$, and this displayed mass tends to zero as $\beta \to \infty$. Finiteness also gives the stronger concentration statement

$$\lim_{\beta\to\infty} \sum_{S(\gamma) > S(\gamma_0)} q_\beta(\gamma) = 0,$$

so the total mass outside the full minimizer set vanishes.

Proof. Expand $D(P\|R) - D(P\|\tau)$. The displayed logarithmic form, normalization, and moment matching make the result independent of P on the constraint surface; evaluating it at $P = \tau$ gives the Pythagorean identity. Nonnegativity of $D(P\|\tau)$ gives minimality. Exponentiating the logarithmic form reverses probability and effective-action order. For the tail bound, the partition sum is at least $e^{-\beta S(\gamma_0)}$, while each displayed numerator is at most that quantity times $e^{-\beta\Delta}$; summing and taking the elementary exponential limit proves the fixed-gap statement. Apply that argument to each strictly nonminimal history and sum over the finite set for the final limit. The statements are checked in Lean [29]; they are instances of standard information-projection mathematics [37] and elementary finite exponential estimates. □

The theorem assumes the exponential form and matching moments. It neither constructs R from the OPH source nor proves that a premise A3 history optimizer exists, has that form, or is unique. Calling R a probability law additionally requires its normalization.

Theorem 5.2 (Real finite-horizon variational helpers)**.** *Let $L : \mathbb{R}^2 \to \mathbb{R}$ be differentiable and define*

$$\mathcal{S}_L(\gamma) = \sum_{n=0}^{N-1} L(\gamma_n, \gamma_{n+1}) \qquad (\gamma : \{0, \dots, N\} \to \mathbb{R}).$$

At an interior junction $k+1$, suppose γ minimizes $\mathcal{S}_L$ under every replacement of γ_{k+1}. Then

$$\partial_2 L(\gamma_k, \gamma_{k+1}) + \partial_1 L(\gamma_{k+1}, \gamma_{k+2}) = 0.$$

If a differentiable one-parameter family T_s satisfies $T_0 x = x$, has generator $\xi(x) = \partial_s T_s x|_{s=0}$, and obeys $L(T_s x, T_s y) = L(x, y)$, then the segment momentum

$$p_n = \partial_2 L(\gamma_n, \gamma_{n+1})\, \xi(\gamma_{n+1})$$

is equal on the two segments adjacent to that junction. More generally, for a path with $M + 2$ records, if the same single-site minimization premise holds at all M interior records, then there is a scalar J such that $p_n = J$ on every one of its $M + 1$ segments.

Proof. Changing one interior record changes exactly the two adjacent summands. Differentiating their sum at its declared minimum gives the first identity. Differentiating the invariance relation at $s = 0$ and combining it with that identity gives $p_k = p_{k+1}$. Applying this identity at every interior junction and inducting along the finite chain gives the scalar J. These are standard discrete variational and Noether calculations [38]; their precise premises are checked in Lean [29]. □

Proposition 5.3 (Finite/real variation obstruction)**.** *Fix a real path $\gamma : \{0, \dots, N\} \to \mathbb{R}$, a site i, and any finite family G of real paths. There is an $x \in \mathbb{R}$ for which the path obtained by replacing γ_i with x does not belong to G. Hence no finite real-path family contains every single-site real variation of even one supplied path.*

Proof. The map from x to the updated path is injective because evaluation at i recovers x. Its image is infinite and therefore cannot be contained in finite G. This elementary cardinality obstruction is machine checked in Lean [29]. □

The two theorem domains compose only through a certified receipt. The first uses paths in a finite state set; the second uses arbitrary real single-site variations, and Proposition 5.3 rules out closure under all such variations inside any finite real-path family. The composition that survives this obstruction is conditional: an undercut receipt asserts that every real single-site variation of an embedded path is undercut in action by some embedded finite path, and under that receipt finite minimality transfers to real local minimality and to the real Euler–Lagrange packet. Membership receipts are impossible (that is the obstruction restated); undercut receipts are attainable, and the committed quadratic witness carries one, while a one-path control family provably carries none and the transferred conclusion fails there. All three layers, transfer, witness, and control, are checked in Lean [29].

Which action the history law carries is itself a theorem rather than a choice.

Theorem 5.4 (Derived action of a source chain)**.** *Let Ω be a finite nonempty state set, P a strictly positive row-stochastic kernel, and π a strictly positive normalized initial law. On paths with n transitions define the log-transition action $S_{\log}(\gamma) = -\sum_i \log P(\gamma_i, \gamma_{i+1})$ and the reference $R(\gamma) = \pi(\gamma_0)\, |\Omega|^{-n}$. Then the Markov path law equals the exponential tilt of R by $S_{\log}$ at multiplier one, with partition constant $|\Omega|^{-n}$; and a pair (S', λ') reproduces the path law as its tilt of R if and only if there is a constant c with $\lambda' S' = S_{\log} + c$ pointwise.*

Proof. Exponentiating the negated action recovers the transition product, and row normalization makes the partition sum telescope to $|\Omega|^{-n}$, which cancels against the reference weight. Uniqueness reduces to the statement that two tilts of one strictly positive reference agree exactly when the multiplier-weighted actions differ by an additive constant, proved by taking logarithms at a common path and cancelling the shared normalizer. Both directions of the gauge are realized: per-step additive constants aggregate to a path constant, and the multiplier trades against the action scale, so the bare-action convention chooses $\lambda = 1$ as a representative. It is the unique multiplier for that unrescaled action when the path action is nonconstant; for a uniform kernel the action is constant and normalization erases the multiplier. The statements are checked in Lean [29]. □

The committed source chain instantiates the theorem exactly: the tilt of its reference by its log-transition action reproduces its path law with partition constant $1/4$, through kernel-decided integer identities, and the committed repair-count action reproduces that law at no multiplier, because it takes equal values on two paths that the reference weighs equally and the chain separates. The action of the realized dynamics is therefore forced up to an additive constant and a multiplier rescaling. This multiplier is an action-normalization gauge for the supplied chain; it is not thereby identified with a gauge-sector coupling coefficient. The separate finite Gibbs-kernel construction for one three-dimensional P-family factor shows that a common rescaling of its supplied quadratic cost can occupy the same formal multiplier slot, but it neither derives that kernel from source dynamics nor determines relative couplings between simple factors.

The retained length-three history packet also has an exact finite operator attachment. Let $p_g = n_g/1754$ be its empirical law on the eight binary paths and let $S_g \in \{0, 1, 2\}$ count the two adjacent state changes. The diagonal matrices

$$\rho_{\mathrm{hist}} = \mathrm{diag}(p_g), \qquad H_{\mathrm{hist}} = \mathrm{diag}(S_g)$$

form a strictly positive normalized density and a positive non-scalar self-adjoint Hamiltonian, with

$$[\rho_{\mathrm{hist}}, H_{\mathrm{hist}}] = 0, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{\mathrm{hist}}) = \frac{197}{1754}.$$

On a one-regulator consensus tower, the selected-state construction places this pair on its completed-colimit GNS Hilbert space. The finite-stage representation is injective, the represented Hamiltonian and one off-diagonal commutator response are nonzero, and the mapped exponential flow is unitary and leaves the density stationary. The cyclic unit has nonzero energy, so it is not a ground state or vacuum for this Hamiltonian.

The eight histories also reindex exactly as three binary tensor slots. Their one-slot algebras and the two adjacent interval algebras are isotone, the first and last slot algebras commute, and the Hamiltonian splits without a remainder into positive local bonds,

$$H_{\mathrm{hist}} = H_{01} + H_{12}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{01}) = \frac{94}{1754}, \qquad \mathrm{Tr}(\rho_{\mathrm{hist}} H_{12}) = \frac{103}{1754}.$$

The first commutator of an endpoint observable lies in its adjacent two-slot algebra. The source functional restricts compatibly along all these algebra inclusions, and the locality, bond, and propagation identities survive the injective selected-GNS representation. This supplies a finite history-coordinate local dynamical system on a selected-GNS carrier built from the source-counted diagonal functional. The retained source packet fixes the diagonal probabilities and repair counts. The extension to the full matrix algebra, the tensor-local reading, off-diagonal observables, and the real flow parameter are mathematical adapters. The history slots are not calibrated physical times or Lorentzian regions, and no spectrum condition, continuum/RG limit, physical field, particle, scattering, or detector interpretation is obtained from them [29].

The rational record layer of the same carrier carries a finite-probability conditioning family. Conditioning the window law on any slot region defines a rational-linear, unital, pointwise-positive, localized, idempotent, tower-compatible map that preserves the empirical mean; the walls are fixed points of their own regional maps, and the exact conditional weights of the late slot are $383/415$ and $2/89$ in two (s_0, s_1) cells of the counts $1149 : 96$ and $2 : 87$. These counts give a narrowly scoped matrix no-go: no complex-linear map from the full matrix algebra into the fixed first two-slot algebra $\mathcal{A}_{01}$ can both obey the $\mathcal{A}_{01}$-bimodule law and preserve the committed state. They do not rule out ordinary AQFT inclusions with state restriction, a different or enlarged

target algebra, ancillary boundary memory, generalized expectations, or non-bimodule positive or completely positive channels [29]. The enlarged-target path is constructed: a unital injective star-embedding reads the slot-two record bit of each history into one qubit ancilla, its range is exactly the ampliated interval algebra joined with the ancilla factor, and it satisfies clause for clause the conjunction the fixed-target no-go kills, with the range clause the single clause whose target changes and the exhibited state a proved non-product extension restricting to the committed source state. Two delimiting theorems establish two necessary features of this escape: a product-form state extension keeps the obstruction for every ancilla matrix, and a map into the ampliated copy alone is obstructed even against the correlated extension. Thus, within this clause family, the ancilla factor and some non-product state extension are necessary. The exhibited record-correlated extension is one witness and is not proved unique. This is a finite-matrix existence statement; regions, time slices, dynamics, and any continuum attachment are not touched by it [29].

The same packet also fixes a bounded overlapping-window action statistic. Among 1754 retained length-three windows, the two domain-wall indicators have count-weighted incidences 94 in the first edge position and 103 in the second, hence mean two-edge action 197/1754. The value 197 counts incidences inside overlapping windows, not distinct underlying repair events. The formal object stores only the eight-path histogram and chosen observables: it contains no window order, successor relation, cadence, or period. Dividing by an externally declared positive real, interpreted as a window duration, is therefore only a conditional unit conversion; that interpretation is not related to the histogram by any theorem. This supplies neither a source-derived clock nor a reduction of the physical calibration input [29].

The time row is located more exactly by two finite results. Accepted repair hosts no periodic clock: every accepted step strictly lowers the mismatch count, so the accepted-repair relation admits no cycle, and the canonical repair iteration admits no periodic orbit other than a fixed point. The committed stationary source chain, by contrast, carries a recurrent process: the first-return law of each state is written in closed form from the kernel, sums to one, and satisfies Kac's identity with the exact mean return times 61511/7155 and 61511/54356 steps, whose ratio 54356/7155 is the same number under every declared tick. The return law assigns positive probability to both one- and two-step returns. Its standard Markov-chain period is therefore one, while no deterministic return interval exists because no return time has probability one. A return count along a trajectory counts distinct visit events, the property the overlapping incidence statistic lacks. One chain step stays a declared tick and no physical duration is identified; the abstract rate non-identifiability result concerns transition systems whose bridge to the repair layer fails, and it forbids nothing here (`SourceRecurrenceClock`).

The field sector, by contrast, hosts periodic processes. Every eigenvector v of the local operator $C^{\mathsf{T}}C$ with eigenvalue λ carries, at a declared nonzero step h with $0 \le h^2\lambda \le 4$, the histories $A_n = \cos(n\theta)\, v$ and $A_n = \sin(n\theta)\, v$ with $\theta = \arccos(1 - h^2\lambda/2)$; they solve the zero-current scaled evolution in the temporal gauge, conserve the staggered form, and are bounded. Kernel-checked integer eigenvectors with eigenvalues 2 and 3 join the committed ones with eigenvalues 5 and $3+\sqrt{5}$. At $h^2 = 1/2$ the eigenvalue-two mode returns after six steps, at $h^2 = 2/3$ the eigenvalue-three mode after four, at $h^2 = 1/3$ after six, each step inside the sharp window $h^2 < 2/\varphi^2$, since $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$; the projector traces $1, 5, 4, 4, 6$ read with rank equal to trace give the multiplicities $1, 5, 4, 4$ and $3+3$. Frequency ratios carry no tick: $h\sqrt{\lambda} \le \theta \le h\sqrt{\lambda}/\sqrt{1 - h^2\lambda/4}$, $\theta/h \to \sqrt{\lambda}$ as the step shrinks, and the small-step ratio of the golden mode to the slowest mode is φ^2 exactly. A period p and the mean return time 61511/7155 of the recurrence clock are counted in one shared step index, with ratio $7155p/61511$ under every declared tick. The step, the gauge, and the kinetic term are declared; the repair-layer no-periodic-orbit theorems and these field-sector periods have disjoint scope (`CarrierModeOscillators`).

The discrete step embeds in a continuous flow. On the amplitude-velocity plane of every eigenvector v with eigenvalue λ and $0 < h^2\lambda < 4$, the zero-current temporal-gauge evolution is the symplectic map $T = \left(\begin{smallmatrix} 1 & h \\ -h\lambda & 1-h^2\lambda \end{smallmatrix}\right)$ of determinant one and trace $2 - h^2\lambda$, conjugate by an explicit matrix Q to the rotation by the mode angle; the family $\Phi_t = Q\,\mathrm{rot}(\theta t/h)\,Q^{-1}$ is a continuous one-parameter group with $\Phi_h = T$ and $\Phi_{nh} = T^n$ that preserves the committed staggered energy, equal to $(\|v\|^2 \sin^2\theta/2)$ times the pulled-back Euclidean form. On the kernel of the local operator, exactly the gradient space, the evolution is the h-step of the shear flow. For any finite family of pairwise orthogonal admissible eigenvectors the componentwise coefficient-state flow is a linear continuous one-parameter group whose h-step is the committed step, instantiated on the eigenvalue-two, three, five, and golden modes and a gradient. The assembled theorem permits zero listed vectors, so a faithful flow on the actual field span additionally requires nonzero linear independence or quotienting redundant coefficient directions. This generic flow theorem alone supplies neither an inner product nor the full-curl selection; the energy construction and explicit nineteen-vector basis below supply those separate inputs. Reading t as physical time remains declared (`CarrierEvolutionFlow`).

Inside the strict Courant window $0 < h^2\lambda < 4$ the staggered energy of a carrier mode is a positive definite quadratic form on the state (a, b), and fails to be one exactly at and beyond the boundary: at $h^2\lambda = 4$ the state $(-h/2, 1)$ has zero energy, beyond it negative energy, and on the gradient sector the amplitude direction is null. Its polarization is an inner product on each curl mode, the committed step and the interpolating flow are isometries, and for a nonzero step every step-invariant symmetric bilinear form is a real multiple of it, the committed energy fixing the multiple. The conjugated quarter turn squares to minus one and is orthogonal for this inner product, so each mode carries a complex line on which the flow is the phase $e^{i\theta t/h}$, with orbit derivative $i(\theta/h)$ times the state; the generator is exhibited, not obtained from Stone's theorem. On a finite orthogonal family of curl modes the direct sum is an inner product whose diagonal is the committed staggered energy at every step, the assembled flow is unitary with diagonal generator θ_i/h. Precisely the static gradient-amplitude states with zero electric component form the radical of the semidefinite extension; gradient velocity/electric directions are non-null. On an equal-eigenvalue block invariance alone does not fix the form and the energy does. The step h is a declared Courant number and t a declared flow parameter (`FieldSectorEnergyInnerProduct`).

The curl sector of the committed seam operator has an explicit orthogonal eigenbasis: nineteen seam vectors with entries in $\mathbb{Z}[\varphi]$, five at eigenvalue 2, four at 3, four at 5, three at $3+\sqrt{5}$ and their Galois conjugates at $3-\sqrt{5}$, kernel-checked as eigenvectors with diagonal Gram matrix. The four committed modes are rows of the list. The nineteen span the range of the codifferential, which is the kernel of the boundary; eleven coboundaries of listed port loads span the gradient sector, and the two sectors are complementary. On this basis the coefficient flow extends to the whole seam space: for a declared step with $h^2(3+\sqrt{5}) < 4$, the two-sided admissibility window of the family, every zero-current temporal-gauge solution is the assembled history of its initial state. On all nineteen nonzero curl modes, the weighted energy form is positive definite, its real and Hermitian forms are flow-invariant, its complex coordinates have the explicit phase generator, its diagonal is the field energy, and the coefficient state is recoverable from potential and electric readouts. After conjugating the coefficient orientation, the packet has an exact finite Stone representation for the selected principal-angle interpolation on a separately constructed $M_{19}(\mathbb{C})$ block: the self-adjoint diagonal Hamiltonian has $H_{ii} = \theta_i/h$, its propagator is e^{-iHt}, and conjugation of the coefficient outer product is exactly the outer product of the evolved coefficient state. The vector intertwiner retains absolute phase; the rank-one outer product loses global phase and retains relative phases only. The energy-weighted outer product has trace equal to the assembled field-energy Hermitian norm. A selected mode is driven by the joined scalar clock at discrete steps under the sufficient

calibrations $\delta = h$ and $m = \theta_i/h$. The 19-index block is not a relabelling of the committed $M_2(\mathbb{C})$ private carrier, the committed witness fails the duration calibration, and the eigenvalue-two and eigenvalue-three principal-branch frequencies are unequal throughout the strict positive window. Thus no one identical scalar rate equals every selected principal-branch mode frequency. That statement does not exclude a shared time parameter, phase aliases, harmonics, finite-horizon decoding, mode-resolved rates, or operator-valued readout. There are five distinct frequencies among the nineteen coordinates; the logarithm branch is selected rather than uniquely derived from the discrete step, and the two displayed calibrations are sufficient rather than necessary modulo 2π aliases. The coefficient input is arbitrary and is not proved to be the joined record's actual Maxwell history. The rank-one projective image is neither onto $M_{19}(\mathbb{C})$ nor an algebra homomorphism, and it is not an equivalence with the committed private algebra. A physical bridge therefore requires a specified compression/channel, a selected lower-dimensional sector, or an enlarged private carrier. This is not a source-selected public representation or Hamiltonian, physical time calibration, photon Hilbert-space identification, or laboratory Schrödinger law. The step and flow parameter remain declared (`CurlSectorEigenbasis`, `CurlStoneClockBridge`).

These modes are sorted by the carrier's own symmetry. The sixty listed icosahedral port permutations induce signed permutations of the seams and orientation-preserving permutations of the faces, the signed face–seam incidence is invariant under the simultaneous action on all $60 \times 20 \times 30$ triples, and so the curvature, the codifferential, the local operator, the five spectral projectors, the mode histories, the scaled Ampère evolution for every step, potentials, and current, and the staggered energy commute with the action, all by kernel checks on the committed tables (the intertwining of the operators, fields, evolution, and energy under the same sixty rows was proved with the carrier dynamics compatibility; new here are the explicit signed actions with their composition laws, the projector and eigenspace invariance, the mode transport, and the characters). The constant face vector spans the fixed eigenvalue-zero line, the element orders are $1, 2, 3, 5$ with class sizes $1, 15, 20, 24$, and the traces of the action on the projector images by element order are $1;\ 5, 1, -1, 0;\ 4, 0, 1, -1;\ 4, 0, 1, -1;\ 6, -2, 0, 1$, with character norms $60, 60, 60, 120$. Read with the orthogonality relations, an inference outside the checked statements, the face space decomposes as $1 \oplus 5 \oplus 4 \oplus 4 \oplus (3 \oplus 3')$, the two four-dimensional eigenspaces carrying one irreducible representation and the golden sector the two three-dimensional ones; the checked characters do not separate the two order-five classes, so the split of the golden sector by eigenvalue is an observation. The field modes are thus sorted by the same group that sorts the matter sectors, with the identification of a mode with a physical oscillation open (`CarrierModeEquivariance`).

The finite six-axis interface also has an abstract group endpoint. The antipodal port bridge is first a pointwise equivalence of indexed action rows. Separately, the canonical center quotient $SL(2, \mathbb{Z}/5) \to PSL(2, \mathbb{Z}/5)$, whose kernel is the center $\{+I, -I\}$ of $SL(2, \mathbb{Z}/5)$, yields a faithful action of $PSL(2, \mathbb{Z}/5)$ on the projective line under the explicit coordinates $[z : 1] \leftrightarrow z$ and $[1 : 0] \leftrightarrow 5$; its image is exactly the committed sixty-element subgroup `A5SixAxes.L60`, and the result is a typed group isomorphism onto that subgroup (`A5PortSixAxesBridge`, `PSL2F5SixAxesBridge`). The sixty committed twelve-port rotations are then packaged as a typed subgroup of the permutations of the twelve ports, with the multiplication and inverse laws checked on the actual port permutations, and the pointwise port bridge is promoted to a group isomorphism between that port subgroup, the six-axis subgroup, and $PSL(2, \mathbb{Z}/5)$. The two golden pieces are then carried as typed representations: the left face action is packaged as a linear representation of the port subgroup on the twenty-dimensional face space, restricted to either rank-three golden projector image, transported along the isomorphism to $PSL(2, \mathbb{Z}/5)$, and pulled back through the center quotient to $SL(2, \mathbb{Z}/5)$, where every central element, in particular $-I$, acts trivially; the row law behind the representation is an exact finite-table check against the certified port-group

multiplication (`GoldenSectorPSL2F5Representations`). This is not an abstract $PSL(2,5) \simeq A_5$ classification, an identification with the binary icosahedral group or McKay E_8, a two-dimensional spinor representation of $SL(2, \mathbb{Z}/5)$, a selection of φ or a mass law, or a physical-rotation statement.

The golden split is exact. On the six-dimensional golden sector, where $N^2 - 6N + 4 = 0$, the two spectral projectors with eigenvalues $3 \pm \sqrt{5} = 2\varphi^{\pm 2}$ have entries in $\frac{1}{20}\mathbb{Z}[\varphi]$, are symmetric orthogonal idempotents summing to the golden projector, each of trace 3, and the minus projector is the entrywise Galois conjugate ($\varphi \mapsto 1 - \varphi$) of the plus one. On the sixty listed automorphisms the plus character takes the values $3, -1, 0$ on the orders $1, 2, 3$ and φ or $1 - \varphi$ on the twenty-four elements of order five, twelve each; the minus character is its Galois conjugate on every element, the two agree exactly off order five, squaring an order-five element exchanges them, both are class functions commuting with every automorphism, and each has norm 60. Read with the orthogonality relations, these data have the usual two-character interpretation, but this character module does not itself prove irreducibility. The separate span certificate below proves that the two real projector images are three-dimensional and irreducible for the listed incidence action. Scalar extension to complex irreducibility, identification with the abstract icosahedral or A_5 character table, and a physical-oscillation reading remain outside the checked statements (`GoldenSectorCharacters`).

Irreducibility of the two golden pieces is a theorem. Each image $W_\pm$ of the golden projectors is invariant under every listed automorphism and has dimension three (an explicit factorization $P = AB$ with $BA = 1$); the real span of the sixty matrices $gP_\pm$ has dimension nine and maps onto all 3×3 real matrices in the basis A (nine listed elements with an explicit two-sided inverse certificate over $\mathbb{Z}[\varphi]$, the minus certificates the Galois conjugates of the plus ones), so every invariant subspace of $W_\pm$ is zero or all of it: both pieces are irreducible real representations of the listed incidence-automorphism action (`GoldenSectorIrreducibility`). The pieces stay irreducible after extension of scalars to $\mathbb{C}$: the nine listed automorphisms whose transported tables span the real endomorphism algebra span the complex one with the same inverse-certificate coefficients, so every invariant complex subspace is zero or the whole piece. The two complexified pieces are inequivalent, since on the listed row of order five their traces are $1 - \varphi$ and φ, every complex intertwiner between them is zero, and the real endomorphism algebra of each piece is the real scalars, so both are of real type (`GoldenSectorComplexIrreducibility`). Identification with the abstract icosahedral or A_5 table, with physical rotations, or with physical oscillations is unproved.

The two attained faces of mechanics are joined by a finite Legendre bridge, also checked in Lean [29]. For discrete two-point Lagrangians the Legendre transform, its involutivity on the strictly convex class, the Fenchel–Young inequality with its equality case, and exact degenerate controls are proved; the discrete Euler–Lagrange condition at a junction holds exactly when one step of the discrete Hamilton flow carries the incoming junction state to the outgoing one, for the quadratic class and for strictly convex Lagrangians with a solver section; single-site minimizers of the log-transition local action coincide with most probable paths of the realized chain, through an exact corner identity; and the constant Noether current of the chain face equals the quadratic Legendre momentum, conserved together with the energy along the free Hamilton orbit that reproduces the committed witness path. One remainder is named: no theorem produces the realized chain from a Hamiltonian flow. What is proved in that direction is narrower. The committed repair-count action reproduces the chain path law at no multiplier and lies outside the additive-constant and rescaling gauge orbit of the derived log-transition action (`sourceRepairAction_no_multiplier`, `sourceRepairAction_not_gauge`), so the naive quadratic-Gibbs identification is refuted on the chain literals, and the derived action above is the only route this package supplies. A direct Hamiltonian-flow derivation of the chain, for example a block Hamiltonian of the Stone converse coarse-grained to the two-state chain, is excluded by no theorem here and is unconstructed.

Theorem 5.5 (Realized histories do not select a real Legendre enrichment)**.** *Let $L_0(x, y)$ be the bilinear real extension of the committed two-state log-transition table. It is affine in y, so its momentum at fixed x has singleton image and admits no global velocity solver. For every $a \in \mathbb{R}$, define*

$$L_a(x, y) = L_0(x, y) + \frac{a}{2}\, y(y - 1).$$

Every L_a agrees with the exact source log-transition action on every realized binary history at every path length. For $a > 0$, however, L_a is strictly convex in y, has an explicit global velocity solver and Legendre transform, and different positive a's give different formulas. In particular the checked $a = 1$ and $a = 2$ members have distinct Lagrangians and Hamiltonians, agree on all source histories but differ by $1/8$ when one middle record is varied to $y = 1/2$.

Proof. The correction $y(y - 1)$ vanishes at both source symbols. Differentiation gives $\partial_y L_a = \partial_y L_0 + ay - a/2$, so $a = 0$ is degenerate and $a \neq 0$ has the displayed affine inverse. Completing the square gives the Legendre transform; the strict-convexity gap is $(a/2)b(1 - b)(y - z)^2$. The path identity, midpoint control, solver, convexity, and the $a = 1, 2$ distinction are checked in Lean. □

The reference measure is characterized within a declared independent-target scrambling normal form: among row-stochastic kernels, invariance under every independent relabeling of transition targets at fixed source, row-constant transition weight, and constant log-transition step action are each exactly equivalent to the uniform kernel, so any independently target-relabeling-invariant Markov reference has the step-uniform path law, and a biased two-state control shows the invariance premise is load-bearing. Ordinary simultaneous source-target relabeling is strictly weaker and admits a positive nonuniform stay-biased control. The stronger condition fixes the reference gauge by an invariance convention, exactly as a coordinate convention is fixed; the invariance principle itself is stated, not source-produced. Feeding the separately declared counting reference and trivial visible datum into the premise A3 conditional-resampling construction reproduces the uniform transition kernel exactly. This is a conditional realization, not source selection of those inputs, the initial distribution, or the complete path reference; constant rescalings of the counting mass are another exact ambiguity. The committed reference/action/multiplier receipts also belong to two distinct Gibbs constructions and do not form one source-selected capstone. Within the repair-count construction, the matching polynomial is strictly increasing on the positive exponential-parameter ray, so the supplied empirical target has one unique matching parameter. The target observable and level are declared postprocessing inputs rather than prospectively source-selected constraints.

A three-record concave-action control has a stationary history that is not a minimum and is assigned less positive Gibbs weight than a unit variation. It proves that a positive Gibbs/MAP rule need not make every stationary point a mode or minimizer. It is not a no-go for constrained saddles, complex stationary phase, path-integral interference, or a source-selected real Lagrangian, all of which remain available. The package identifies neither A_{eff} with the local action nor the finite scalar chain constant with a physical current, and supplies no amplitude, clock, or continuum path measure. The theorem above also shows that the realized two-state history law, on the committed chain literals, does not select the off-alphabet curvature needed for a regular Legendre system; finer alphabets of the same run, refinement limits, other source data, and declared principles are untested by that theorem and stay live routes to the curvature. It derives the finite action and conditional bridges at the representation level; the real enrichment and physical attachments are not supplied under the attachment doctrine.

Two composition results sharpen this boundary. First, the two halves of the mechanics package form one theorem (`source_to_hamiltonian_composed`): under the registered path reference, real enrichment, and supplied-dynamics rows, a fixed-endpoint real-extremal embedded history realizes

the derived corner action of the same kernel whose exponential tilt over the registered reference is the path law, is an interior most-probable update of that law at every interior site, and satisfies the discrete Hamilton equations. The separate `source_to_hamiltonian_composed_noether` theorem makes the segment momenta constant under supplied symmetry data; the mode/minimizer scoping and the enrichment no-go ride along as conjuncts of the same statement. Second, while the realized law cannot select the enrichment, a declared mode-extremality principle fixes one member inside the registered one-parameter curvature ansatz on the committed chain: interior stationarity of the embedded constant-record history forces the curvature $a_\star = 2\log(W_{11}^2/(W_{10}W_{01})) > 0$, positivity being exactly strict interior mode dominance, and at $a_\star$ that history is a genuine fixed-endpoint single-site minimizer at every interior junction and length, so the composed conditional is inhabited on the committed kernel rather than vacuous. This is not uniqueness among regular corner-preserving enrichments, and the residual freedom is exactly characterized inside one declared grammar, in the module `EnrichmentCharacterization`. Within the quadratic two-slot grammar, a sentence vanishes at the four binary corners exactly when it is $\varepsilon\, x(x-1)+\zeta\, y(y-1)$, so the corner-invisible enrichments are exactly the two-parameter family $L_{a,c} = L_0 + a\, y(y-1)/2 + c\, x(x-1)/2$, with injective parameters and no third corner-invisible quadratic direction (`quadPoly_cornerInvisible_iff`). The committed interior-junction stationarity holds on this family exactly on the affine line $a+c = a_\star$ (`chainTwoSlot_stationary_line`); members of that line share the binary corners, the constant-history stationarity, and the single-site variation minimum, while the regular member $a = c = a_\star/2$ has a different velocity curvature. One named velocity-only clause, that the added sentence is a function of the velocity slot alone, restores uniqueness at $(a, c) = (a_\star, 0)$, the committed one-parameter rule with all six grammar coefficients pinned (`quadPoly_selection_unique`). Each clause is load-bearing: the counterfamily point $a = c = a_\star/2$ satisfies every clause except velocity-only and carries a different momentum map (`velocityOnly_clause_necessary`); an explicit velocity-only cubic satisfies corner invisibility and the junction stationarity equation, so quadraticity is load-bearing under stationarity alone (`quadraticity_clause_necessary`); and the committed real fixed-endpoint single-site minimality excludes those cubics and returns the committed rule exactly (`chainCubic_realMin_forces_quadratic`). This cubic exclusion is not global selection. For every $0 < \lambda < a_\star$, the velocity-only quartic continuation

$$L_\lambda(x, y) = L_{a_\star}(x, y) + \lambda y^2(y-1)^2$$

is corner-invisible, has the same junction stationarity, and keeps the constant-one history globally minimal under every fixed-endpoint single-site variation, with exact gap $(x-1)^2(a_\star/2 + \lambda x^2)$. Its velocity Hessian is $(a_\star - \lambda) + 12\lambda(y - 1/2)^2 > 0$, so its momentum map is strictly increasing, surjective, and globally invertible, yet it is neither the committed rule nor any member of the quadratic grammar (`quartic_continuation_blocks_global_selection`). Thus the tested clauses are not a proved-minimal or globally selecting axiom system.

The quartic boundary is sharpened by one further named clause, tested in `PolynomialEnrichmentSelection` against the full two-slot polynomial grammar at every degree bound. A two-point sentence is a discrete null Lagrangian when no single-site interior replacement changes the sum of the two adjacent terms; by the committed difference identity such sentences contribute only telescoping and boundary terms to every fixed-endpoint local action. The literal boost clause, that every shift residue $q(x+u, y+v) - q(x, y)$ is null, fails at the committed increment itself (`committed_increment_not_boostNull`): a lattice-anchored quadratic has a linear, non-telescoping boost residue. The named candidate clause is second order: every boost difference of a boost difference is null (`BoostNullCovariant`). Every quadratic-grammar sentence satisfies it. Among velocity-only polynomial enrichments of every degree bound the clause holds exactly for the quadratic-affine members (`velocityPoly_boostNull_iff`), and the quartic continuation fails it at

an explicit finite witness (`quarticIncrement_not_boostNullCovariant`). In the full polynomial grammar, corner invisibility, the velocity-only clause, the committed stationarity calibration, and the boost clause select exactly the committed point (`polyGrammar_selection_eq_committed`), and the committed point satisfies all four clauses, so the axiom system is inhabited. Every clause is load-bearing: dropping the boost clause readmits the quartic, and in the full two-slot grammar the boost clause does not force quadraticity, since the antisymmetric cubic $(x-y)^3-(x-y)$ is corner-invisible, satisfies the boost clause, is neutral for the committed stationarity calibration, and equals no sentence of the committed quadratic grammar (`boost_clause_does_not_force_quadraticity`); the velocity-only clause is what removes this direction. The boost clause is a declared candidate axiom rather than a derived symmetry: no boost action on realized histories and no invariance of the source law is constructed, and smooth non-polynomial enrichments are outside the theorem.

The grammar, the mode-extremality principle, and the velocity-only clause are declared rather than source-derived; nothing here states that the source selects the enrichment, and no real enrichment is supplied. The realized history law alone selects no real enrichment (Theorem 5.5).

6 Thermodynamics from Conditional Repair

On one finite regulator stage, the four laws form a conditional theorem package once one repaired distinction and two compatible instantiations of premise A3 are supplied. Five named source and physical receipts are unsupplied. The Lean development carries three composed named-premise surfaces for these packages. `FourLawAdequacySurface` proves one theorem, `fourLaws_composed`: one typed antecedent bundle whose fields are exactly the registered repair law, the clock-and-energy calibration, and the refinement-uniform gap family, and one conclusion record whose clauses all consume that bundle: the kernel receipts, both zeroth-law forms, the modular and diagonal-heat-channel first laws, the second-law and fluctuation statements, the finite and refinement-uniform third laws threaded to the same calibrated energy through an explicit member identification, and the Landauer bound. The same surface carries an exact dissipation identity: the mean fluctuating entropy production of one repair step equals the relative entropy from the input state to its repaired image, is strictly positive whenever the step changes a strictly positive state, and vanishes exactly on the fibre-conditional-reference fixed points, so one state-changing projection has a strictly positive dissipation. Because this repair kernel is idempotent, repeating the identical step stalls on the fixed manifold after that first projection; a sustained macroscopic arrow requires source dynamics. `HorizonThermalitySurface` carries the cap Clausius inequality and the capacity identities as the horizon-facing composition. `MechanicsAdequacySurface` re-exports the composed source-to-Hamiltonian theorem of Section 5. Three maps stay distinct. The strict-descent normalizer settles public facts; it is terminating and idempotent and carries no entropy inequality, and the two-point normalizer $a \mapsto b$, $b \mapsto b$ sends the uniform law to a deterministic law, dropping Shannon entropy from $\log 2$ to zero. The equilibrium repair operator resamples unresolved degrees inside the fibre of the complete repaired visible datum. The modular flow of a faithful reference supplies the reversible observer ordering and preserves relative entropy. Only the middle map carries the arrow.

The refinement-uniform gap family is the module `CofinalSpectralTailFamily`. It carries per-stage finite spaces and energies, refine maps obeying identity, composition, energy and ground-compatibility laws, one uniform spectral-tail envelope, a nonempty ambient regulator preorder, an order-reflecting cofinal stage embedding, no maximal ambient regulator, and the data-level condition

$$\forall r\ \exists s > r, \qquad |X_r| < |X_s|.$$

The represented stage type is therefore infinite, and a constant finite carrier cannot inhabit the

interface; a nonconstant $\mathbb{N}$-indexed ladder with strict carrier growth supplies a nonvacuous inhabitant. The envelope gives one inverse-temperature threshold per tolerance for the off-minimum Gibbs mass uniformly over all stages and, for every ambient regulator, over the complete cofinal tail above it (`uniform_concentration`). The ladder family also proves that a spectral-tail envelope does not imply a uniform total-cardinality bound on the same carriers and energies (`ladderFamily`, `ladder_uniformGap_isEmpty`). The theorem `fourLaws_composed_cofinal` constructs the refinement-independent finite-law core directly from the declared repair and calibration, and every regulator-family conclusion refers to the same supplied cofinal family. A separately supplied calibrated-stage energy identification yields exact off-minimum-mass equality and the same envelope at that member; the calibration is not derived. Strict carrier-cardinality growth is one sufficient genuine-refinement route, not a necessary one. No OPH source produces this family or identifies its regulator, energy, inverse temperature, repair or continuum meaning.

Premise A3 applies separately to states and to transition distributions, and both instantiations are solved exactly. On the state side, with faithful global reference σ_0 and observer-visible conserved quantities Q_a, assume the constrained moment surface contains a faithful feasible state. On this interior branch, the information-projection Pythagorean identity $D(\rho\|\sigma_0) = D(\rho\|\tau_\lambda) + D(\tau_\lambda\|\sigma_0)$ holds on the constrained moment surface, so the optimizer, unique by strict convexity of relative entropy, has the finite-multiplier exponential form $\tau_\lambda = Z^{-1}\exp(\log\sigma_0 - \sum_a \lambda_a Q_a)$; on the tracial branch this is the Gibbs family. Without the faithful-feasibility premise, a boundary constraint can force a rank-deficient optimizer with no finite full-algebra multiplier; it needs a support-restricted or limiting treatment. The exact zeroth-law receipt is temperature identifiability: on one nondegenerate finite spectrum, two Gibbs distributions that agree have equal β. It is not a contact-dynamics or equilibration theorem. On the transition side, the feasible set at x is the fibre $F_C(x) = \{y : b_C(y) = b_C(x)\}$ of the complete repaired visible datum b_C, and the minimizer of $D(q\|\pi)$ over the fibre, unique by the same strict convexity, is weighted conditional resampling,

$$P_C(x,y) = \mathbf{1}_{\{b_C(y)=b_C(x)\}}\,\frac{\pi(y)}{\pi(F_C(x))},$$

which is the weighted observation-fiber resampling projector, and any physical identification of a concrete repair law with it passes through the support, equal-fiber-row, and weighted detailed-balance receipt of the observation-normal-form construction [16]. The kernel is stochastic, idempotent, reversible, and stationary for π, and fixes every fibre-measurable observable, which is dynamical conservation of every protected charge and the content side of the first law. For a finite joint update, the exact bookkeeping identity is $\Delta U = \delta Q + \delta W + \mathrm{Re}\,\mathrm{tr}(\delta\rho\,\delta H)$. The two-term form is first-order, or exact for an ordered stroke with one of H or ρ held fixed; the composed repair heat stroke holds H fixed, so its work and cross terms vanish. Data processing then gives the transition-side second law $D(pP_C\|\pi) \le D(p\|\pi)$, and the exact split $D(p\|\tau) = \langle K\rangle_p - S(p)$ with $K = -\log\tau$ turns it into the modular Clausius inequality $\Delta S \ge \Delta\langle K\rangle$ for every τ-preserving repair channel, which specializes to $\Delta S \ge \beta Q$ on the identified energy branch. The Landauer bound is a one-line corollary: a repair channel that lowers the record entropy by c expels at least c/β of energy, so erasing one bit costs at least $k_B T\log 2$. At every finite regulator the excited Gibbs mass obeys $q_\beta \le \frac{d-g_0}{g_0}e^{-\beta\Delta}$, with the quantitative threshold that any target mass ε is reached once $\beta\Delta$ exceeds $\log\frac{d-g_0}{g_0\varepsilon}$, and the entropy limit $k_B\log g_0$ is the standard finite consequence of the displayed mass bound and finite continuity of entropy; the composed Lean record contains the mass bound, not a separate entropy-limit field. The repair kernel preserves full support, since every diagonal entry is positive, while the zero-temperature state is rank deficient for $g_0 < d$; finitely many repair or pinching steps therefore cannot reach it, which is the fixed-regulator third law. Public quantum-record pinching and averaging have explicit normalized Kraus families and preserve trace. A separate module

proves these channels completely positive and trace preserving through the Kraus criterion; this result's own packet reports Kraus and trace identities. The record tower $S(\rho) \le S(\mathcal{E}_\pi \rho) \le S(\mathcal{A}_\pi \rho)$ separates conditioned learning from the nonselective commit, so observers convert uncertainty into correlations and durable records without violating the second law.

The second law refines to an exact identity. For one repair step the fluctuating entropy production $\sigma(x,y) = \log \frac{p(x)}{\pi(x)} - \log \frac{(pP_C)(y)}{\pi(y)}$ obeys three exact relations. The integral identity $\langle e^{-\sigma} \rangle = 1$ holds for every kernel that preserves the reference; under detailed balance the pointwise relation $p(x)\, P_C(x,y) = e^{\sigma(x,y)}\, (pP_C)(y)\, P_C(y,x)$ holds at every pair of states; and summing it over a level set gives the forward weight at entropy production s as e^s times the reversed weight. The mean of σ is exactly the certified descent $D(p\|\pi) - D(pP_C\|\pi)$, so the second law is the Jensen shadow of an identity rather than a primitive inequality. Detailed balance also gives finite Onsager reciprocity. Because $e^{-\sigma}$ is a ratio of rationals, the fluctuation certificate closes exactly over the rationals, with no numerical logarithm.

The reversible symmetry and correlation representation below are finite relatives of standard Onsager and Green–Kubo theory [35, 36]. The OPH-specific statement is the resulting model-selection obstruction for the full-fibre publicization projector.

Theorem 6.1 (Finite transport kernel and repair-projector obstruction)**.** *Let K be a finite stochastic kernel reversible under a strictly positive weight π, let $L = I - K$, and let R be a linear solver of $LRj = j$ on the π-centered currents. For any finite current family $\{j_a\}$,*

$$\Gamma_{ab} = \langle j_a, R j_b \rangle_\pi$$

is symmetric and positive semidefinite. At every finite cutoff N,

$$\langle j, Rk \rangle_\pi = \sum_{n=0}^{N} \langle j, K^n k \rangle_\pi + \langle j, K^{N+1} R k \rangle_\pi.$$

Thus a Green–Kubo coefficient is a correlation sum plus an exact displayed remainder, rather than a silently assumed infinite series.

For the full-fibre conditional-resampling kernel P_C, idempotence gives $P_C^n = P_C$ for every $n \ge 1$. Positive-lag correlations are therefore constant. A fibre-centered current with $P_C j = 0$ has no positive-lag memory, while a nonzero one-step correlation makes consecutive partial sums differ by the same nonzero amount. One full-fibre repair projector cannot be a decaying, nonzero-memory transport dynamics.

On a finite closed oriented graph, typed positive cell volumes and heat capacities convert concentration to particle amount and temperature to energy. Fick and Fourier fluxes then define canonical one-step updates whose total changes equal the clock increment times the total source and hence vanish when the source sums vanish. Nonnegative conductance makes the flux–gradient pairing nonpositive; explicit negative-conductance controls reverse that sign.

Proof. Detailed balance makes L self-adjoint, and the exact Dirichlet identity expresses $\langle f, Lf \rangle_\pi$ as a nonnegative weighted sum of squared differences. Applying this to linear combinations of the Poisson solutions proves symmetry and the full finite quadratic-form inequality. The cutoff formula telescopes the Poisson equation. Projector idempotence gives the positive-lag dichotomy. Graph summation by parts and cancellation of internal oriented edges give the constitutive and global-balance statements. Every step, including two-state and two-vertex witnesses and negative controls, is checked in Lean [29]. □

The theorem does not select a physical transition generator, distance, clock, boundary condition, conductance, or readout. It proves no stability, hydrodynamic limit, or measured transport coefficient. In particular, the projector obstruction requires a source-derived nonidempotent local or random-scan effective evolution before OPH can make a long-memory transport claim.

6.1 Protected records with decaying repair memory

A conditional expectation describing equilibrium need not be the transition operator. This distinction admits conserved records and dissipative memory on one finite state space. The following construction supplies a local stochastic law and its faithful reference; neither is identified with a particular physical source.

Let $X = \coprod_c(\{c\} \times U_c \times V_c)$ be finite, with nonempty coordinate sets and a strictly positive probability law π. On the real space $L^2(\pi)$, let P be conditional expectation given c, E_u conditional expectation given (c, v), and E_v conditional expectation given (c, u). Their stochastic interpretations resample only the omitted coordinate from the same reference. Define

$$T = \tfrac{1}{2}I + \tfrac{1}{4}E_u + \tfrac{1}{4}E_v. \tag{3}$$

Thus bounded patches carry local state (c, u, v), a protected record c, coordinate readback ports and conditional repair moves. The step index is a declared update count, with no laboratory time calibration.

Proposition 6.2 (A protected-record dynamics with finite correlation tails)**.** *The kernel T is stochastic, reversible with respect to π, and fixes every function of c. It satisfies $TP = PT = P$. There is $0 < \varepsilon < 1$ such that the entrywise inequality $T^2 \geq \varepsilon P$ holds. Consequently, on* $\ker P$*,*

$$\|T^n\| \leq (1-\varepsilon)^{\lfloor n/2 \rfloor}, \qquad \operatorname{gap}(I - T) \geq \varepsilon/2. \tag{4}$$

The unique fibre-centred Poisson solver is $R = (I - T + P)^{-1} - P$. For $Pf = Pg = 0$,

$$\mathcal{G}(f, g) = \langle f, Rg\rangle_\pi = \sum_{n=0}^{\infty} \langle f, T^n g\rangle_\pi \tag{5}$$

is symmetric and positive semidefinite. After retaining lags 0 through N, its exact remainder and a uniform bound are

$$\begin{aligned} \mathcal{G}(f, g) - \sum_{n=0}^{N} \langle f, T^n g\rangle_\pi &= \langle f, T^{N+1} Rg\rangle_\pi, \\ \left| \mathcal{G}(f, g) - \sum_{n=0}^{N} \langle f, T^n g\rangle_\pi \right| &\leq \frac{\|f\|_\pi^2 + \|g\|_\pi^2}{\varepsilon} (1-\varepsilon)^{\lfloor (N+1)/2 \rfloor}. \end{aligned} \tag{6}$$

Proof. Conditional expectations are orthogonal projections in $L^2(\pi)$; their kernels satisfy detailed balance and preserve c. The tower property gives $TP = PT = P$, and $T \geq I/2$ as an operator. Two consecutive repairs, first one coordinate and then the other, can reach every state in the same fibre with positive probability. Finiteness and faithfulness therefore give the displayed minorization. The remainder $Q = (T^2 - \varepsilon P)/(1 - \varepsilon)$ is stochastic and stationary for π. Jensen's inequality gives $\|Q\| \leq 1$. On $\ker P$ this implies (4), using $1 - \sqrt{1-\varepsilon} \geq \varepsilon/2$ for the gap. The Neumann series on that subspace defines R, while R vanishes on $\operatorname{ran} P$. Reversibility makes R self-adjoint, and its eigenvalues there are $(1-\lambda)^{-1} > 0$. The finite geometric-series identity gives the exact remainder. Sum $|\langle f, T^n g\rangle| \leq \|f\|\|g\|(1-\varepsilon)^{\lfloor n/2 \rfloor}$ from $N+1$ onwards, grouping successive pairs and using $2\|f\|\|g\| \leq \|f\|^2 + \|g\|^2$. □

An exact eight-state instance takes $c, u, v \in \{0, 1\}$, fibre masses $(1/3, 2/3)$, and conditional reference weights $(1, 2, 3, 4)/10$ and $(4, 1, 2, 3)/10$ in lexicographic (u, v) order. Both references are nonproduct laws. Its sharp two-step minorization is $\varepsilon = 7/96$, giving a certified gap lower bound $7/192$. For $f = u - Pu$ and $g = v - Pv$, the one-sided coefficient matrix is exactly

$$\begin{pmatrix} 11417/9375 & 1832/3125 \\ 1832/3125 & 3816/3125 \end{pmatrix},$$

and the positive-lag autocorrelation $\langle f, Tf \rangle = 443/2400$ is nonzero. An independent rational verifier reconstructs the coordinate kernels, checks both Poisson inverse identities, and replays 65 correlation lags, the exact cutoff remainder and eight distribution readouts.

Fibre centering is essential: $c - 2/3$ has global mean zero but is fixed by T, with persistent correlation $2/9$ and no centred Poisson solution. The limiting distribution retains its initial fibre masses. This is not global ergodicity, an identification of T with the full-fibre projector, or a dynamic intertwiner recovering a recurrent mode from that projector. It leaves the obstruction for the distinct source-counted transition artifact intact. The local stochastic law and reference are supplied; physical source attachment, transport normalization, clock calibration and a continuum limit require additional arguments.

Maintained records in a noisy coherent source. A declared coherent-code source realizes bounded self-reading patches with local state, ports, readback, noisy feedback and actual public record banks. Complete transfer/reset services reduce carrier leakage, and a protected quantum computation exports into a maintained classical archive. With a fixed positive local service error below a derived library bound, $O(\log \log q)$ quantum protection levels, staged decoding and expander refresh of $O(\log q)$-bit words make the joint decoded quantum and live record error tend to zero. The archive remains readable throughout each refresh; no perfect bare decision bit or fault-free runtime controller is assumed. The analytical argument imports computation-code fault tolerance and an explicit expander bound; finite tests check its interfaces and fault controls, without executing the enormous all-size apparatus. Coherent code gluing, the local noise law, external schedule and cofinal control/inventory resources are hypotheses. They are not consequences of classical record agreement. Logical accounting, physical energy, calibrated time and source-selected matter dynamics are distinct.

For a separate deterministically prepared filled vacuum, the same massive walk has a positive normal-ordered excitation Hamiltonian with exactly its integer-time Fock channels. Both carrier bands enter the local covariance limit. Complete resolved-mode number reads retain the occupied background and have bounded mean excitation cost in the stated smooth carrier class; a finite analytical clock witness has probability error below 0.038 and swing above 0.923. Its enormous three-dimensional apparatus is not simulated. The longer sea preparation has its own charged protection and storage bounds, so the empty-vacuum resource estimates do not apply to it. High-band Floquet shifts preserve these integer-time reads and infrared limits while changing ultraviolet excitation costs. This positive stroboscopic generator identifies neither laboratory energy nor the work of the supplied controls.

A native stationary history source. Fair integer seam repair continued within a balanced fixed-total sector supplies a concrete nonidempotent evolution. On a connected weighted graph of p ports with $0 < r < p$ raised loads, it is the symmetric exclusion chain with generator $Gz = -Lz/2$ for $z = n - (r/p)\mathbf{1}$. Under the uniform stationary law, $\mathrm{Cov}(z) = \kappa\Pi$, where $\kappa = r(p - r)/(p(p - 1))$

and $\Pi = I - \mathbf{1}\mathbf{1}^{\mathsf{T}}/p$. Consequently a fixed linear local readout B has history limit

$$T^{-1/2} \int_0^T Bz(t)\, dt \Longrightarrow \mathcal{N}(0, 4\kappa B L^{+} B^{\mathsf{T}}).$$

This is a fixed-graph additive-functional limit for independent Poisson attempt clocks. Its ordinary time average has mode variance $\kappa h(\lambda T/2)$, with $h(u) = 2(u - 1 + e^{-u})/u^2$, rather than a freely normalized Gaussian source. The screen-microphysics paper gives the martingale proof, the nonlinear readout decomposition and the correction for pre-attempt incident-event records. Exact finite-state witnesses and independent numerical controls are retained in the public repository [29]. Occupancy, record gain, elapsed-time convention and the physical meaning of the readout require specification. The result supplies neither a uniform refinement limit nor a measured primordial amplitude or tilt.

The same package carries the entropy-balance shape used by the gravitational branch. For any state and faithful reference the exact identity $S(p) - S(\tau) = \langle K \rangle_p - \langle K \rangle_\tau - D(p\|\tau)$ holds, so the entanglement first law $\delta S = \delta\langle K \rangle$ is this identity with the quadratic-order deficit dropped; through the central split $K = 2\pi B + Z$ it reads

$$\Delta S = 2\pi \Delta\langle B \rangle + \Delta\langle Z \rangle - \Delta D(\,\cdot\,\|\tau),$$

where $\Delta D = D(p'\|\tau) - D(p\|\tau)$. For the reference-preserving repair step, data processing gives $\Delta D \leq 0$, and one repair step with a central charge measurable through the repaired visible datum obeys the cap Clausius inequality $2\pi\Delta\langle B \rangle \leq \Delta S$, since the kernel also conserves every fibre-measurable charge and hence $\Delta\langle Z \rangle = 0$. The finite first-law premise package of the Einstein branch is discharged on this model: on the tangent space of the probability simplex the entropy differential at the reference equals the modular pairing as a theorem, the modular pairing distributes through the central split, and the central pairing is the edge differential, so the branch's finite bulk-edge-central first law applies with no assumed premise beyond the split of the reference. Repair displacements are mass preserving with vanishing central pairing, so the first-order entropy change of one repair step is purely bulk-modular. The physical attachment of the split stays with the energy-clock receipt.

The Lean development carries the exact finite statements: relative entropy nonnegativity, the log-sum inequality, data processing, the conditional-resampling optimizer and its kernel package, the Gibbs Pythagorean identity and minimizer, the first-law split, the low-temperature bound, the fluctuation identities with reciprocity, the finite Green–Kubo matrix and exact cutoff remainder, the repair-projector memory obstruction, the typed Fick and Fourier updates, the exact-remainder cap first law, the discharged Einstein-branch first-law premises, and the Kraus completeness of partition pinching; the modules are `FiniteConditionalRepair`, `FirstLawIdentity`, `FluctuationTheorems`, `GreenKubo`, `GraphDiffusion`, `CapFirstLaw`, `EinsteinPremiseLink`, `StationaryRealization`, and `PartitionPinchingCP` in the Lean thermodynamics and event-algebra libraries, and the exact-rational certificate replays the kernel algebra and fluctuation identities and records the entropy checks [29]. The source-realization interface needs less than reversibility: every stochastic kernel that preserves a faithful reference contracts relative entropy to that reference. Detailed balance is not a premise. The exact lazy directed three-cycle is stationary and violates detailed balance, yet satisfies this second law; the Lean module `StationaryRealization` proves the separation. Detailed fluctuation relations and Onsager reciprocity therefore remain additional receipts rather than consequences of entropy descent.

The pinned source-counted collar table has also been exhausted within its declared coordinate-projection grammar. All $2^4 - 1 = 15$ syntactic maps obtained by retaining a nonempty subset of its four committed packet fields were audited on 1028 observers and 31744 counted transitions, without

an additional observer-patch run. Because the record-family and S_3-sector fields are constant on this run, those maps induce only four distinct partitions. The repair-load count aggregation is an eight-state stochastic kernel that is irreducible and aperiodic and has a full-support stationary law. It is not a certified Markov quotient of the fine chain: its strong-lumpability defect, the largest outgoing target-block mass difference between two fine states in one repair-load block, is 0.911587983 on the pinned floating-point table. On that table its detailed-balance defect is $2.009924089 \times 10^{-4}$; from the uniform initial law the sampled relative entropy falls from 5.00947 to 1.84009 over sixteen steps. This is a nonreversible finite H-theorem probe, not a microscopic-reversibility receipt. It does not close the realization: the only observed record-family value makes protected-charge preservation vacuous, and the stationary law is not identified with the state optimizer's source reference. On the fine twenty-state quotient the only closed communicating class is the singleton freezeout state, so the alternative recurrent-class restriction is a trivial equilibrium rather than a nontrivial relaxation model. These results exhaust only the declared coordinate maps and the fine-chain recurrent restriction. They do not enumerate arbitrary partitions or nonlinear, statistical, stochastic, weakly lumpable, or history-dependent quotient maps of the retained artifact. The audit excludes none of those routes, nor changed source dynamics or a bounded run with a nonconstant protected record.

The state table/reference was locally pre-specified and hash-pinned; the recurrent transition chain was extracted post hoc from retained run data. This does not repair the failure. The state-side conditional-resampling action is idempotent, whereas the exact two-state recurrent transition action has the nonconstant eigenpair

$$Pv = \frac{665437}{726948}v, \qquad v = (54356, -7155), \qquad 0 < \frac{665437}{726948} < 1.$$

If an idempotent action H and this transition action P were related by $HT = TP$, applying idempotence twice gives $\lambda(\lambda - 1)Tv = 0$, hence $Tv = 0$. Thus every such intertwiner erases the exact extracted mixing mode. The transition stationary mass 7155/61511 also lies strictly between the adjacent empirical masses 1905/16384 and 1906/16384; no deterministic coarse graining of the 16384-sample state reference produces it. A freely chosen stochastic coupling could evade the denominator test, but would be a new premise rather than source evidence. These exact obstructions exclude the two audited direct bindings (a mixing-mode-preserving linear intertwiner into the idempotent heat bath and a deterministic empirical pushforward) on the committed artifact. They do not exclude stochastic, nonlinear, reverse-direction, or enriched-source constructions. The committed random-scan preflight decides the random-scan instance of the stochastic route within its certified grammar: on every computed subset (the eleven subsets of at least two of the four committed packet fields on the 26-state alphabet, under the visit-count reference, and the record/companion pair on the realized 256-state structure, under the occupation reference), the random-scan mixture of conditional-resampling projectors under both declared schedulers is row-stochastic and exactly stationary for the one shared reference of its arena. The committed uniform-scheduler mixtures are additionally certified non-idempotent, so they escape the idempotent spectral obstruction; the exact fixed space of every computed mixture under either scheduler is one-dimensional, so the only observables it protects are constants. The remaining step-field subsets contain a certified-constant field; their mixtures carry the strictly positive full-space resampler as a component, which forces the same one-dimensional fixed space without any claim of non-idempotence. The only retained step field beyond the packet fields is certified constant on the run, so the retained step record admits no enrichment; the retained non-step cell fields are outside the certified grammar. The recorded continuations are a fresh export of a dynamically conserved label whose join with a committed field disconnects, or a dilated construction outside the random-scan grammar.

The conditional package retains five typed requirements: the global representation of the

weighted local objective for both optimizer instantiations; the common source-derived reference shared by state and transition optimizers, which forbids manufacturing the transition matrix from a desired equilibrium output; the identification of the actual source-derived collar transition matrix with the conditional-resampling kernel through the existing matrix receipt; stochasticity, stationarity, and protected-charge preservation alone certify only the weaker stationary H-theorem branch and do not discharge that full premise-A3 receipt, while detailed balance is additionally required wherever microscopic reversibility is claimed; the identification of one modular charge with physical energy and a calibrated clock, of which the finite central-interface split $K_C = 2\pi B_C + Z_C$ is the attained cap-branch part; and uniform low-temperature spectral-tail control on a genuinely cofinal family for a continuum third law. The committed cofinal interface provides the ambient order, cofinal embedding, no-terminal-stage theorem, strict carrier growth, and same-family concentration theorem, while supplying no source or physical continuum interpretation. Thus a source-produced family inhabiting that interface and its physical regulator attachment are unsupplied. The finite gap-plus-cardinality theorem is one sufficient route rather than a necessary condition. Every proof in the package is short and elementary once the framework supplies its two inputs, the repair fibre as the transition feasible set and the reference shared with the state optimizer. The four-law identities then require no additional thermodynamic premise or deep theorem; the work lies in the decomposition, which the certified strict-descent counterexample shows is necessary. On a source object discharging the named receipts, ordinary finite thermodynamic identities follow from the instantiated observer model rather than being postulated separately.

6.2 Elementary corollaries of the record and screen structure

The finite record algebra, the celestial sphere and the information projection carry several textbook laws as direct corollaries. Each is stated below with the exact identity it rests on and with the identification it does not supply. Exact rational controls for every statement, pinned to this section, are part of the repository receipts [29].

Repeated reads and the Zeno limit.

Proposition 6.3 (Repeatability and the Zeno limit)**.** *Let ρ be a state on the finite accessible algebra of one completed slice, P an event projector with $w = \mathrm{Tr}(\rho P) > 0$, and $\rho_P = P\rho P/w$ the declared Lüders update. Then $\mathrm{Tr}(\rho_P P) = 1$, and the update is idempotent. Let H be a Hermitian generator on the same space, $U_s = e^{-isH}$, and let the observer reread P after each of n equal steps of total parameter t. For ρ supported in the range of P, the probability p_n that all n rereads return P satisfies*

$$1 - p_n \le \frac{2t^2\|H\|^2}{n},$$

so $p_n \to 1$ as $n \to \infty$, and the conditioned state converges to $e^{-itPHP}\rho\, e^{itPHP}$.

Proof. Repeatability and idempotence are the Lüders theorems of the finite event-algebra library [19, 29]. Put $s = t/n$, $A = PU_sP$ and $B = Pe^{-isPHP}P$; both are contractions on the range of P. For Hermitian X the exponential series gives $\|e^{-isX} - (1 - isX)\| \le s^2\|X\|^2/2$, hence $\|A - B\| \le s^2\|H\|^2/2 + s^2\|PHP\|^2/2 \le s^2\|H\|^2$. Telescoping, $\|A^n - B^n\| \le n\|A - B\| \le t^2\|H\|^2/n$. The joint probability of n returns is $p_n = \mathrm{Tr}(A^n\rho A^{n\dagger})$, while $B^n = Pe^{-itPHP}P$ is unitary on the range of P, so $\mathrm{Tr}(B^n\rho B^{n\dagger}) = 1$. For contractions $|\mathrm{Tr}(X\rho X^\dagger) - \mathrm{Tr}(Y\rho Y^\dagger)| \le 2\|X - Y\|$, which gives the bound; the convergence of the conditioned state follows from the same estimate. □

A rational control replaces the exponential by the rotation with $\tan(\theta_n/2) = 1/n$, so $\cos\theta_n = (n^2-1)/(n^2+1)$, read through a rank-one projector after every step. The survival with reads is $\cos^{2n}\theta_n$, which obeys $1-\cos^{2n}\theta_n \le 4/n$, while the survival without reads is $\cos^2(n\theta_n)$ and stays away from one. The step parameter is a flow parameter and H is supplied; neither is a physical clock or a source-selected generator.

Distinguishable records through a port.

Proposition 6.4 (Port capacity)**.** *Let the interface algebra of port p act on a Hilbert space of dimension d_p. A family of states that one read through the port distinguishes without error has at most d_p members. One read therefore carries at most $\log_2 d_p$ bits, and one read of all twelve ports of a carrier at most $\sum_p \log_2 d_p$ bits.*

Proof. States distinguished without error by one measurement have pairwise orthogonal supports, and pairwise orthogonal nonzero vectors are linearly independent, so there are at most d_p of them. The port bound adds over the ports because the boundary space is their tensor product. □

Holevo's bound, an imported result, gives the same ceiling for reads with error. Neither statement assigns an energy to a record: the energy of one distinction enters only through the clock and energy calibration, and the bound itself is a count.

Celestial kinematics.

Proposition 6.5 (Aberration, the null scale factor and rapidity addition)**.** *Let the celestial sphere carry the action of $SL(2,\mathbb{C})$ by Möbius maps of the spacetime and Einstein derivation [17], with stereographic coordinate ζ, and let each null direction carry the null vector $q = (1,\Omega)$. Realize the Lorentz group on Hermitian 2×2 matrices by $X \mapsto AXA^\dagger$, under which q is the rank-one matrix $\kappa\kappa^\dagger$ with $\kappa = (1,\zeta)$. For $A = \left(\begin{smallmatrix} a & b \\ c & d \end{smallmatrix}\right) \in SL(2,\mathbb{C})$:*

1. *the direction transforms as $\zeta \mapsto (c+d\zeta)/(a+b\zeta)$;*
2. *the time component of q is multiplied by $\lambda_A(\zeta) = (|a+b\zeta|^2 + |c+d\zeta|^2)/(1+|\zeta|^2)$, and $\lambda_A \equiv 1$ for every $A \in SU(2)$;*
3. *for $A_k = \mathrm{diag}(k^{-1/2}, k^{1/2})$ with $k>0$, put $v = (k-1/k)/(k+1/k)$, $\gamma = (k+1/k)/2$ and $\cos\theta = (1-|\zeta|^2)/(1+|\zeta|^2)$. Then $\lambda_{A_k} = \gamma(1-v\cos\theta)$ and the transformed direction obeys $\cos\theta' = (\cos\theta - v)/(1-v\cos\theta)$;*
4. *$A_{k_1}A_{k_2} = A_{k_1k_2}$, so $v(k_1k_2) = (v_1+v_2)/(1+v_1v_2)$.*

Proof. With $X = \kappa\kappa^\dagger$ the time component is $(|\kappa_0|^2+|\kappa_1|^2)/2$ and the axial component is $(|\kappa_0|^2 - |\kappa_1|^2)/2$, which gives $\cos\theta$. Under A, $\kappa \mapsto A\kappa = (a+b\zeta, c+d\zeta)$, which is (1) and (2); $SU(2)$ preserves $|\kappa|$. For A_k, $A\kappa = (k^{-1/2}, k^{1/2}\zeta)$, so $\lambda = (1/k + k|\zeta|^2)/(1+|\zeta|^2)$ and $\zeta' = k\zeta$; substituting $|\zeta|^2 = (1-\cos\theta)/(1+\cos\theta)$ gives (3). Diagonal matrices multiply entrywise, which is (4). □

Read with θ the angle between the propagation direction of a null ray and the axis, and v the axial velocity of a second frame, (3) is the aberration law and λ is the Doppler factor of that frame; (4) is the relativistic addition of collinear velocities. The scale factor is a statement about the null vector on the declared sphere. It becomes a frequency ratio only through the clock calibration and the physical spacetime attachment, which this section does not supply.

The canonical distribution as the information projection.

Proposition 6.6 (Gibbs law from one linear constraint)**.** *Let τ be a faithful reference law on a finite record set, E a real function on it, and $g_\beta = \tau e^{-\beta E}/Z_\beta$. For every law p with $\langle E\rangle_p = \langle E\rangle_{g_\beta}$,*

$$D(p\|\tau) = D(p\|g_\beta) + D(g_\beta\|\tau).$$

Hence g_β is the unique minimizer of $D(\cdot\|\tau)$ among laws with that expected value: the information projection of the third axiom under one linear constraint is the canonical distribution, and β is the multiplier of the constraint.

Proof. $\log(g_\beta/\tau) = -\beta E - \log Z_\beta$ is affine in E, so $D(p\|\tau) - D(p\|g_\beta) = \sum_x p(x)\log(g_\beta/\tau)(x) = -\beta\langle E\rangle_p - \log Z_\beta$, which by the constraint equals the same expression under g_β, namely $D(g_\beta\|\tau)$. Uniqueness follows from $D(p\|g_\beta) = 0$ only at $p = g_\beta$. □

The history-level form of this identity and the exponential form of the projected path law are machine-checked in the information-projection library [29]. The energy ledger E is declared, and β is free against the modular Hamiltonian alone; a temperature reading consumes the calibration row.

Occupation laws under a declared exchange sector.

Proposition 6.7 (Fermi–Dirac, Bose–Einstein and truncated occupation)**.** *Consider one mode of energy ε at chemical potential μ and inverse temperature β, and put $x = e^{-\beta(\varepsilon-\mu)}$. The Gibbs law of Proposition 6.6 on the occupation ledger, with weight x^n for occupation n, has mean occupation $x/(1+x) = 1/(e^{\beta(\varepsilon-\mu)}+1)$ when the mode admits the occupations $\{0,1\}$, $x/(1-x) = 1/(e^{\beta(\varepsilon-\mu)}-1)$ when it admits every $n \geq 0$ and $x < 1$, and $\sum_{n\leq m} nx^n / \sum_{n\leq m} x^n$ when it is cut at m; the last equals the first at $m = 1$ and increases to the second as $m \to \infty$.*

Proof. Geometric sums: $\sum_{n\geq 0} x^n = 1/(1-x)$ and $\sum_{n\geq 0} nx^n = x/(1-x)^2$; the two-term case and the truncated case are the same sums cut off. Monotonicity in m holds because each added term has occupation above the previous mean. □

The exchange sector is a declared premise of this proposition. Nothing in it selects the sector, and the two laws stand as the two admissible sectors of one information projection.

Coarse-graining of the record law.

Proposition 6.8 (Record identification does not raise the entropy of the record law)**.** *Let $r\colon \Sigma \to \Sigma'$ be a map on public records that identifies some of them, and let p be a law on Σ with push-forward r_*p. Then $H(r_*p) \leq H(p)$ for the Shannon entropy and $\sum_y (r_*p)(y)^2 \geq \sum_x p(x)^2$, with equality in both exactly when r is injective on the support of p. If K is a stochastic repair kernel with stationary law π, then $D_f(pK\|\pi) \leq D_f(p\|\pi)$ for every f-divergence, in particular for the relative entropy and the χ^2 divergence.*

Proof. $H(r_*p) = H(p) - H(p \mid r_*p)$ and the conditional entropy is nonnegative, vanishing exactly when r is injective on the support. For the collision sum, $(a+b)^2 \geq a^2+b^2$ for $a, b \geq 0$, with equality only when one vanishes. The kernel statement is the data-processing inequality for f-divergences applied to the pair (p, π) with $\pi K = \pi$. □

This is a statement about the law of the public records under repair. It does not identify that entropy with a thermodynamic entropy; the four-law package makes that identification under its declared repair law and calibration.

None of the six statements supplies a physical clock, energy, frequency, exchange sector or source selection. Their standing in the observation ledger is recorded row by row with the premises each consumes.

7 The Closure Equations

Closure asks observers inside one history to read back the specification of that history. Two dimensionless equations make this concrete at the observer layer. A local equation asks one screen cell to agree with an electromagnetic reading of the same cell; a global equation asks a trial universe to reproduce its own logarithmic correctable public-record capacity. Each equation tests an independently produced readback map. If both hold with their declared selection rules, the branch admits at most one coordinate pair (P, N). This does not prove that exactly one physical universe exists, and neither equation is a physical identity before its same-quantity bridge is constructed. This paper is the canonical place where both equations are defined, so the other papers of the tower cite one derivation.

7.1 Local closure for the pixel ratio

The construction has six steps.

1. **Outer pixel ratio.** The outer-side local screen datum is the dimensionless area ratio

$$P := \frac{a_{\mathrm{cell}}}{\ell_\star^2}, \qquad \gamma_\star = \frac{\ell_\star \nu_{\mathrm{Cs}}}{c}, \qquad B_\star = \frac{3\pi}{\ell_\star^2}.$$

   Here $\ell_\star^2 = 3\pi/B_\star$ is supplied by the selected scale certificate. After that scale is emitted, it is displayed in SI language as the Planck area.

2. **Exact equilibrium benchmark.** Writing

$$x(C) := \frac{S_{\mathrm{gen}}(C)}{S_{\mathrm{bulk}}(C)} = 1 + \frac{\langle L_C \rangle}{S_{\mathrm{bulk}}(C)},$$

   the self-similar balance condition $S_{\mathrm{gen}}/S_{\mathrm{bulk}} = S_{\mathrm{bulk}}/\langle L_C \rangle$ gives $x^2 - x - 1 = 0$ and hence $x = \varphi = (1+\sqrt{5})/2$. The realized branch must sit near, and not exactly at, this point, because exact equilibrium is too symmetric to support durable records, structure and dynamics.

3. **Outer detuning variable.** On the declared closure surface the outer-side detuning is

$$\alpha_{\mathrm{ext}}(P) := \frac{P - \varphi}{\sqrt{\pi}}, \qquad \text{equivalently} \qquad P = \varphi + \alpha_{\mathrm{ext}}(P)\sqrt{\pi}.$$

4. **Declared quantitative anchor on the trial domain.** Let $I \subset \mathbb{R}_+$ be the declared branch interval and let $F_{\mathrm{EW}} : I \to \mathcal{D}_{\mathrm{EW}}$ denote the declared electroweak source map. For any trial $P \in I$, the forward electroweak branch emits

$$P \longmapsto \alpha_U(P) \longmapsto (t_U(P), t_{\mathrm{tr}}(P)) \longmapsto (t_2(P), t_3(P), v(P)) \longmapsto A_Z(P) = \alpha_{\mathrm{em}}^{-1}(m_Z^2; P).$$

5. **Ward-projected Thomson transport.** Let T_Q be the required Ward-projected $\mathrm{U}(1)_Q$ transport operator from the m_Z-anchor to the Thomson limit, and

$$A_T(P) := T_Q\big(A_Z(P), F_{\mathrm{EW}}(P)\big) = \alpha_{\mathrm{Th}}^{-1}(P), \qquad \alpha_{\mathrm{in}}(P) := \frac{1}{A_T(P)}.$$

   The complete source-only operator T_Q is not emitted.

6. **Conditional closure.** If the typed same-quantity bridge, the target-independent map-selection premise and same-scheme endpoint transport hold, the physical pixel ratio is the fixed point for which the outer detuning equals the inner observation coupling:

$$\boxed{P = \varphi + \frac{\sqrt{\pi}}{A_T(P)}}, \qquad H(P) := P - \varphi - \frac{\sqrt{\pi}}{A_T(P)} = 0.$$

On any declared branch interval where $G(P) = \varphi + \sqrt{\pi}/A_T(P)$ is a self-map and a contraction, its fixed point is unique. Outward-rounded interval certificates verify these hypotheses, enclose one root for each declared map and exclude a second root across the full analytic domain: a domain-global certificate bounds $\sup|G'| < 1$ on every piece of a 256-piece subdivision of the declared numerical domain ($\alpha^{-1} \in [100, 200]$, both readout maps, empty exceptional set). No measured value is inserted into either solve. Observer-facing data may localize the interval; the measured endpoint cannot replace the solve. Conditional on the bridge, one screen cell carries two coupled roles: its displacement above φ sets the outer detuning, and the returned electromagnetic coupling is another reading of the same cell.

Four fine-structure coordinates. The certified source/root witness of the declared map is $\alpha_{\mathrm{root}}^{-1} = 136.994835177413\ldots$, with enclosure width 7.2×10^{-24}; no derivation identifies it with a physical fine-structure endpoint. The certified self-consistent gauge-width fixed point of the declared map is $\alpha^{-1} = 137.035660136946577\ldots$. The mixed-provenance no-hadron diagnostic $137.0359595136\ldots$ combines the inner value at the certified forward pixel $P_{\mathrm{fwd}} = 1.630972095858897\ldots$ with α_U at the comparison pixel, is a fixed point of no single declared map, and is excluded from the physical output set. An executed empirical closure, using external $e^+e^- \to$ hadrons spectral data, records

$$\Delta\alpha_{\mathrm{had}}^{(5)}(M_Z) = 0.027609 \pm 0.000112, \qquad \alpha_{\mathrm{emp}}^{-1}(0) = 136.3827548175$$

on $[136.3670480603, 136.3984651934]$; the measured endpoint 137.035999177(21) lies outside that interval. The same-scheme anchor correction is $[0.6198609041, 0.6505569679]$ inverse-alpha units, and both the standard on-shell reference deficit 0.631 and the exact closure value 0.6379 lie inside it. The comparison pixel $P_C \simeq 1.6309682094$ is inferred from the measured endpoint solely for comparison. Measured against it, the loop residual is 3.0×10^{-4} relative on the source chain and 2.5×10^{-6} relative (about 1.6×10^4 measurement sigma) for the certified gauge-width fixed point. The Ward-projected hadronic transport gives $S_{\mathrm{hadronic}} \in [0.5578, 1.0543]$ for the two electromagnetic endpoint residuals; the interval contains the zero-electroweak diagnostic 0.8954, spans about 1.16×10^8 declared tolerance widths and is evaluated at a pixel different from the comparison pixel. These residuals have no valid target-blind score: the exposed target conflates total and residual coordinates and no decision threshold is active. A physical source-only endpoint requires a target-independent map choice, the typed same-quantity bridge, and a source-derived hadronic spectral backend with source QCD parameters, a quotient ensemble, Ward-normalized currents, the two-current spectral

marginal, the same-scheme remainder and no-target-leak receipts. The two-current measure covers the running-α marginal; light-by-light and rare-decay rows require higher-point and transition spectral sectors from the same source law.

Scale reading of the closure. If the two declared-map outputs acquire their physical identifications, $P_\star$ and N_{CRC} fix $\Lambda_\star \ell_\star^2 = 3\pi/N_{\mathrm{CRC}}$ and $\Lambda_\star a_{\mathrm{cell}} = 3\pi P_\star/N_{\mathrm{CRC}}$, and the selected scale certificate supplies the SI scale product $B_\star = \Lambda_\star N_{\mathrm{CRC}} = 3\pi/\ell_\star^2$. Under the same identifications the pixel together with the Λ-located working capacity $N_\Lambda = 3.31 \times 10^{122}$ gives the equal-area chart count

$$K_{\mathrm{cell}} = \frac{A_{\mathrm{screen}}}{a_{\mathrm{cell}}} = \frac{4N_\Lambda}{P_\star} \simeq 8.12 \times 10^{122},$$

and 8.66×10^{122} at the conditional bridge capacity. This is a measured-side comparison display that enters neither closure. The quantity $P_\star/4 \simeq 0.408$ nats, about 0.588 bits, is not $\log \dim \mathcal{H}_{\mathrm{cell}}$ of an autonomous tensor factor: OPH capacity lives on shared cuts, edge centers and constrained overlap data, so no fixed primitive observer algebra follows from $P_\star$ and N_{CRC} alone. The gravity normalization comes from $\ell_\star^2$, and P cancels in the Newton area-law readout.

7.2 The unified-coupling proof record and the clock hierarchy

The electroweak hierarchy used by the scale branch depends on a source readout of the unified coupling $\alpha_U(P_\star)$. For a trial pixel P define $M_U(P) = E_\star e^{-2\pi} P^{1/6}$ and $E_{\mathrm{cell}}(P) = E_\star/\sqrt{P}$. On the color branch $N_c = 3$ and $\beta_{\mathrm{EW}} := N_c + 1 = 4$, and the transmutation scale for a candidate coupling a is $v(P,a) = E_{\mathrm{cell}}(P)\exp[-2\pi/(\beta_{\mathrm{EW}} a)]$. The exterior representation theorem fixes the weak-doublet multiplicity $3 + 1 = 4$; its use as the transmutation coefficient requires the common screen/electroweak load-carrier identification and is declared. The SU(2) one-loop coefficient of the one-Higgs branch is 19/6 and the declared unification packet's coefficient is 1; neither equals 4. The source running family is

$$\alpha_i^{-1}(\mu; P, a) = a^{-1} + \frac{b_i}{2\pi}\log\!\left(\frac{M_U(P)}{\mu}\right), \qquad i = 1,2,3, \qquad \alpha_Y = \tfrac{3}{5}\alpha_1,$$

and the source Z-scale is the fixed point $\mu_Z = \frac{1}{2}v\sqrt{4\pi\alpha_2(\mu_Z) + 4\pi\alpha_Y(\mu_Z)}$. With $Z_G(t) = \sum_R d_R e^{-tC_2(R)}$, $\bar{\ell}_G(t) = Z_G(t)^{-1}\sum_R d_R e^{-tC_2(R)}\log d_R$ and $t_{2,3} = 4\pi^2\alpha_{2,3}(\mu_Z)$, the unified-coupling residual is

$$\Phi_U(P,a) := \bar{\ell}_{\mathrm{SU(2)}}(t_2) + \bar{\ell}_{\mathrm{SU(3)}}(t_3) - \frac{P}{4}, \qquad \Phi_U(P, \alpha_U(P)) = 0.$$

Definition 7.1 (No-hidden-free-variable admissibility)**.** *The unified-coupling proof record is admissible only when the representation cutoffs N_2, N_3, the beta packet (b_1, b_2, b_3), the matching prescription, the threshold packet and the renormalization convention are frozen before comparison with measured electroweak or gravitational data, and none is selected by minimizing an error against M_W, M_Z, $\alpha_s(M_Z)$, $\sin^2\theta_W$, v_{exp} or G_{exp}. Tuning any of them after such a comparison makes the hierarchy row a calibration.*

Theorem 7.2 (No-measured-input unified-coupling proof record)**.** *Let*

$$\mathcal{R}_U = (P_\star, N_2, N_3, I_U, \Phi_U, K_U, a_U^{(0)}, r_U, m_U, \mathrm{DAG}_U)$$

be an admissible record, with $I_U \subset \mathbb{R}_+$ the declared interval for α_U, K_U an interval-Newton or Krawczyk operator, $a_U^{(0)}$ the displayed candidate, r_U a residual bound, m_U a lower derivative bound

and DAG_U *the dependency graph. Assume* $K_U(I_U) \subset \mathrm{int}(I_U)$, $|\Phi_U(P_\star, a_U^{(0)})| \le r_U$, $|\partial_a \Phi_U(P_\star, a)| \ge m_U > 0$ *on* I_U, *and that* DAG_U *has no directed path from measured electroweak masses, measured low-energy gauge couplings, measured* G, *Planck area or mass, measured* Λ, *or any equivalent gravity- or electroweak-calibrated scale. Then there is a unique* $\alpha_U(P_\star) \in I_U$ *with* $\Phi_U(P_\star, \alpha_U(P_\star)) = 0$ *and* $|\alpha_U(P_\star) - a_U^{(0)}| \le r_U/m_U$, *and the hierarchy*

$$\frac{v}{E_\star} = P_\star^{-1/2} \exp\left[-\frac{2\pi}{(N_c+1)\alpha_U(P_\star)}\right]$$

is no-measured-input on this record.

Proof. The residual is built from $P_\star$, the declared cutoffs and the source transmutation, running and Z-scale laws, and the dependency condition excludes measured electroweak data and gravity-calibrated scales. The Krawczyk inclusion gives existence and uniqueness of a zero in I_U, and the mean-value theorem with the derivative bound gives the displayed distance. Substitution into the transmutation law gives the hierarchy. □

The theorem fixes a dimensionless ratio; it supplies no independently physical $E_\star$, mass in GeV or complex propagator pole. The public comparison-branch certificate at P_C gives $\alpha_U(P_C) = 0.041124336195630495$ and $v/E_\star = 2.0199803239725553 \times 10^{-17}$, with the Krawczyk image of $[0.041123336195630494,\ 0.041125336195630496]$ strictly inside that interval. The source-side diagnostic branch excludes the public Thomson endpoint upstream; it is a witness, not the strict interval root of the Ward-projected self-map. The same exponent has the local/global capacity form

$$\frac{v}{E_{\mathrm{cell}}} = \left(\frac{N_{\mathrm{CRC}}^{\mathrm{EW}}}{\pi}\right)^{-P_\star/12}, \qquad N_{\mathrm{CRC}}^{\mathrm{EW}} = \pi \exp\left[\frac{6\pi}{P_\star \alpha_U(P_\star)}\right],$$

about 3.53235×10^{122} at P_C and $N_0 = 3.5321315434 \times 10^{122}$ interval-certified at P_{fwd}.

Lemma 7.3 (Exponential sensitivity of the hierarchy row)**.** *On the transmutation branch with* $\beta_{\mathrm{EW}} = 4$, *and on a clock row whose dominant scale dependence enters through* $m_e^2 \propto v^2$ *in the leading hyperfine scaling* $\varepsilon_{\mathrm{Cs}} \sim \mathcal{C}_{\mathrm{Cs}} \alpha_\star^4 m_e^2/(m_p E_\star)$, *the induced gravity readout with* $G \propto \varepsilon_{\mathrm{Cs}}^2$ *satisfies* $\partial \ln G/\partial \alpha_U = 2\pi/\alpha_U^2$, *about* 3.7×10^3 *at* $\alpha_U \simeq 0.0411$. *A certified relative gravity precision* η_G *therefore requires* $|\delta\alpha_U| \lesssim \eta_G/(3.7 \times 10^3)$.

Proof. $\partial \ln(v/E_\star)/\partial \alpha_U = \pi/(2\alpha_U^2)$; the clock gap carries v^2 at fixed Yukawa, fine-structure and strong-sector ratios, and the gravity row is quadratic in the clock gap. □

No-G clock hierarchy. The gravity row uses the dimensionless cesium gap $\varepsilon_{\mathrm{Cs}} := \hbar\omega_{\mathrm{Cs}}/E_\star$, with $E_\star = \hbar c/\ell_\star$ and $\omega_{\mathrm{Cs}} = 2\pi\nu_{\mathrm{Cs}}$. A source-only cesium packet $\mathcal{D}_{\mathrm{Cs}}^{\mathrm{src}} = (\alpha_\star, m_e/E_\star, \Lambda_{\mathrm{QCD}}/E_\star, \{y_q\}, \mathcal{N}_{133}, \mathcal{B}_{\mathrm{atom}}^{\mathrm{Cs}})$ collects dimensionless couplings, the ^{133}Cs nuclear data $\mathcal{N}_{133}$ and the atomic many-body, relativistic, QED, recoil and finite-nucleus correction package; the dimensionless OPH cesium Hamiltonian $\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}}$ is the sum of the Dirac–Coulomb, Breit, QED, recoil, nuclear and many-body terms in units of $E_\star$.

Theorem 7.4 (No-G OPH clock-hierarchy theorem)**.** *Assume the local pixel proof selects* $P_\star$; *the record* $\mathcal{R}_U$ *of Theorem 7.2 emits* $\alpha_U(P_\star)$ *and* $v/E_\star$; *the electromagnetic branch* $\mathcal{R}_\alpha$ *emits* $\alpha_\star$; *the charged-lepton branch emits* $m_e/E_\star = (y_e/\sqrt{2})\, v/E_\star$; *the QCD/nuclear branch emits* $\mathcal{N}_{133}$; *and the atomic branch supplies disjoint isolated spectral enclosures* $J_{F=3}, J_{F=4}$ *for* $\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}}$, *so that* $\varepsilon_{\mathrm{Cs}} = \lambda_{F=4} - \lambda_{F=3}$ *is certified. If the combined dependency graph has no directed path from* G_{exp}, ℓ_P^2,

m_P, t_P, Λ_{exp}, $3\pi c^3/(\hbar G)$ or an algebraically equivalent gravity-calibrated scale, then $\varepsilon_{\mathrm{Cs}}$ is a no-G readout, $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c = \varepsilon_{\mathrm{Cs}}/(2\pi)$ is non-circular, and

$$G_{\mathrm{SI}} = \frac{c^5}{4\pi^2 \hbar \nu_{\mathrm{Cs}}^2}\, \varepsilon_{\mathrm{Cs}}^2$$

has no measured-G ancestor.

Proof. Every coefficient of $\widehat{H}_{\mathrm{Cs}}^{\mathrm{OPH}}$ is emitted by the listed records and so has no measured-G ancestor; the isolated enclosures make the gap well defined and it inherits the dependency condition. Since $\varepsilon_{\mathrm{Cs}} = 2\pi\nu_{\mathrm{Cs}}\ell_\star/c$, substitution into $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$ gives the display. □

Printing $\varepsilon_{\mathrm{Cs}} = 3.1139305134\ldots \times 10^{-33}$ does not discharge the theorem. The decimal is source-predictive only when emitted by the five records; computed as $2\pi\nu_{\mathrm{Cs}}\sqrt{\hbar G_{\mathrm{exp}}/c^5}$ it is a gravity-side calibration checksum. The source-only completion requires the hadronic same-scheme endpoint payload in $\mathcal{R}_\alpha$, the charged-lepton absolute-scale premise and the source-side cesium QCD/nuclear packet, for example from a Ward-projected nonperturbative backend with manifest provenance and systematics. The printed digits of G_{SI} may not exceed the precision these records certify.

7.3 Global closure for total capacity

The global counterpart of the local P-closure begins with a supplied positive integer capacity-carrier dimension

$$D = \dim \mathcal{H}_{\mathrm{cap},r,D}, \qquad N = \log D.$$

The universe-level readback equation is

$$\boxed{N = \log M_0(\mathfrak{U}_N)}.$$

Here the argument fixes the type. In every occurrence, M_0 is a multiplicative code size. Once the complete terminal fiber scalarizes, the universe-level notation means

$$M_0(\mathfrak{U}_N) := \widehat{F}_{r,0}(e^N).$$

Thus N is the logarithmic capacity read back from the trial universe, while $M_0(\mathfrak{U}_N)$ is the number of correctable public records. The equation is the logarithmic form of stable direct capacity closure, not a new producer or an appeal to a measured target. The carrier type is frozen in the branch contract. Boundary, total, edge-center, and other Hilbert spaces are not interchangeable. Let $\widetilde{\Omega}_{r,D}$ be the terminal physical quotient fiber reached from the source-derived trial universe $\mathfrak{U}_{r,D}$. Its members are repair-normal and pass the declared local, obstruction, and provenance gates. Membership contains no capacity-equals-D predicate.

For $q \in \widetilde{\Omega}_{r,D}$, observer O has a finite commutative record algebra $\mathcal{R}_O(q)$ with atom set $X_O(q) = \mathrm{At}\,\mathcal{R}_O(q)$. Each shared interface e has an atom set $X_e(q)$ and a source-derived record-atom readout

$$r_{Oe} : X_O(q) \longrightarrow X_e(q).$$

This atom-level construction is used because the generic visible maps on full observer algebras need not be unital *-homomorphisms in the direction required by an algebra equalizer. The compatible public global sections are

$$X_{\mathrm{pub}}(q) = \{(x_O)_O : r_{Oe}(x_O) = r_{O'e}(x_{O'}) \text{ on every shared interface}\}, \qquad \mathcal{R}_{\mathrm{pub}}(q) = C(X_{\mathrm{pub}}(q)).$$

Theorem 7.5 (Public global sections)**.** *For every finite observer record diagram, $X_{\mathrm{pub}}(q)$ is finite and $C(X_{\mathrm{pub}}(q))$ is a finite commutative unital C^*-algebra. Every isomorphism of observer record diagrams induces a bijection of public global sections and a *-isomorphism of their function algebras.*

Proof. The public sections form the subset of the finite product $\prod_O X_O(q)$ satisfying finitely many equalities. A diagram isomorphism transports the atom sets and interface maps, hence compatible tuples bijectively; pullback transports their function algebras. □

Only endogenously writable records count. Define

$$X_{\mathrm{reach}}(q) = \{x \in X_{\mathrm{pub}}(q) : x \text{ terminates an admissible endogenous semantic history}\}.$$

The history may use declared source data, accepted repair, semantic continuation, and quotient-visible writes. It may not use a target record, supplied-capacity metadata, measured Λ, or the electroweak bridge. The publicness policy is a frozen nonempty family

$$\mathfrak{P}(q) \subseteq 2^{\mathcal{O}(q)} \setminus \{\varnothing\}$$

of authorized observer subsets. Collective, universal-local, and quorum publicness are distinct theorem branches.

For every $x \in X_{\mathrm{reach}}(q)$, authorized set $A \in \mathfrak{P}(q)$, and allowed same-interface semantic continuation κ, the source construction must emit a joint stochastic kernel

$$K_{A,\kappa}(y \mid x), \qquad y \in Y_{A,\kappa},$$

whose observer marginals agree with the local checkpoint packets. The joint kernel is an independent receipt. It cannot be manufactured from local marginals.

Theorem 7.6 (Local marginals do not determine public capacity)**.** *There are two binary-input joint checkpoint channels with identical observer-by-observer marginals and different public capacities.*

Proof. Let U be uniform. The channel $(Y_1, Y_2) = (U, U \oplus x)$ has uniform input-independent marginals, while the joint parity recovers x. Independent uniform outputs have the same marginals and carry no information about x. Thus the joint capacities are two and one, respectively. □

For a channel K, write $S_K(x) = \{y : K(y \mid x) > 0\}$. A code $C \subseteq X_{\mathrm{reach}}(q)$ is zero-error correctable for K when a decoder d_K satisfies

$$K(d_K^{-1}(x) \mid x) = 1 \qquad (x \in C).$$

Theorem 7.7 (Support criterion and compound zero-error capacity)**.** *A code is zero-error correctable for K exactly when the supports $S_K(x)$, $x \in C$, are pairwise disjoint. Let $\mathfrak{K}(q)$ contain every authorized checkpoint channel and every declared finite composition. Define the compound confusability graph G_q on $X_{\mathrm{reach}}(q)$ by joining $x \neq x'$ whenever their output supports overlap for at least one $K \in \mathfrak{K}(q)$. Then the exact public-record capacity is*

$$\boxed{M_0(q) = \alpha(G_q)},$$

where α is the graph independence number.

Proof. A successful decoder cannot assign one output to two inputs, which proves the support criterion in one direction. Pairwise disjoint supports admit the decoder that returns the unique supporting input. A code valid for every declared channel is therefore exactly an independent set in the union of the channel confusability graphs. □

For exact indefinite continuation, replace each kernel by its support relation $R_K = \{(x, y) : K(y \mid x) > 0\}$. Boolean relation composition generates at most $2^{|X_{\mathrm{reach}}|^2}$ relations, so the support semigroup and its compound confusability graph close after finitely many steps.

Theorem 7.8 (Reversible fast branch)**.** *If every authorized continuation is a known injective deterministic map on $X_{\mathrm{reach}}(q)$, then*

$$M_0(q) = |X_{\mathrm{reach}}(q)|.$$

Proof. Injectivity keeps every pair of inputs distinguishable for every continuation, so G_q has no edges. □

Checkpoint invariance is not the capacity definition. A cyclic permutation of m labels has a one-dimensional fixed function algebra, while its inverse decodes all m labels. Thus a fixed-projector count can undercount the correctable capacity arbitrarily.

For a frozen finite channel family or continuation horizon and tolerance ε, let $M_\varepsilon(q)$ be the largest code for which every declared channel has a decoder with worst-input error at most ε.

Theorem 7.9 (Approximate-capacity stability and zero-error discontinuity)**.** *If corresponding channel rows obey*

$$\sup_{K,x} d_{\mathrm{TV}}(K(\cdot \mid x), \widehat{K}(\cdot \mid x)) \le \delta,$$

then

$$M_{\varepsilon+\delta}(\widehat{\mathfrak{K}}) \ge M_\varepsilon(\mathfrak{K}).$$

Zero-error capacity is discontinuous at every identity channel on more than one symbol: adding arbitrarily small full-support noise collapses M_0 to one.

Proof. Total-variation control changes the probability of each correct-decoding event by at most δ. For the second statement, $(1-\delta)I + \delta U$, with U full support, makes every pair of inputs confusable for every $\delta > 0$. □

The capacity carrier is an exact finite Hilbert space $\mathcal{H}_{\mathrm{cap},r,D}$ of dimension D. Each reachable public record class has a nonzero projection P_x on this carrier, and distinct records have orthogonal projections.

Theorem 7.10 (Carrier bound and saturation rigidity)**.** *Every correctable public code C satisfies*

$$|C| \le D, \qquad \log M_\varepsilon(q) \le \log D = N.$$

If $M_0(q) = D$, the code projections are rank one and sum to the identity.

Proof. The ranks of nonzero orthogonal projections add and their sum has rank at most D. Equality with D codewords forces every rank to be one and the sum to have full rank. □

The primary finite readback remains set-valued:

$$\boxed{\mathfrak{F}_{r,\varepsilon}(D) = \{M_\varepsilon(q) : q \in \widetilde{\Omega}_{r,D}\}.}$$

A scalar $\widehat{F}_{r,\varepsilon}(D)$ exists exactly when the nonempty terminal fiber has one common defined value. The evaluator must distinguish an empty readback, an ambiguous readback, and a singleton scalar readback. Existential closure means $D \in \mathfrak{F}_{r,0}(D)$; selector-free stable closure means

$$\boxed{\mathfrak{F}_{r,0}(D) = \{D\}.}$$

Equivalently, the unclosed kernel

$$K_r(D, m) = |\{q \in \widetilde{\Omega}_{r,D} : M_0(q) = m\}|$$

has support only at $m = D$. One saturated branch is insufficient when the terminal fiber is ambiguous.

The full capacity packet consists of observer lineage, record atoms, interface maps, reachable public sections, publicness policy, global kernels, and carrier projections. An isomorphism preserving that packet transports every exact confusability edge, approximate decoder error, and projection rank. Hence M_ε is implementation invariant under packet isomorphism.

Once scalarized, the active map $f(D) = \widehat{F}_{r,0}(D)$ is deflationary. It cannot attract a positive fixed point from below. If f is total on a declared finite admissible chain, monotone, and deflationary, iteration from its top element stabilizes at the greatest fixed point. This is a conditional order theorem, not a uniqueness or cutoff-independence theorem.

Theorem 7.11 (Bounded capacity-continuation counterfamily)**.** *At rung k, each of the twenty-four visible base records has k copies, so $D_{\mathrm{pub}}(k) = 24k$. In the spectator control the raw carrier is $D_{\mathrm{raw}}(k, s) = 24ks$. A target-clean counterfamily shares base agreement, positivity, the carrier bound, and executable finite incidence, publicness, reversible-action, semantic-projection, composition, extension, and oriented-record fiber-product controls. Its reversible identity completion has*

$$M_{\mathrm{id}}(k) = 24k.$$

Copy collapse and a two-class completion have

$$M_{\mathrm{erase}}(k) = 24, \qquad M_2(k) = 24\min(k, 2).$$

A hidden spectator of multiplicity s leaves the public capacity at $24k$. The corresponding slack-zero sets are every positive k, $\{1\}$, $\{1, 2\}$, and every positive k only for $s = 1$. The shared bounded structure does not entail a unique zero.

Proof. Each collapse channel yields a disjoint union of complete confusability components. One representative per component is an independent set, and completeness of each component gives the matching upper bound. The collapse maps are idempotent, commute with the forty reversible slot actions, and preserve the old induced graph under rung extension. Lean proves the all-rung arithmetic consequence of typed base agreement, positivity, and the carrier bound, including the disagreement between the displayed zero sets. The executable certificate checks the finite incidence, action, projection, composition, extension, and sewing controls. A separate simulator implementation reconstructs held-out finite sample graphs without importing the producer. □

This bounded counterfamily is not a complete source-class nonidentifiability theorem. The complete A1–A3 terminal fibers, observer and interface atom maps, joint checkpoint kernels, meaning maps, feasible sets, and regulator controls are absent across capacity. The result does not reject strange-loop self-identification. Equality follows after the simulating and simulated readings are proved to be readings of the same invariant quantity.

Capacity extension and regulator refinement are distinct. For $D \le D'$, an injection of reachable records that reflects confusability,

$$e_{DD'}(x) \sim_{G_{D'}} e_{DD'}(x') \Longrightarrow x \sim_{G_D} x',$$

maps every coarse independent set to a fine independent set and gives $M_0(D) \le M_0(D')$. At fixed D, refinement injections with the same property make $M_{0,r}(D)$ a nondecreasing integer bounded by D, hence eventually constant along each cofinal sequence.

For regions A, B sewn along public seam S, exact public records form the fiber product

$$X_{A\cup B} = X_A \times_{X_S} X_B, \qquad |X_{A\cup B}| = \sum_{z\in X_S} |r_A^{-1}(z)|\,|r_B^{-1}(z)|.$$

This gives an exact CSP, tensor-network, or model-counting route on the reversible branch. A cosmological conclusion additionally requires an exact transfer recurrence, exact fiber-product recursion, or a seam theorem with a certified subleading term.

Define the finite-size slack

$$\boxed{s_r(D) = \log D - \log M_{0,r}(D).}$$

If $\limsup_{D\to\infty} \log M_0(D)/\log D < 1$, closure is excluded at all sufficiently large dimensions. Unit asymptotic density is insufficient: $M(D) = D$ and $M(D) = D - 1$ have the same limiting density but opposite fixed-point behavior. Full physical N-closure requires a source-derived finite-size law proving

$$s(D_\star) = 0, \qquad s(D) > 0 \quad (D \neq D_\star)$$

on the declared physical domain, or an equally explicit physical selector. Such a law is a separately named physical premise unless it is derived from a stronger source architecture. The common $D = 24$ screen packet is not a universe-level selection.

Two independent physical identifications follow only after stable direct closure. The first hypothesis identifies the same capacity carrier with the de Sitter horizon record:

$$N_\star = \log D_\star = \frac{A_{\rm dS}}{4\ell_\star^2}, \qquad A_{\rm dS} = \frac{12\pi}{\Lambda} \quad\Longrightarrow\quad \boxed{\Lambda\ell_\star^2 = \frac{3\pi}{N_\star}}.$$

The second hypothesis identifies the screen load with the electroweak load by a positive, unital, refinement-natural map

$$\Xi_r : \mathcal{L}_{{\rm screen},r} \longrightarrow \mathcal{L}_{{\rm EW},r}.$$

Given the independent screen-sieve and electroweak source laws,

$$\Gamma_{\rm scr} = \frac{P}{12}\log\frac{N_\star}{\pi}, \qquad \log\frac{E_{\rm cell}}{v} = \frac{\pi}{2\alpha_U(P)}, \qquad \Xi_r(\Gamma_{\rm scr}) = \log\frac{E_{\rm cell}}{v},$$

it gives the conditional Higgs/electroweak residual and coordinate

$$\boxed{R_{\rm EW} = \alpha_U(P)\log\frac{N_\star}{\pi} - \frac{6\pi}{P} = 0, \qquad N_{\rm bridge} = \pi\exp\!\left[\frac{6\pi}{P\alpha_U(P)}\right].}$$

Neither physical identification constructs the public-record map. The independently frozen operational-resolution residual

$$R_\rho = \log M_0 - \frac{\pi}{\rho_{\rm op}^2}$$

is likewise a downstream test and never a producer definition.

The finite global-section, correctable-code, compound-capacity, approximate-stability, carrier-bound, scalarization, order, extension, refinement, and sewing implications are theorems. The

fixed-$D = 24$ packet realizes its complete terminal fiber and the associated atom, publicness, joint-kernel, carrier, scalarization, extension, and refinement controls. Physical closure requires a complete capacity-indexed lift of those objects, including the A2 meaning maps, A3 feasible sets, and regulator controls; an exact finite-size slack law with one physical zero; the universe-level carrier attachment; the horizon–record identification; and the common screen/electroweak load-carrier identification. No QCD or hadronic backend is a dependency of this program.

Capacity comparison coordinates. The Gibbons–Hawking entropy and the Planck-2018 late-time de Sitter benchmark give a concrete normalization [61, 63]: with $R_{\mathrm{dS}} \simeq 1.66 \times 10^{26}\,\mathrm{m}$ and $\ell_P \simeq 1.616 \times 10^{-35}\,\mathrm{m}$, the bare horizon ratio is $N_{\mathrm{patch}} = (R_{\mathrm{dS}}/\ell_P)^2 \simeq 1.05 \times 10^{122}$, the entropy capacity is $N_{\mathrm{scr}} = S_{\mathrm{dS}} = \pi N_{\mathrm{patch}} \simeq 3.31 \times 10^{122}$, and $\Lambda\ell_P^2 \simeq 2.85 \times 10^{-122}$. The weighted Planck base-ΛCDM read-off is $3.3129\ldots \times 10^{122}$; the certified electroweak bridge value $3.5321315434\ldots \times 10^{122}$ at P_{fwd} solves the declared bridge equation, not the unconstructed universe-level public-record producer, and is about 6.6 percent above it. The separately premised finite-presence and Poisson carriers give $3.2920978773\ldots \times 10^{122}$ and $3.3000722254\ldots \times 10^{122}$, retrospective residuals of -0.63 and -0.39 percent. Exact neutral and multiplicative global-capacity completions share the same local survival datum and satisfy positive disconnected-cut composition and finite cut-count regrouping, yet disagree after one cut, so the finite source selects neither expression as a global action. Four manifest-authenticated official DESI DR2 BAO+CMB base-ΛCDM chains, transformed sample by sample, give $\Lambda\ell_P^2 = (2.96770 \pm 0.03978) \times 10^{-122}$ with weighted 2.5–97.5 percent interval $[2.88997, 3.04530] \times 10^{-122}$ [31, 29]; this retains the sampled H_0–Ω_Λ covariance and is a retrospective, target-informed posterior display, not an OPH likelihood or predictive pull. These comparisons are retrospective, and none establishes the common-carrier or horizon identification.

The declared candidate family. The declared finite candidate family comprises the seventeen-row menu for the executed capacity families, the two declared P maps, one direct fixed-cutoff control, the common-load baseline, reserve maps, RC-LOAD, the construction aggregate and five hierarchy packets. Fifteen of the thirty entries record exact fixed-point progress: eight carry certified fixed points, one is conditional on its three coupling premises, and six are sublattice or aggregate rows. None supplies target-independent selection, a constructed same-quantity identification and a complete source return map together, and because the input catalogue, nuisance quotient, coordinate and scheme catalogue and candidate grammar are not frozen, no entry qualifies as a source-only candidate. The four multiplicative capacity-P maps fix whichever positive normalization seed is supplied and do not select it; the two additive rows have no positive fixed point. The conditional RC-LOAD equation has positive discriminant, maps its declared macroscopic domain into itself, is contractive there, has one macroscopic fixed point and excludes its microscopic root; its pricing law, same-quantity identity and target-independent selection are premises. The family does not exhaust other source laws.

7.4 Comparison coordinates that are not outputs

Several numerical rows organized by P are comparison coordinates and are excluded from the physical particle outputs; the particle paper gives their tables and verification [21]. The declared, not source-selected, zero-selector map gives the target-free running-chart coordinates $(80.330, 91.119)$ GeV, not physical pole masses. The selected carrier, conditional value law and reference-fitted inverse adapter give $(80.3862916924, 91.1829044467)$, $(80.3770000154, 91.1879780779)$ and $(80.3625, 91.1879)$ GeV; the target-conditioned closure chart has central coordinates $(80.373315, 91.193124)$ GeV. For the PDG-2026 reference pair the exact

convention map gives energy-pole masses $(80.3411410, 91.1623040)$ GeV, while the legacy $\sqrt{\mathrm{Re}\, s}$ coordinates are $(80.3340218, 91.1537725)$ GeV. A separate strict one-loop fixture excludes scalar zeros in declared principal-sheet boxes and isolates one simple scalar zero for each of W and Z in channel-specific lower-half pole boxes; no certified continuation joins it to the OPH chart. The fixed 96-entry one-loop electroweak menu excludes its own entries and says nothing about prescriptions outside it.

The Higgs/top pair is read on the double-criticality branch, whose comparison-exposed boundary-scale candidate $E_\star e^{-\pi} P^{-1/6}$ gives $(m_H, m_t) = (125.77, 172.63)$ GeV at two loops, with $m_H = 125.72$ GeV on the fit-free curve at the measured top; the declared calibration surface (125.1995304097, 172.3523553288) GeV is a target-anchored fit. The quark reciprocal-ray candidate fails its common-scale audit with a 21.556% held-out minimax error. The register-Clebsch candidate supplies the light-family coefficient-ratio menu $\{1/9, 1/3, 3, 9\}$; both FLAG 2024 rows reject every assignment under the retrospective conservative experimental-only rule, which excludes only the declared common-transport assignment family. Twelve fixed flavor candidates are likewise excluded without exhausting the broader selector family.

On the stated local extension surface the gravity row reads $G = 6.674299995910528 \times 10^{-11}\ \mathrm{m^3\,kg^{-1}\,s^{-2}}$ through $\gamma_\star = \ell_\star \nu_{\mathrm{Cs}}/c$, $B_\star = 3\pi/\ell_\star^2$, $a_{\mathrm{cell}} = P\ell_\star^2$, $\bar{\ell}_{\mathrm{shared}} = P/4$ and $G_{\mathrm{SI}} = c^3\ell_\star^2/\hbar$; the pixel fixes the shared cell and edge identity and cancels in the Newton area law. The invariant speed is structural from the Lorentz branch, and 299792458 m/s is its SI convention. Familiar units are display conventions: $L_{\mathrm{loc}} = \sqrt{a_{\mathrm{cell}}}\,\widehat{L}(P)$, $t_{\mathrm{loc}} = (\sqrt{a_{\mathrm{cell}}}/c)\,\widehat{T}(P)$, $E_{\mathrm{loc}} = (\hbar c/\sqrt{a_{\mathrm{cell}}})\,\widehat{E}(P)$ and $\Theta_{\mathrm{loc}} = (\hbar c/(k_B\sqrt{a_{\mathrm{cell}}}))\,\widehat{\Theta}(P)$ with dimensionless hatted functions.

8 Comparison with other unification approaches

The case for OPH is the explanatory reach of its observer architecture. A useful comparison records the starting objects, the physical structures forced by them, and the experiments that could distinguish the resulting models. Mathematical consistency, recovery of known physics and a successful prospective prediction answer different questions. The constructions below illustrate these distinctions without assigning a numerical probability of physical truth to a research program.

String theory and holographic duality. String theory has concrete results that a unification proposal must take seriously. Strominger and Vafa count microscopic states reproducing the area entropy of a class of five-dimensional extremal black holes [100]. Maldacena's construction relates specified superconformal theories to anti-de Sitter gravity and proposes their full duality [101]. Thus a fixed bulk spacetime is not a universal primitive of string-based descriptions. OPH asks a different reconstruction question: which public records, symmetries and effective physical laws follow from bounded patches and their agreement constraints? Its finite gauge result fixes a Lie type under complete response and endogenous transport. That result must be compared with a specified string construction, including its compactification and matter inputs, rather than with an unrestricted menu of vacua.

Loop quantum gravity. The canonical loop construction quantizes general relativity using connection variables on a differentiable manifold, without choosing a fixed background metric [102]. Discrete area and volume operators are substantive quantum-geometric results [103]. OPH's finite antecedent instead starts with patch records and authenticated causal dependence. Its rank-three response quotient admits conservative record populations whose complete local reads and declared model clock have flat $1 + 3$ causal and count-volume limits [17]. Counts on fixed interior time-like intervals recover proper-time ratios up to a reference unit. Operational rotation and boost covariance select the Lorentz cone within the nonzero closed convex pointed cone class. Physi-

cal selection of that population, read law and covariance is a separate condition. A meaningful comparison concerns the controlled recovery of geometry and matter on each construction's stated domain.

Wolfram's rewriting approach. Hypergraph and multiway rewriting supply executable discrete structures, and observers are explicitly treated as systems within those structures [104]. Gorard, Namuduri and Arsiwalla embed quantum ZX-diagram rewrites into multiway systems and prove compatibility of the associated monoidal structures [105]. OPH's specific construction uses bounded self-reading patches, protected records and typed overlap repair. Its confluence theorem makes accepted public readouts independent of authorized repair scheduling, while authenticated read-from relations define record order. The comparison therefore concerns these exact observer and update semantics and their physical realization, rather than whether either approach mentions observers or performs computation.

Geometric Unity. Eric Weinstein's 2021 draft starts its Einsteinian construction with a smooth four-manifold and the bundle of metrics over it, using differential geometry to organize gravitational and gauge data [106]. Its observer maps are part of that geometric construction. OPH's primitive observer carries bounded local state, a boundary and writable records, with geometric readouts attached through additional maps. These are different starting objects. Their physical content must be compared after specifying dynamics, quantization and the measured observables represented by each construction.

Hopf fibrations and geometric state representations. A Hopf fibration is a mathematical construction used in several physical settings. For example, Mosseri and Dandoloff use it to represent qubit phases and two-qubit entanglement [107]. It does not name a unique competing theory of everything. OPH also uses familiar bundle and representation geometry. The substantive question is which dynamics, probabilities and physical identifications a particular use of that geometry derives. A symmetry or fibration identity alone supplies no laboratory law.

Causal and operational reconstruction. Causal-set theory starts from locally finite order [1, 2]; OPH's authenticated record relation constructs one finite informational order from specified transactions. Both physical readings require a justified relation between order, number and continuum geometry. Operational quantum reconstructions and entanglement-based gravity also share parts of OPH's mathematical vocabulary. Their results are credited in the relevant derivations. Finite discreteness, relational observables and emergent geometry are therefore shared research ideas; the OPH contribution is the particular construction joining them and the implications it proves.

Concrete reasons to investigate OPH. Protected overlap agreement gives an explicit criterion for a public fact and, under termination, local-diamond and completeness premises, a unique normal form from each initial quotient state. The same architecture constrains internal symmetry: complete reversible response and endogenous transport force the local Standard Model gauge Lie algebra. The fermionic matter representation and its charge normalization are additional inputs to the conditional hypercharge and common $\mathbb{Z}_6$ kernel theorem. Lie type, matter content and physical current identification are separately checkable statements.

One supplied charged-scalar/Maxwell action also connects finite dynamics to ordinary calculational objects. Its invariant real zero-current sector has first-order spatial trajectory convergence for smooth Neumann references and Ritz initial data with positive mass squared and nonnegative quartic coupling. At fixed mesh and nonzero charge, the full reduced kinetic metric is complete; the declared Laplace–Beltrami quantization has a unique self-adjoint closure and a neutral Hilbert sector. These results concern the same supplied action. Their physical couplings, preparation, geometry and quantization prescription retain their stated input status.

The gravity construction gives an explicit implication from realized null balance, stress conservation and continuum control to the Einstein equation up to a cosmological constant. Finite null

tomography isolates the metric-proportional ambiguity. Subsystem boundaries are treated with algebras and edge sectors, a problem also addressed directly in gauge theory and gravity [108]. Small conditional mutual information controls specified recovery estimates. Local modular stress, alignment and refinement require the additional hypotheses printed in the corresponding theorems; small conditional mutual information alone does not supply them.

Proofs, exact certificates and independent replay make these implications inspectable. The primitive-port dispersion branch adds a prospective test: its linked coefficients and first anisotropic harmonic are fixed before an eligible comparison. A resolved violation rejects that physical branch; extension to the whole framework requires the branch to be forced and exclusive. Target-informed closure diagnostics retain a different evidential role. Theorem counts and numerical proximity do not substitute for this test.

The resulting argument for OPH is constructive: a specified observer architecture supports linked reconstruction results and explicit ways to check them. Demonstrating one physical realization carrying the records, geometry, matter and quantum predictions is the decisive test of the proposed unification. The standard applies equally to every candidate.

9 Conclusion

The observer layer has an inhabited finite definition, and its quantum and thermodynamic identities are exact finite theorems once their displayed inputs are supplied. Observerhood at one finite cut, the common origin of an event's support, record and chart coordinate, the Hamiltonian form of continuous private symmetry flows, the Born form of every additive effect valuation and the four-law package with its exact dissipation identity hold as stated. The decisive inputs are the same in every case: a native process must produce the complete effect and instrument web, the shared state and transition reference, the clock and energy dictionary and the capacity source antecedent. The closure equations make the remaining physical claim quantitative. The local equation has interval-certified unique roots for each declared map, and the global equation has an exact finite capacity theory with a bounded counterfamily that shows its base structure does not select a zero. Turning either into a physical identity requires the same-quantity bridges, target-independent map selection and the source-derived hadronic and capacity producers named above.

Companion Papers

The detailed results that this paper consumes are proved in:

1. *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems* [16], which gives the presentation-invariant same-source/cross-source quantifier split, endpoint and refinement bounds, collar-repair obstructions, and the finite conditional-resampling receipt used by the repair papers.

2. *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency* [17], which gives the exact port-record metric completion together with the support-incidence, modular, source-causal continuum, stress, generalized-entropy, and Einstein derivation chain. Every finite, continuum, scaling, and physical premise is stated at the arrow where it enters.

3. *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises* [18], which gives the categorical and finite-carrier routes. It proves the complete-response Lie-type theorem and gives the conditional matrix-current, matter, and tensor-kernel theorems. Identification of the two routes requires a physical current intertwiner.

4. *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics* [19], which carries the finite patch-net fixed-point, defect, quotient, and operator-record consensus construction. Its semantic-complete transaction theorem derives the local diamond from coherent aggregate gluing and full acceptance dependencies; repair completeness is a separate premise. Quantum error-correcting code/min-cut, spectral, Byzantine fault-tolerance, and hardware-speedup statements require their respective certificates.

5. *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography* [20], which carries the regulated federated patch-carrier architecture, the fixed-cutoff edge heat-kernel / Casimir theorem, exact finite modular and rate statements, the $A_5 \times \mathbb{Z}_2$ register classification, and the measurement and observer checkpoint/restoration packages.

6. *Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions* [21], which carries the particle derivations, masses, couplings, and sector calculations.

7. *Observer-Patch Holography and the Dark Sector* [22], which studies modular charge on overlap collars under a physical stress attachment, with conditional recovery bounds, compact-source saturation and deep-galaxy scaling laws.

8. *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt* [23], which isolates the controlled compact-gauge repair mechanism and states the continuum certificate needed before the finite repair gap becomes a Clay-facing four-dimensional Yang–Mills gap.

9. *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins* [26], which states the executable specification, its acceptance criteria, its certified components and the premise map, and reports its executed finite results.

Scope of the gauge and W/Z results used here. The finite icosahedral package used by this paper contains a Lie-type theorem under adopted premises and an exact conditional recognition result, both proved in the gauge paper [18]. Complete reversible response and endogenous transport imply the local Standard Model gauge Lie algebra. Incidence determines the antipode J, and under the explicit contract that an admissible response is a signed central involutive graph automorphism implementing inverse-port readback, the responses are exactly $\pm J$. Conditional also on the matrix current and the rank-15 matter contract, the anomaly and tensor-descent certificates fix the $3+2$ block structure, hypercharge lattice, common $\mathbb{Z}_6$ kernel and maximal faithful matter image. The source does not select the matrix current, matter action or physical global quotient. The finite local action, the parallel measure and Hamiltonian branches, and the strict finite-order perturbative W/Z pole theorem state what follows from typed action and quantization packets; they do not show that the target-free source emits those packets, and no source-native dimensionless or physical-unit W/Z pole follows from the external validation packet.

A Finite Field Histories Read by Observers

This appendix gives the finite field histories on which the observer operations of Sections 1–3 are exercised: exact classical record feedback reads a declared Maxwell history without disturbing it, and a supplied charged-scalar/Maxwell action on a declared solid has controlled real continuum

trajectories, an interacting Hilbert space on its fixed mesh, record-restored execution and a configuration clock. The geometry, matter law and couplings are supplied; the gauge and spacetime papers own the physical field content [18, 17], and the screen-microphysics paper owns the primitive-port and source-seam propagation branches [20].

The first refinement layer admits an exact active internal readout. The twelve inherited values Qx, together with one labeled pair-averaging response for each of the thirty midpoint values, form an invertible 42×42 rational map. Its decoder is $x_m = 2(QE_{u,m}x)_u - (Qx)_u$. The minimal thirty-event selector is not A_5-invariant. Retaining both endpoint choices gives a sixty-event family stable under the proper A_5 action and rank 42 once Q is included. This is a different protocol from the passive history $(Q, QL_f, \dots, QL_f^{41})$, whose rank is 29. The active responses must share one pre-event state; a classical record-assisted implementation satisfies this condition by restoring each probed pair before the next probe. The symmetry statement concerns the scalar probe support and asserts no equivariance of mixed scalar and seam-potential measurements under signed cochain transformations. No source-selected physical instrument, spatial translation, laboratory observable or sky map follows [28].

Theorem A.1 (Serial feedback readout and finite Maxwell action)**.** *Assume exact classical scalar records and writable coordinates on $\{1, \dots, 12\} \sqcup \{1, \dots, 30\}$. Retain the inherited baseline $b_u = x_u$. For a labeled endpoint–midpoint pair (u, m), apply pair averaging, record its exposed endpoint response r, and replace the pair by $(b_u, 2r - b_u)$, fixing every other coordinate. Every finite sequence of these cycles preserves x, including repeated and overlapping pairs. If each midpoint is probed, the records reconstruct x exactly.*

Explicitly type a packet as $x_n = (\phi_n, A_n)$, with twelve scalar potentials and thirty oriented seam potentials. Serial decoding at each slice commutes with

$$E_n = -\frac{A_{n+1} - A_n}{h} - D\phi_n, \qquad B_n = CA_n,$$

where $h \neq 0$, D is the committed port coboundary and C its oriented face incidence. It preserves the entire declared neutral-pair action, including its values on varied fields and paths.

Proof. The response is $r = (b_u + x_m)/2$, so $2r - b_u = x_m$. The feedback restores both written coordinates and leaves the rest fixed. Induction on the probe word gives preservation at every prefix; each retained response therefore refers to the identical initial state. The explicit inverse then reconstructs every coordinate. Applied separately at each potential slice, this identity gives the displayed fields and equality of any action evaluated on the decoded potentials. It holds for every input, hence for every allowed variation. These identities and their application to the rational stationary neutral-pair history are kernel-checked in `SerialMaxwellReadout`. □

The executable instrument uses one evolving 42-slot state, retained baseline and response records, port-local feedback writes, and a public evidence bundle. It reads all sixty endpoint–midpoint pairs at each of three slices. Only A_0, A_1 are supplied as initial seam data. At $h = 1/2$, the decoded records and the current of the opposite charged paths produce

$$A_2 = 2A_1 - A_0 - h^2 C^{\mathsf{T}} C A_1 + h^2 J_0 - hD(\phi_1 - \phi_0).$$

An independent implementation advances E by Ampère's equation and then recovers A_2; it obtains the same rational stationary history. It checks all 54 free field derivatives of the finite action, all eight admissible closed two-step path replacements, Gauss, Faraday and source continuity. Each of two gauge-related executions has 585 semantic events and 180 completed feedback cycles. Replay

reconstructs each operation's read set and checks its pre-write values and writer versions; declared parent lists cannot manufacture causal reads. The gauge control changes raw potentials and records while preserving fields and, with zero endpoint gauge, the coupled action. The exact execution receipt and its independent verifier are reproducible from the accompanying code [29].

This instrument assumes classical read/write access and exact memory. A baseline error ϵ_b and response error ϵ_r give midpoint error $2\epsilon_r - \epsilon_b$ in one cycle; this is not a stability bound for a long noisy execution. Pair averaging alone does not implement the feedback, and quantum state copying is not asserted. The feedback-cycle counter counts executed operations. Its relation to the separately declared Lorentz clock unit $\tau = 3$, laboratory time, source-selected geometry or a spatial continuum refinement is not supplied. A scalar first-refinement slot becomes a potential only under the displayed typing assumption.

From decoded cochains to fields on a solid. The twelve vertices, thirty edges and twenty faces form a closed surface. A concrete volume reconstruction is obtained by coning its declared Euclidean realization to the centre. Use the registered coordinates $(0, \pm 1, \pm\varphi)$, $(\pm\varphi, 0, \pm 1)$, $(\pm 1, \pm\varphi, 0)$, with their fixed port labels and $\varphi = (1 + \sqrt{5})/2$. Each boundary edge has length two; each of the twenty positively oriented tetrahedra has volume $(3 + \sqrt{5})/6$. The cone has $13, 42, 50, 20$ simplices in degrees zero through three. This geometric realization and its numerical length unit are supplied.

Theorem A.2 (Gauge-covariant cone reconstruction)**.** *Let D and C denote the signed boundary gradient and curl. For $\Pi_m = \mathbf{1}_m \mathbf{1}_m^{\mathsf{T}}/m$, set*

$$G = (D^{\mathsf{T}} D + \Pi_{12})^{-1} - \Pi_{12}, \qquad H = (CC^{\mathsf{T}} + \Pi_{20})^{-1} - \Pi_{20}.$$

Order cone edges radially first and cone faces by boundary faces first. Then its incidence maps and trace-preserving extensions are

$$\widetilde{D} = \begin{pmatrix} -\mathbf{1}_{12} & I_{12} \\ 0 & D \end{pmatrix}, \qquad \widetilde{C} = \begin{pmatrix} 0 & C \\ -D & I_{30} \end{pmatrix}, \qquad \widetilde{B} = \begin{pmatrix} I_{20} & -C \end{pmatrix},$$
$$P_0 = \begin{pmatrix} \mathbf{1}_{12}^{\mathsf{T}}/12 \\ I_{12} \end{pmatrix}, \qquad P_1 = \begin{pmatrix} GD^{\mathsf{T}} \\ I_{30} \end{pmatrix}, \qquad P_2 = \begin{pmatrix} I_{20} \\ C^{\mathsf{T}} H \end{pmatrix}.$$

They satisfy

$$\widetilde{D} P_0 = P_1 D, \qquad \widetilde{C} P_1 = P_2 C, \qquad \widetilde{B} P_2 = \Pi_{20}.$$

A boundary two-cochain has a closed cone extension if and only if its total flux vanishes. In particular, every $B = CA$ has such an extension.

Let W_k be Whitney interpolation on this geometric cone. On a uniform time grid $t_n = nh$, $h > 0$, interpolate $P_1 A_n$ linearly and $P_0 \phi_n$ constantly on each open slab. For $s = (t - t_n)/h$, the fields are

$$\mathcal{E}(t) = W_1 P_1 \left[-(A_{n+1} - A_n)/h - D\phi_n \right],$$
$$\mathcal{B}(t) = W_2 P_2 \left[(1 - s) C A_n + s C A_{n+1} \right].$$

The discrete gauge rule lifts to the ordinary potential gauge rule. On the interior of the supplied solid and time window, $\partial_t \mathcal{B} + d\mathcal{E} = 0$ and $d\mathcal{B} = 0$ hold distributionally.

Proof. The exact Green identity gives $GD^{\mathsf{T}} D = I - \Pi_{12}$. The verified boundary complex has $\ker C = \operatorname{im} D$ and $\operatorname{im} C = \mathbf{1}_{20}^{\perp}$, hence $C^{\mathsf{T}} H C = I - DGD^{\mathsf{T}}$ and $CC^{\mathsf{T}} H = I - \Pi_{20}$. Block multiplication proves the three displayed identities. Closedness of a cone two-cochain with boundary value B

requires $B = Cr$ for its radial component r, equivalent to zero total flux. The radial potential $GD^{\mathsf{T}}A$ has zero mean and changes by $\chi - \overline{\chi}$ under $A \mapsto A + D\chi$, exactly matching the apex scalar value $\overline{\chi}$.

In barycentric coordinates, the local forms are

$$w_i = \lambda_i, \qquad w_{ij} = \lambda_i\, d\lambda_j - \lambda_j\, d\lambda_i,$$
$$w_{ijk} = 2 \sum_{\mathrm{cyc}(i,j,k)} \lambda_i\, d\lambda_j \wedge d\lambda_k.$$

Their exterior derivatives commute with signed incidence. Their matching traces give the assembled identities in the weak sense [45]. Vector potentials continuous and piecewise affine in time, with constant slab scalar potentials, give the displayed fields. For affine $\chi(t)$, the scalar gauge change is $-\partial_t \chi$; differentiation and $d^2 = 0$ prove the claims. The magnetic field is continuous in time, so its derivative acquires no additional time-interface impulse. □

The radial reconstruction is global on the boundary graph: its 12×30 matrix has 240 nonzero entries. It is a reconstruction rule, not a derivation of local radial transport. Its cone coefficients are determined by the boundary data and introduce no independent radial degrees of freedom. The finite cochain and zero-flux statements are formalized in `ConeCochainBridge`. That implementation uses a dual-tree section followed by the cycle projector for the radial two-form; uniqueness among co-closed radial solutions proves agreement with $C^{\mathsf{T}}H$ on the zero-flux domain. The Whitney field and distributional statements are the paper argument above.

The metric and time defects of the actual history. Let M_1, M_2 be the Gram matrices of W_1, W_2 in the supplied Euclidean metric, with unit constitutive coefficients. Their exact integrals use $\int_T \lambda_i \lambda_j = |T|(1 + \delta_{ij})/20$. These positive matrices specify geometric field energy. They are distinct from counting inner products. Extend source covectors as $\widetilde{\rho}_n = (0, \rho_n)$ and $\widetilde{J}_n = (0_{12}, J_n)$, preserving the source pairing. Slabwise constant charge has jumps at the knots. Its compatible temporal current functional is $\sum_n h\widetilde{J}_n \delta_{t_{n+1}}$, since $\rho_{n+1} - \rho_n = -hD^{\mathsf{T}} J_n$. This identity uses interior test functions; an extension beyond the window uses the initial charge before zero and the terminal charge after the last current impulse. Ordinary constant slab current has a different pairing.

Write $\widetilde{A}_n = P_1 A_n$, $\widetilde{\phi}_n = P_0 \phi_n$, $\widetilde{E}_n = P_1 E_n$, $\widetilde{B}_n = P_2 C A_n$, and $Q(b) = b^{\mathsf{T}} M_2 b$. The exactly integrated field-plus-source action is

$$S_{\mathrm{pr}} = \sum_{n=0}^{N-1} \Big\{ \frac{h}{2} \widetilde{E}_n^{\mathsf{T}} M_1 \widetilde{E}_n - \frac{h}{6} \Big[Q(\widetilde{B}_n) + \widetilde{B}_n^{\mathsf{T}} M_2 \widetilde{B}_{n+1} + Q(\widetilde{B}_{n+1}) \Big] + h\widetilde{J}_n^{\mathsf{T}} \widetilde{A}_{n+1} + h\widetilde{\rho}_n^{\mathsf{T}} \widetilde{\phi}_n \Big\}.$$

Let S_{right} use the same electric and source terms but right-endpoint magnetic quadrature. Direct integration and telescoping give

$$S_{\mathrm{pr}} - S_{\mathrm{right}} = \frac{h}{4} \Big[Q(\widetilde{B}_N) - Q(\widetilde{B}_0) \Big] + \frac{h}{12} \sum_{n=0}^{N-1} Q(\widetilde{B}_{n+1} - \widetilde{B}_n). \tag{7}$$

The second term equals $h^2 \int Q(\partial_t \widetilde{B})\, dt/12$. It is second order on a fixed time window only under uniform temporal regularity; the total action defect includes the displayed endpoint term. Fixed-endpoint variations remove that term. Nonuniform time grids need additional terms. The separate

spatial defect relative to the counting action is

$$S_{\mathrm{right}} - S_{\mathrm{count}} = \frac{h}{2} \sum_n \left[E_n^{\mathsf{T}} (P_1^{\mathsf{T}} M_1 P_1 - I) E_n - B_{n+1}^{\mathsf{T}} (P_2^{\mathsf{T}} M_2 P_2 - I) B_{n+1} \right].$$

The clock action is excluded from all three field-plus-source actions.

For the derivatives, allow every cone potential coefficient to vary independently and evaluate afterward at the reconstructed history; variations need not stay in the images of P_1 and P_0. With fixed vector-potential endpoints and free spatial traces, the full finite-element derivatives of S_{pr} are

$$\begin{aligned}
\frac{\partial S_{\mathrm{pr}}}{\partial \widetilde{\phi}_n} &= h \left(\widetilde{\rho}_n - \widetilde{D}^{\mathsf{T}} M_1 \widetilde{E}_n \right), \\
\frac{\partial S_{\mathrm{pr}}}{\partial \widetilde{A}_m} &= M_1 (\widetilde{E}_m - \widetilde{E}_{m-1}) + h \widetilde{J}_{m-1} \\
&\quad - \frac{h}{6} \widetilde{C}^{\mathsf{T}} M_2 (\widetilde{B}_{m-1} + 4\widetilde{B}_m + \widetilde{B}_{m+1}), \qquad 0 < m < N.
\end{aligned}$$

The second line differs from the metric right-endpoint derivative by $h\widetilde{C}^{\mathsf{T}} M_2 (2\widetilde{B}_m - \widetilde{B}_{m-1} - \widetilde{B}_{m+1})/6$. This temporal correction does not remove the spatial metric/source defect. The coefficient identities and telescoping are formalized in `WhitneyTimeBridge`; analytic integration and distributional source interpretation are stated here separately.

The authenticated serial history gives the following dimensionless values for both endpoint-fixed gauge representatives:

Counting field-plus-source action	−14.42982115034
Volume action, right-endpoint magnetic term	−14.25989506948
Exactly time-integrated volume action	−10.21278655268
Spatial action defect	0.16992608086
Temporal action defect	4.04710851680
Maximum absolute scalar residual, divided by h	0.64544633271
Maximum absolute interior vector residual	1.57903507514

The rational cochain identities are exact. The volume Gram matrices and displayed values use double-precision analytic moments and independent tetrahedral quadrature, compared with absolute and relative tolerances 10^{-10}; they are not interval certificates. The independent verifier differentiates all $42 + 2 \cdot 13 = 68$ free coefficients of the action, including the twelve radial vector variations omitted by a boundary-only test. A separate exact counting-metric control gives temporal action defect 663629599/63369648 and first vector residual 44749/55152. These are mathematical controls on recorded inputs, not measurements.

Whitney one-forms have matching tangential traces; their normal jumps can contribute to electric divergence. Magnetic tangential jumps and temporal electric jumps likewise contribute to the sourced equations. The assembled weak derivatives retain these effects; element-interior tests are insufficient. The source covectors specify finite-element test functionals, not arbitrary-test physical densities. Under the conventional smooth potential signs, D^{T} represents weak negative divergence, so the coupling $+J \cdot A + \rho_{\mathrm{load}} \phi$ corresponds to $\rho_{\mathrm{physical}} = -\rho_{\mathrm{load}}$ if a density identification is made. The supplied cone yields continuous-coordinate fields and explicit defects. It supplies neither sourced stationary volume dynamics nor a source-selected geometry, physical clock, spatial refinement family, intertwiner with the translation-symbol field group, or laboratory law.

Stationary evolution in the full volume. The preceding interpolation need not be stationary. A different, explicitly declared initial-value problem uses every volume edge as an independent coordinate. In this paragraph write $D = \widetilde{D}$, $C = \widetilde{C}$, $M = M_1$, $K = C^{\mathsf{T}} M_2 C$ and $F = M + h^2 K/6$. These are the geometric matrices above, not counting pairings. Since $M > 0$, $K \geq 0$ and $CD = 0$, one has $F > 0$ and $KD = 0$. Sources have the same covector and temporal impulse interpretation as before.

Theorem A.3 (Full-volume prism evolution and constraint preservation)**.** *For $h > 0$, the interior vector-potential equations of the exactly integrated prism action are equivalent to*

$$F A_{n+1} = \left(2M - \frac{2h^2}{3} K\right) A_n - F A_{n-1} + h^2 J_{n-1} - hMD(\phi_n - \phi_{n-1}). \tag{8}$$

They determine a unique next vector potential for every supplied scalar potential and pair of previous vector potentials. With $E_n = -(A_{n+1} - A_n)/h - D\phi_n$, the Gauss residual $g_n = D^{\mathsf{T}} M E_n - \rho_n$ satisfies

$$g_n - g_{n-1} = -hD^{\mathsf{T}} J_{n-1} - (\rho_n - \rho_{n-1}).$$

Thus conserved sources propagate the initial constraint, including any initial error. For neutral ρ_0, a given initial electric field E_{old} can instead be replaced by

$$E_0 = E_{\mathrm{old}} + Dz, \qquad z = (D^{\mathsf{T}} M D + \Pi_{13})^{-1} (\rho_0 - D^{\mathsf{T}} M E_{\mathrm{old}}). \tag{9}$$

This is the minimum M-norm correction satisfying Gauss's law. It preserves the curl of the initial electric field and defines a new initial condition.

Proof. Substitute the definitions of E_n and $B_n = CA_n$ in the full action derivatives above and collect the next-potential coefficient. Positive definiteness of F proves existence and uniqueness in the finite-dimensional edge space. Applying D^{T} to the vector equation removes every magnetic term and gives the displayed residual identity. The connected cone has $\ker D = \mathrm{span}\{\mathbf{1}\}$, so $D^{\mathsf{T}} M D$ is invertible on mean-zero vertex vectors. Neutrality places the right-hand side of (9) in that subspace. Any competing correction differs from Dz by w with $D^{\mathsf{T}} M w = 0$; hence $(Dz)^{\mathsf{T}} M w = 0$. Pythagoras proves minimality. Finally, $CDz = 0$. □

Theorem A.4 (Work identity and sharp positive-mode stability)**.** *Let $\overline{B}_n = (B_n + B_{n+1})/2$. Along the full recurrence,*

$$H_n = \frac{1}{2} E_n^{\mathsf{T}} \left(M - \frac{h^2}{12} K\right) E_n + \frac{1}{2} \overline{B}_n^{\mathsf{T}} M_2 \overline{B}_n,$$

$$H_n - H_{n-1} = -\frac{h}{2} J_{n-1}^{\mathsf{T}} (E_n + E_{n-1}).$$

For a positive generalized mode $Kv = \lambda M v$, all source-free modal solutions are bounded exactly when $0 < h^2\lambda < 12$. At $h^2\lambda = 12$, the double characteristic root -1 admits linearly growing alternating solutions. Above twelve an exponentially growing solution exists. Source-free zero modes have constant electric field and can have linearly drifting gauge potentials.

Proof. The recurrence also reads $F(E_n - E_{n-1}) = hKA_n - hJ_{n-1}$; here $KD = 0$ removes the scalar-potential terms. Faraday's identity is $B_{n+1} - B_n = -hCE_n$. Polarize the two quadratic terms, use these identities, and cancel the magnetic cross terms. Equivalently, $H_n = E_n^{\mathsf{T}} F E_n/2 + B_n^{\mathsf{T}} M_2 B_{n+1}/2$, which makes the telescoping cancellation immediate. This proves the work law in every gauge.

For $z = h^2\lambda$, the scalar characteristic equation is

$$r^2 - 2a_z r + 1 = 0, \qquad a_z = \frac{1 - z/3}{1 + z/6}.$$

For positive z, the strict inequality $|a_z| < 1$ holds exactly on $z < 12$. Its two distinct unit-circle roots give bounded solutions. At twelve, $(-1)^{n+1}n$ is a solution and its electric difference is unbounded. Above twelve, the negative real reciprocal roots include one with modulus greater than one. In temporal gauge the zero-mode recurrence is a second difference, giving affine potentials. These facts do not bound gauge potentials by field energy. □

The finite action expansions, weak Noether identity, work law, uniqueness and sharp scalar alternatives are formalized in `Lean/Screen/WhitneyMaxwellDynamics.lean`. Stable initial-value evolution does not imply a nonsingular fixed-endpoint boundary problem: for two slabs and one mode with $h^2\lambda = 3$, zero endpoint potentials leave the interior potential arbitrary when its source is zero, and are incompatible with a nonzero interior source. The initial-value coefficient $1 + h^2\lambda/6$ remains positive there.

An exact bound for the supplied cone. Every tetrahedron has three apex rays with Gram matrix $2I_3 + \varphi\mathbf{1}\mathbf{1}^{\mathsf{T}}$. Divide each local Whitney mass and curl-energy matrix by its positive tetrahedron volume. Exact arithmetic in $\mathbb{Q}(\sqrt{5})$ gives an LDL^{T} factorization of $24M_T - K_T$ with strictly positive diagonal. The Lean construction decodes the committed coordinates and signed incidence tables, derives barycentric gradients and exact local mass forms, and checks the factorization with a proved sign decision in $\mathbb{Q}(\sqrt{5})$. Its finite-dimensional checker is sound and admits a certificate for every positive-definite matrix over that field. The emitted witness binds the certificate payload to the kernel-checked definitions. Independent numerical quadrature reconstructs the global forms and checks their agreement with the exact local forms after signed assembly. The twenty signed local-to-global edge maps cover every global edge, so assembly proves $K < 24M$, without a floating-point eigenvalue bound. At $h = 1/2$, therefore,

$$M - h^2K/12 > M/2, \qquad E_n^{\mathsf{T}} M E_n \le 4H_n, \qquad B_n^{\mathsf{T}} M_2 B_n \le 16H_n$$

on a source-free interval. The magnetic estimate also holds at the other endpoint of each slab: use $B_n = \overline{B}_n + hCE_n/2$ and $h^2 E_n^{\mathsf{T}} K E_n \le 6E_n^{\mathsf{T}} M E_n$. The source-derived positive mass forms also construct the complete thirty-mode frame used by the classical action and polynomial quantum Hamiltonian. Signed local-coordinate changes transport exact certificate existence, and the global face-order permutation separately preserves the real action, constraint and quantum consumers. These are two proved reexpression policies; their collection in one bundle supplies no theorem identifying them as one local/global runtime transport. Source-manifest and payload checks in a Lean-free environment authenticate the committed witness; replaying the kernel and consumer proofs requires Lean. The factorization is an exact algebraic certificate; the trajectory below is a separate numerical calculation.

A recorded stationary volume history. Retain the inherited initial vector potential and prescribed neutral source covectors, apply (9), and use (8). The correction changes the initial electric field by approximately 0.71279026 in M-norm. Consequently this history is distinct from the earlier counting-action history and its cone interpolation. Each of three slices occupies thirteen scalar and forty-two edge registers. For each edge, read both endpoint pair averages, retain scalar baselines, and restore the registers by $(b, 2r - b)$. The resulting bounded observer-like software patch has

writable ports, readback, records, feedback and a public evidence bundle. Its 805 recorded events include 252 feedback cycles. Every read names its last writer and exact value; the volume advance consumes the decoded registers and source covectors. The dense metric projection and evolution rule are supplied software operations, not consequences of pair averaging or local causal ancestry.

The two endpoint-fixed gauge representatives have field-plus-source action approximately -1.14785515542. Independent spacetime quadrature and complex-step differentiation verify all $42 + 2 \cdot 13 = 68$ free action derivatives, including radial edges and the apex. Their absolute residuals are below 10^{-12}. The metric evolution uses double precision and declared 10^{-9} comparison tolerances; register values are exact rational encodings of the numerical outputs, and feedback restoration is exact for those encodings. No interval or exact-real trajectory certificate is asserted.

A separate 64-slab numerical continuation starts from the decoded temporal-gauge data. Its maximum Gauss, vector-equation and work residuals are below 10^{-12}, as is the source-free energy drift. Its departure from the fixed radial extension reaches approximately 0.45218, showing that the volume recurrence exercises independent radial degrees of freedom. Only the first three slices have the serial execution trace. The remaining continuation is a numerical field calculation. Prescribed currents are not dynamical charged matter. Stationarity against all finite-element coordinates also does not imply stationarity against arbitrary smooth continuum tests. Spatial refinement, physical clock and metric selection, and laboratory identification require further input.

The Hilbert space of the same radiative modes. Consider the source-free neutral sector of this same geometric action, in continuous time and temporal gauge, with the spatial mesh fixed. The constraint is $D^{\mathsf{T}} M \dot{A} = 0$. A time-independent gauge transformation removes the constant longitudinal potential, giving the transverse space $V_\perp = \ker(D^{\mathsf{T}} M)$. The cone is contractible: its explicit curl blocks imply $\ker C = \operatorname{im} D$, of dimension twelve. Thus $V_\perp$ has dimension thirty and K is positive definite there. The finite-dimensional spectral theorem gives a complete real frame $v_1, \dots, v_{30}$ with

$$v_i^{\mathsf{T}} M v_j = \delta_{ij}, \qquad K v_i = \omega_i^2 M v_i, \qquad \omega_i > 0.$$

For $A = \sum_i q_i v_i$, the exact spatially semidiscrete action is

$$S_0 = \frac{1}{2} \int \left(\dot{A}^{\mathsf{T}} M \dot{A} - A^{\mathsf{T}} K A \right) dt = \frac{1}{2} \int \sum_i (\dot{q}_i^2 - \omega_i^2 q_i^2)\, dt.$$

The spatial edge-element action is standard [46], as are the finite normal-mode theorem and canonical field quantization. The attachment here is to these geometric matrices.

Theorem A.5 (Canonical quantum realization of the volume radiative sector)**.** *Supply canonical bosonic quantization and $\hbar > 0$. The preceding sector has Hilbert space $\mathcal{H} = \ell^2(\mathbb{N}^{30})$. On its finite-occupation domain, let*

$$a_i |n\rangle = \sqrt{n_i}\, |n - e_i\rangle, \qquad a_i^\dagger |n\rangle = \sqrt{n_i + 1}\, |n + e_i\rangle,$$

$$\widehat{q}_i = \sqrt{\frac{\hbar}{2\omega_i}} (a_i + a_i^\dagger), \qquad \widehat{p}_i = i \sqrt{\frac{\hbar \omega_i}{2}} (a_i^\dagger - a_i).$$

Then $[\widehat{q}_i, \widehat{p}_j] = i\hbar \delta_{ij} I$, and

$$\widehat{H} = \frac{1}{2} \sum_i (\widehat{p}_i^2 + \omega_i^2 \widehat{q}_i^2), \qquad E_n = \hbar \sum_i \omega_i (n_i + 1/2).$$

The diagonal Hamiltonian on $\mathcal{D}(\widehat{H}) = \{\psi : \sum_n E_n^2|\psi_n|^2 < \infty\}$ *is self-adjoint and generates the strongly continuous unitary group* $(U_t\psi)_n = e^{-itE_n/\hbar}\psi_n$. *The Heisenberg equations are* $\dot{\widehat{q}}_i = \widehat{p}_i$ *and* $\dot{\widehat{p}}_i = -\omega_i^2\widehat{q}_i$. *Consequently* $\widehat{A} = \sum_i v_i\widehat{q}_i$, $\widehat{E} = -\sum_i v_i\widehat{p}_i$ *and* $\widehat{B} = C\widehat{A}$ *use the same classical mode shapes and satisfy the source-free semidiscrete Maxwell equations on the common invariant domain.*

Proof. Start with polynomials in thirty complex variables. Give monomials the inner product $\langle X^n, X^m\rangle = n!\delta_{nm}$, where $n! = \prod_i n_i!$. Multiplication by X_i and differentiation by X_i are adjoint on this domain and satisfy $[a_i, a_j^\dagger] = \delta_{ij}I$. Normalizing monomials gives the displayed orthonormal occupation basis and its ℓ^2 completion. The quadratic Hamiltonian and commutators follow by expansion. Finite occupation vectors are analytic for each position and momentum operator. If N bounds a vector's occupation support, its k-th iterate has norm at most $C_\psi C^k\sqrt{(N+k)!/N!}$, with fixed constants C_ψ, C. This makes the analytic-vector series converge for sufficiently small argument. Thus these symmetric operators have unique self-adjoint closures by the analytic-vector theorem [47].

The adjoint of the real diagonal operator has exactly the stated weighted square-summability domain, proving self-adjointness of $\widehat{H}$. Truncation in the occupation basis proves that finite-support vectors are a core. The same finite-tail argument proves strong continuity of U_t. Finally $[\widehat{H}, \widehat{q}_i] = -i\hbar\widehat{p}_i$ and $[\widehat{H}, \widehat{p}_i] = i\hbar\omega_i^2\widehat{q}_i$ give the equations, and the complete spectral frame pulls them back to the actual mass and stiffness forms. □

`Lean/Screen/WhitneyQuantumBridge.lean` formalizes the complete normal-frame action pullback, polynomial commutation relations, occupation energies and Heisenberg identities. The modules `Lean/Screen/WhitneyNormalModeConstruction.lean`, `Lean/Screen/WhitneyConeModes.lean`, and `Lean/Screen/WhitneyConeMass.lean` construct the complete positive 30-mode frame from the cone cochain maps and supplied symmetric positive-definite Whitney mass forms. The frame-naturality modules prove that the classical action and the full polynomial Hamiltonian, including the zero-point term, are unchanged under mass-overlap transport between any two admissible frames; the transition and its polynomial lift obey identity and composition. The Hilbert completion and self-adjoint-domain argument are the analytic proof just given, not Lean results. This is canonical quantization of the free radiative sector. The recorded history with nonzero prescribed charge additionally needs an affine longitudinal sector and source coupling. Classical readback does not select a quantum state, Born statistics, $\hbar$ or a physical clock, and the finite mesh supplies no relativistic quantum field net.

A controlled temporal limit. The finite-step prism dynamics and the continuous-time oscillator have different frequencies. For one mode, put $z = h^2\lambda$, $f = 1 + z/6$ and choose $0 < z < 12$. The discrete Legendre momenta give the exact symplectic map

$$\begin{pmatrix} q' \\ p' \end{pmatrix} = \frac{1}{f}\begin{pmatrix} 1 - z/3 & h \\ -h\lambda(1 - z/12) & 1 - z/3 \end{pmatrix}\begin{pmatrix} q \\ p \end{pmatrix}.$$

Define $\theta \in (0, \pi)$ by $\cos\theta = (1 - z/3)/f$. Since $\sin^2\theta = z(1 - z/12)/f^2$, this is the time-$h$ flow of the positive modified Hamiltonian

$$H_h = \frac{\theta}{2f\sin\theta}\left[p^2 + \lambda(1 - z/12)q^2\right].$$

On any fixed bounded positive spectral window, $\theta/h = \sqrt{\lambda} + O(h^2)$ and the quadratic coefficients of $H_h - H_0$ are $O(h^2)$, uniformly as $h \to 0$. On bounded time intervals the corresponding classical

propagators therefore converge at order two for bounded canonical initial data. The initial pair of potentials must be obtained from the discrete Legendre map for those data; an arbitrary first-order initial displacement does not provide the same order claim. This is a finite-dimensional temporal convergence argument with the spatial mesh fixed. Canonical quantization of H_h supplies its own unitary oscillator flow, with frequency θ/h; its continuous Hamiltonian path fixes the metaplectic lift. Neither its generator nor its frequency equals $\widehat{H}$ at fixed h. No uniform operator-norm error bound for the unbounded quantum generators, spatial convergence theorem or physical time calibration follows from this calculation.

A joint action with dynamical charged matter. Prescribed currents can be replaced by a dynamical complex scalar on the same solid. This requires an additional matter action and coupling; neither is selected by the classical readback protocol. Gauge-invariant simplicial actions have established precedents [48]. The construction below specifies a potential-dependent scalar trial space on this particular volume and retains the complete dependence of the scalar interpolation on the gauge potential.

Let $a \in \mathbb{R}^{42}$, $\phi \in \mathbb{R}^{13}$ and $\psi \in \mathbb{C}^{13}$ be variable coefficients, and supply a real charge e. On a tetrahedron, write $a_{ij} = -a_{ji}$ for the oriented edge coefficient, set $a_{ii} = 0$, and define

$$A_h = W_1 a, \qquad \phi_h = W_0 \phi, \qquad \chi_h = W_0 \chi,$$
$$\theta_i(x) = \int_{v_i}^{x} A_h = \sum_j a_{ij} \lambda_j(x), \qquad \Psi_a(x) = \sum_i \lambda_i(x) e^{ie\theta_i(x)} \psi_i.$$

The paths in θ_i are straight segments within the tetrahedron. The coefficients a_{ij} are real unwrapped integrals; compact link phases alone do not determine these interpolants. Let $D_t\Psi = \partial_t\Psi + ie\phi_h\Psi$, $D_x\Psi = \nabla\Psi - ieA_h\Psi$, and $E = -\dot{a} - D\phi$. With supplied $m^2, g \geq 0$, use the autonomous action

$$S = \int dt \left\{ \frac{1}{2} E^{\mathsf{T}} M E - \frac{1}{2} a^{\mathsf{T}} K a + \int_\Omega \left[|D_t\Psi_a|^2 - |D_x\Psi_a|^2 - m^2|\Psi_a|^2 - \frac{g}{2}|\Psi_a|^4 \right] dx \right\}. \tag{10}$$

Every occurrence of Ψ_a, including its time derivative, is differentiated when varying a. The field-dependent trial space makes this requirement substantive.

Theorem A.6 (Gauge-covariant charged field and joint evolution)**.** *On the fixed nondegenerate cone, Ψ_a is a continuous $H^1(\Omega)$ field with nodal values ψ_i. The action (10) is invariant under every smooth nodal gauge transformation*

$$a \longmapsto a + D\chi, \qquad \phi \longmapsto \phi - \dot{\chi}, \qquad \psi_i \longmapsto e^{ie\chi_i}\psi_i.$$

In temporal gauge, its joint real Euler–Lagrange equations for $a, \operatorname{Re}\psi, \operatorname{Im}\psi$ have a unique global classical solution for every finite initial position and velocity. If all thirteen Gauss equations hold initially, they hold for all time. Nonzero locally charged, globally neutral compatible initial data exist whenever $e \neq 0$. These statements concern the full nonlinear finite-dimensional action and do not assume a prescribed external current.

Proof. Pullback of a Whitney edge form to the straight segment from v_i to x gives the displayed formula for θ_i. Since $\sum_j \lambda_j = 1$, a gauge transformation changes it by $\chi_h(x) - \chi_i$, and consequently $\Psi_{a+D\chi}[e^{ie\chi}\psi] = e^{ie\chi_h}\Psi_a[\psi]$. Both covariant derivatives acquire this same unit phase. The electric and magnetic fields are unchanged, proving action invariance. At a vertex only its own basis function survives. On a common face only its vertices contribute, and the barycentric functions

and tangential Whitney field have identical traces from either tetrahedron. The scalar traces therefore agree. Piecewise smoothness on a finite mesh proves H^1 conformity.

Write $\psi = u + iv$ and define the Euler expressions with the sign convention $\mathcal{E}_y = \partial L/\partial y - d(\partial L/\partial \dot{y})/dt$, while $\mathcal{E}_\phi = \partial L/\partial \phi$. Gauge invariance, integrated against an arbitrary compactly supported $\chi(t)$, gives the off-shell identity

$$\dot{\mathcal{E}}_\phi + D^{\mathsf{T}}\mathcal{E}_a + e\big(u \odot \mathcal{E}_v - v \odot \mathcal{E}_u\big) = 0. \tag{11}$$

Here $\odot$ means coordinatewise multiplication. In particular,

$$(\mathcal{E}_\phi)_i = -(D^{\mathsf{T}}ME)_i + \rho_i, \qquad \rho_i = 2e\int_\Omega \lambda_i \operatorname{Im}(\overline{\Psi_a} D_t \Psi_a)\, dx.$$

Thus the joint matter and vector equations propagate Gauss's law. Free variation of every scalar coefficient requires $\sum_i \rho_i = 0$ on this closed-load convention; a nonzero total charge would require a different boundary-flux or source prescription. As for the prescribed source action, conventional physical charge has the opposite load sign.

To prove existence, fix temporal gauge. Write the scalar interpolation as $W(a)\psi$. Its time derivative is $W(a)\dot{\psi} + S(a,\psi)\dot{a}$, with S the derivative of W in the gauge-potential direction. The quadratic velocity Hessian is

$$\delta\dot{a}^{\mathsf{T}} M \delta\dot{a} + 2\|W(a)\delta\dot{\psi} + S(a,\psi)\delta\dot{a}\|_{L^2}^2.$$

It is positive definite: vanishing first forces $\delta\dot{a} = 0$, and then nodal interpolation forces $\delta\dot{\psi} = 0$. All coefficients are smooth real functions, so the Euler equations form a regular smooth first-order system in positions and velocities and have unique local solutions.

The conserved energy is the sum of the positive electric and magnetic energies and $\|\partial_t \Psi_a\|_{L^2}^2 + \|D_x\Psi_a\|_{L^2}^2 + m^2\|\Psi_a\|_{L^2}^2 + g\|\Psi_a\|_{L^4}^4/2$. It bounds $|\dot{a}|$, hence keeps a in a compact set on each finite time interval. On that set the positive scalar Gram matrix $W(a)^*W(a)$ has a uniform positive lower bound. Smoothness and linearity in ψ give $\|S(a,\psi)\dot{a}\|_{L^2} \le C|\psi||\dot{a}|$. Consequently $|\dot{\psi}| \le C_T(\sqrt{H} + |\psi|)$, where H is the initial energy. Gronwall's inequality bounds ψ and $\dot{\psi}$ on every finite interval. Smooth Hessian inversion then excludes finite-time escape, proving global existence. This argument does not assert a time-uniform bound on gauge potentials or require a globally smooth quotient by the gauge group.

For an explicit charged initial condition, take $a_0 = 0$, $\psi_i(0) = 1$, and $\dot{\psi}_i(0) = if_i$, where the real nonzero nodal function $f_h = \sum_i f_i\lambda_i$ has zero integral. Antisymmetry gives $\sum_{ij}\lambda_i\lambda_j\dot{a}_{ij} = 0$, so the dressing contribution to $\partial_t\Psi_a$ cancels at these data for every $\dot{a}$. Hence $\rho_i = 2e\int \lambda_i f_h$ has zero total and is nonzero by positivity of the scalar mass matrix. Solve $D^{\mathsf{T}}MDz = \rho$ with mean-zero z, and set $E_0 = Dz$, $\dot{a}_0 = -Dz$. This satisfies all Gauss equations without a circular source definition. On the regular cone an explicit choice is $f_0 = 3$ at the centre and $f_i = -1$ at all twelve boundary vertices. If $V_T = (3+\sqrt{5})/6$ is the common tetrahedron volume, then $\rho_0 = 6eV_T$ and $\rho_i = -eV_T/2$. Writing $\beta = e(2+3\varphi)/10$, take $z_0 = 12\beta/13$ and $z_i = -\beta/13$. Each radial electric coefficient is $-\beta$ and every boundary electric coefficient is zero. The apex gradient norm $|\nabla\lambda_0|^2 = 3/(2+3\varphi)$, simplex incidence and the exact barycentric moments verify all thirteen Gauss equations. □

The finite holonomy and interpolation gauge algebra is formalized in `Lean/Screen/WhitneyChargedMatter.lean`. Conformity, the variational identity and global existence are the analytic argument above. In particular, the action-derived edge force includes the change of Ψ_a when a varies. The Whitney projection of the continuum scalar current generally omits these terms even when the nodal matter equations hold. This finite approximation has dynamical matter and

a joint stationary action. Its geometry, scalar species, charge, mass, interaction coefficient and continuum matter Lagrangian are supplied. The analytic classical theorem does not itself certify a numerical history or select physical matter content. Its finite equations, spatial action consistency and interacting Hilbert realization have distinct numerical and analytic statements below. The free radiative Hilbert space describes a sector; quantization of the coupled action requires its full kinetic metric. No Standard Model identification follows from the supplied scalar law.

An executed charged solution of the same finite action. The preceding initial data lie in a useful invariant sector of (10). Let every outward radial edge have coefficient α, every boundary edge have coefficient zero, and let the scalar coefficient be $c \in \mathbb{C}$ at the centre and $b \in \mathbb{C}$ at all boundary vertices. This sector has five real configuration coordinates. On each tetrahedron write $s = \lambda_0$, $C = e^{ie\alpha}c$, and $U = C - b$. Then

$$\begin{aligned} A_h &= -\alpha \nabla s, & \Psi_a &= e^{-ie\alpha s}(sC + (1-s)b), \\ D_x \Psi_a &= e^{-ie\alpha s} U \nabla s, & \partial_t \Psi_a &= e^{-ie\alpha s}\left[s e^{ie\alpha}\dot{c} + (1-s)\dot{b} + ie\dot{\alpha} s(1-s)U \right]. \end{aligned} \tag{12}$$

The last term retains the time derivative of the scalar basis. Although A_h is a spatial gradient and the magnetic field vanishes, the electric field $\dot{\alpha}\nabla s$, local charge and matter motion are nonzero. Removing this electric field by a time-dependent spatial gauge introduces a scalar potential; it does not remove the interaction.

Proposition A.7 (Full variational lift of the symmetric sector)**.** *The five-coordinate Euler–Lagrange equations obtained by restricting* (10) *in temporal gauge imply all sixty-eight real temporal-gauge Euler–Lagrange equations. Initial satisfaction of Gauss's law implies all thirteen Gauss equations along this solution. On this sector the action integrands are polynomials of degree at most four in barycentric coordinates after cancellation of their common unit phase. Their unrestricted first variations, and the full Euler expressions evaluated on a twice differentiable symmetric path, have polynomial degree at most five.*

Proof. The sixty proper rotations of the regular icosahedron act on nodal coefficients by permutation and on oriented edge coefficients by signed permutation. They preserve tetrahedral volume, the Euclidean metric and straight paths, hence preserve the complete dressed action. Vertex transitivity fixes one boundary scalar coefficient. The radial edges form one orbit. Each boundary edge is reversed by a half-turn, forcing its invariant coefficient to vanish. Thus the fixed configuration space is precisely the stated five-dimensional real space.

At an invariant position, velocity and acceleration the full Euler covector is invariant. Its value on any variation equals its value on the average of that variation over the finite rotation group. This average lies in the five-dimensional fixed space. Consequently vanishing of the restricted Euler covector implies vanishing on every real variation. Equivalently, regularity and uniqueness of the full Euler system make its evolution preserve this fixed space. Gauss propagation follows from (11) with the compatible initial data, without discarding the scalar variations.

Antisymmetry gives the first line of (12); direct differentiation gives the second line. After removal of the common phase, the scalar field has degree one and its time derivative degree at most two. The spatial covariant derivative has degree zero. An unrestricted edge variation of the scalar field has degree at most two, its time derivative at most three, and its spatial covariant derivative at most two. These degree counts give the claims for the action and its first variation. The same counts apply after the time integration by parts: the scalar acceleration has degree at most three. Four-point tensor Gauss–Legendre quadrature after a Duffy map integrates these degree-five

tetrahedral polynomials exactly in real arithmetic. The Jacobian adds at most two powers in any quadrature coordinate. For the restricted action one can instead integrate over s with the exact marginal weight $3(1-s)^2$. □

A bounded numerical evolution supplies $e = 1/4$, $m^2 = 1/2$ and $g = 1/4$, starts from the charged initial condition above, and evolves the supplied dimensionless action parameter over $[0, 2]$. It stores 81 samples of all position, velocity and acceleration coefficients, electric and magnetic cochains, nodal charge loads, and scalar and covariant-derivative fields at all twenty tetrahedral centroids. The independent verifier reconstructs the oriented mesh and unrestricted scalar Jacobians, checks all 68 Euler and 13 Gauss equations at every sample, and re-integrates the trajectory using full local Jacobians pulled back to the fixed space. This reconstruction does not use the producer's one-dimensional phase-cancelled formulas.

The largest full Euler residual is below 6×10^{-14}, the largest Gauss residual below 4×10^{-15}, and energy drift below 7×10^{-14}. Independent trajectory reconstruction agrees within 3×10^{-14}. Fixed-step fourth-order Runge–Kutta endpoint errors at 80, 160 and 320 steps are approximately 6.61×10^{-6}, 4.32×10^{-7} and 2.75×10^{-8}, respectively, against a tighter reference. Four- and five-point tetrahedral quadratures agree within 5×10^{-14}; underintegration with three points instead gives a full Euler defect of about 3×10^{-4}. Freezing the scalar basis during variation gives a defect exceeding 0.34. The maximum nodal charge-load change exceeds 2.25, so this is a coupled charged trajectory, not repetition of the initial constraint check. These are observed floating-point residuals and refinement comparisons, not a rigorous enclosure of trajectory error.

The producer and independent verifier are `code/electromagnetism/whitney_charged_dynamics.py` and `code/electromagnetism/verify_whitney_charged_dynamics.py`; the versioned data are in `code/electromagnetism/runtime/whitney_charged_dynamics_receipt.json`. The data identify the mesh, oriented basis, gauge, complex encoding, couplings, clock, source hashes and field sampling locations. They contain no authenticated observer readback or quantum state. The construction executes the supplied classical action on one fixed geometry; it does not identify a physical clock, select the matter theory from source histories, or establish a spatial continuum limit.

A rigorous time enclosure of the charged trajectory. The finite trajectory admits a rigorous error bound on the whole supplied action-time interval $[0, 2]$. The enclosure uses the same charged initial data and full dressed action as the numerical execution. It controls time integration on the fixed cone, separately from spatial convergence, observer provenance and physical clock calibration.

Put $r = \kappa/|\Omega| = 6/(7 + 3\sqrt{5})$, and divide the action by $|\Omega|$, which does not change its Euler equations. In the symmetric sector set $C = e^{ie\alpha}c$, $x = (\operatorname{Re} C, \operatorname{Im} C, \operatorname{Re} b, \operatorname{Im} b)$, and $Z(s) = sC + (1-s)b$. The time kinetic term is

$$\tfrac{1}{2} r \dot{\alpha}^2 + \int_0^1 3(1-s)^2 \left| \dot{Z} - ie\dot{\alpha} s Z \right|^2 ds.$$

Thus α is cyclic. The constant scalar velocity block and its inverse are

$$H = \begin{pmatrix} 1/5 & 3/10 \\ 3/10 & 6/5 \end{pmatrix} \otimes I_2, \qquad H^{-1} = \begin{pmatrix} 8 & -2 \\ -2 & 4/3 \end{pmatrix} \otimes I_2.$$

Writing the full kinetic metric as $\left(\begin{smallmatrix} d & \ell^{\mathsf{T}} \\ \ell & H \end{smallmatrix}\right)$, one obtains

$$\begin{aligned} \ell &= \frac{e}{10}(C_i + b_i, -C_r - b_r, C_i + 2b_i, -C_r - 2b_r), \\ w = H^{-1}\ell &= (e(3C_i + 2b_i)/5, -e(3C_r + 2b_r)/5, e(b_i - C_i)/15, e(C_r - b_r)/15), \\ D = d - \ell^{\mathsf{T}} H^{-1}\ell &= r + \frac{2e^2}{525}|C - b|^2 \ \geq r > \frac{2}{5}. \end{aligned}$$

Before volume normalization, this is the positive denominator $\kappa + 2e^2|\Omega||C - b|^2/525 \geq \kappa$. All coefficients follow from the exact moments $\int_0^1 3s^n(1-s)^2 ds = 6/[(n+1)(n+2)(n+3)]$.

Let $p \in \mathbb{R}^4$ be the scalar canonical momenta of the normalized action. The initial cyclic momentum is exactly zero and remains zero. The normalized potential is

$$W(x) = r|C - b|^2 + \tfrac{1}{2}\langle|Z|^2\rangle + \tfrac{1}{8}\langle|Z|^4\rangle, \qquad \langle f\rangle = \int_0^1 3(1-s)^2 f(s)\, ds.$$

For $X = |C|^2$, $Y = \mathrm{Re}(\overline{C}b)$, $B = |b|^2$, the moments are

$$\langle|Z|^2\rangle = X/10 + 3Y/10 + 3B/5, \qquad \langle|Z|^4\rangle = \frac{X^2 + 3XY + 2XB + 4Y^2 + 10YB + 15B^2}{35}.$$

The reduced Hamiltonian is $\mathcal{H} = \frac{1}{2}p^{\mathsf{T}}H^{-1}p + (w^{\mathsf{T}}p)^2/(2D) + W$. Together with cyclic-coordinate reconstruction, its equations are the rational system

$$\begin{aligned} &a = -w^{\mathsf{T}}p/D, && \dot{\alpha} = a, && \dot{x} = H^{-1}p - wa, \\ &\dot{p}_j = a\,\partial_j(w^{\mathsf{T}}p) + \tfrac{1}{2}a^2\partial_j D - \partial_j W. \end{aligned} \tag{13}$$

Its nine initial coordinates are

$$y(0) = (\alpha, x, p) = (0, 1, 0, 1, 0, 0, 3/10, 0, -3/10).$$

They give $\dot{\alpha}(0) = 3/(40r)$, $\dot{C}(0) = i(3 + e\dot{\alpha}(0))$ and $\dot{b}(0) = -i$, exactly the preceding charged temporal-gauge initial condition. The normalized conserved energy is $49/40 + 9/(3200r)$.

Proposition A.8 (Uniform time enclosure for the finite charged action)**.** *For $e = 1/4$, $m^2 = 1/2$, $g = 1/4$, the exact initial-value solution of (13) exists uniquely on $[0, 2]$. The supplied piecewise degree-32 polynomial P, on eighty intervals of exact length $1/40$, satisfies*

$$\sup_{0 \leq t \leq 2} \|y(t) - P(t)\|_\infty \leq 10^{-20}. \tag{14}$$

At shared interval endpoints either adjacent polynomial may be used; the approximants need not join exactly. Under the inverse phase map, the enclosed solution lifts to all 68 real temporal-gauge Euler equations and all 13 Gauss constraints. At the 81 nominal times $j/40$, its five original positions and five original velocities differ from the stored binary64 samples by at most 10^{-10} in each coordinate.

Proof. The Hamiltonian reduction above follows by completing the scalar kinetic square. Its positive Schur denominator makes the rational vector field smooth on all real canonical coordinates. The exact initial cyclic momentum is zero. Canonical equivalence and the full symmetry lift therefore identify the enclosed solution with the same finite-action solution; the compatible initial Gauss law propagates by the action's Noether identity.

Here is the numerical certificate argument, including continuous times. On a step starting from interval enclosure Y_j, the verifier checks a closed box B_j and the strict Picard inclusion

$$Y_j + [0,h]F(B_j) \subset \operatorname{int} B_j, \qquad h = 1/40.$$

Local existence, smoothness and the first-exit argument keep every relevant solution in this box for the whole step. A contraction estimate is not required. Formal power-series substitution in $\dot{y} = F(y)$ produces intervals $A_k(Y_j)$ containing the normalized derivatives $y^{(k)}(t_j)/k!$. The same recurrence initialized on the entire box B_j bounds $y^{(33)}(t)/33!$ at every time in the step. Taylor's theorem consequently gives, for every $u \in [0,h]$,

$$y(t_j + u) \in \sum_{k=0}^{32} A_k(Y_j)u^k + A_{33}(B_j)u^{33}.$$

At $u = h$ this interval is checked to lie in the next recorded enclosure. For each supplied polynomial coefficient $P_{j,k}$, the verifier also bounds $|A_k(Y_j) - P_{j,k}|$ coordinatewise and sums those radii times h^k, together with the remainder radius times h^{33}. Every resulting bound is at most 10^{-20}. Induction over the eighty steps proves existence and the displayed uniform error estimate, including both choices of approximant at a join.

Interval endpoints are integers divided by 2^{256}; all elementary operations round outwards by integer division. The exact rational step is enclosed rather than rounded to a point. An integer square-root inequality encloses $\sqrt{5}$, and each division checks separation from zero. Polynomial coefficients are exact multiples of 2^{-96}. The verifier independently reconstructs the scalar mass block, Schur complement and potential from rational simplex moments, and does not import the producer or a floating-point ODE solver.

For the historical comparison, each stored binary64 value is interpreted as its exact dyadic rational. Interval sine and cosine evaluations use degree-40 Taylor polynomials and the real Lagrange remainder $|\theta|^{41}/41!$. Applying these to $c = e^{-ie\alpha}C$ and its time derivative encloses the original position and velocity coordinates at each nominal sample time. Their checked differences give the stated 10^{-10} bound. The historical samples play no role in constructing or proving the trajectory enclosure. □

The producer, independent verifier and exact interval data are `code/electromagnetism/whitney_charged_enclosure.py`, `code/electromagnetism/verify_whitney_charged_enclosure.py` and `code/electromagnetism/runtime/whitney_charged_enclosure_receipt.json`. Source identities, coefficient encodings, initial data and interpretation are part of the checked packet. The enclosure certifies this finite classical initial-value problem, not the truth of its supplied physical inputs. It neither calibrates the action parameter in seconds nor certifies the separate polygonal Jacobi-clock quadrature, charged spatial continuum limit, observer execution or quantum evolution. The interval argument is implemented and independently replayed; it is not a Lean formalization.

A self-reading execution of the charged action. The five-coordinate invariant sector also admits a bounded software execution with its own local states, ports, records and feedback. Each patch holds one coordinate and its velocity. Five ring ports connect these computational patches; their locations are not identified with physical observers or with points of the supplied cone. The numerical integrator reads the preceding decoded records before each advance. Historical trajectory samples enter only a subsequent comparison.

Proposition A.9 (Exact record-assisted restoration)**.** *Let each patch hold a pair in $\mathbb{Q}^2$. Record its initial pair, then probe a ring edge by replacing both endpoint pairs x, y with $r = (x + y)/2$. Retain the response r, and restore the endpoints using the first endpoint's retained baseline b as $(b, 2r - b)$. Processing the five edges serially restores every patch exactly. Each patch's pair can be recovered from its predecessor's response and baseline and agrees with its own retained baseline.*

Proof. At the first probe, $b = x$, hence $2r - b = y$. Inductively every probe starts from the original endpoint pairs and restores them, so the same identity holds on every edge. Each patch has one predecessor in the ring, which proves the stated decoding rule. □

The implementation in `code/electromagnetism/whitney_charged_instrument.py` encodes numerical coordinates as exact rational representations of binary64 values. Its destructive probes, responses and feedback therefore obey the proposition exactly, while its dynamical advances remain numerical. The recorded execution has 81 decoded states, 405 completed repair cycles and 1782 events. The action time step 1/40, couplings and initial data are supplied. Every advance consumes the previous decoded state; a distinct verifier reconstructs event read versions, write values and hash links, then independently integrates the action from those records. It checks all 68 configuration equations, all 13 Gauss equations, energy and the full coupled field readouts on the decoded samples. These checks retain the scalar basis derivatives.

The public frames expand the decoded coordinates by the proved symmetry map and provide the Whitney connection, electric cochains, complex matter coefficients, local charge and centroid covariant derivatives. They support synchronized visualization of this software computation and its declared field interpretation. The repair-cycle counter measures completed protocol operations. The conversion from a sweep to an action-time step is supplied; it is neither a laboratory calibration nor a derivation of physical time. Hash-pinned replay establishes internal software provenance, without an external witness or an identification of the patches with physical observers. The exact register restoration is separate from numerical trajectory error, spatial convergence and quantum-state evolution.

Certified observer checkpoints. The trajectory enclosure and the authenticated execution admit a direct error comparison at their common declared action times. Let $z(t) \in \mathbb{R}^{10}$ comprise the five original positions and velocities of the exact charged solution, and let d_j be the state decoded from the retained observer events at $t_j = j/40$, $0 \le j \le 80$.

Proposition A.10 (Error of the decoded charged state)**.** *For the supplied action and initial data of Proposition A.8,*

$$\max_{0 \le j \le 80} \|d_j - z(j/40)\|_\infty \le 10^{-10} + \frac{21}{2^{51}} < 1.0001 \times 10^{-10}.$$

Proof. Write s_j for the stored binary64 reference samples, interpreted as exact dyadic rationals. Fresh interval verification proves $\|s_j - z(j/40)\|_\infty \le 10^{-10}$. Independently replaying all 1782 events reconstructs each decoded coordinate from its actual read versions, retained responses and feedback writes. Exact rational comparison over all 810 decoded coordinates gives $\max_j \|d_j - s_j\|_\infty = 21/2^{51}$. The triangle inequality proves the first bound; integer cross-multiplication proves the second. □

The reproducible certificate and independent verifier are `code/electromagnetism/runtime/whitney_charged_checkpoint_receipt.json` and `code/electromagnetism/verify_whitney_charged_checkpoint.py`. Verification recomputes both parent arguments; copied display frames or cached numerical diagnostics cannot supply the decoded state. The bound applies to

decoded position/velocity checkpoints. Intermediate destructive probes need not approximate a physical trajectory, and no continuous observer-state, nonlinear field-readout, Jacobi-quadrature, laboratory-clock or quantum-evolution error bound is asserted.

A clock read from the changing field configuration. A repair count and a time coordinate are different objects. The same autonomous action does, however, supply a dynamical duration functional. This is an application of the classical Jacobi–Maupertuis construction [40], not a new quantization rule. Write a natural mechanical Lagrangian as $L = \frac{1}{2}G_q(\dot{q},\dot{q}) - V(q)$, using the complete coupled kinetic metric, or its Gauss-reduced metric on an admissible horizontal path. Fix an energy E and an oriented regular path in the open region $V < E$. Define

$$d\tau = \sqrt{\frac{G_q(dq,dq)}{2(E-V(q))}}, \qquad J_E[q] = \int \sqrt{2(E-V(q))G_q(q',q')}\, ds. \tag{15}$$

Proposition A.11 (Dynamical duration on a nonturning path)**.** *For smooth positive-definite G and smooth V, the duration above is invariant under orientation-preserving regular reparameterization. A regular stationary path of J_E, parameterized by τ, solves the Euler–Lagrange equations of L with energy E. Conversely, every solution of that energy with positive kinetic energy is such a path, and its τ increments equal its original action-time increments. For fixed G, V, E, positive energy-compatible timing of that oriented path is unique up to its origin.*

Proof. Set $T_s = G_q(q',q')/2$ and $N = \sqrt{T_s/(E-V)}$. Positive reparameterization multiplies both q' and N by the same positive derivative, proving duration invariance. The lapse action

$$I[q,N] = \int \left(T_s/N + N(E-V)\right) ds$$

has lapse equation $T_s/N^2 = E - V$. Its unique positive solution gives $I[q, N(q,q')] = J_E[q]$. Substitution preserves the configuration Euler equation because the lapse variation vanishes. With $d\tau = N\, ds$, that equation is

$$\frac{d}{d\tau}(G_{ij}\dot{q}^j) - \frac{1}{2}\partial_i G_{jk}\dot{q}^j\dot{q}^k + \partial_i V = 0.$$

The lapse equation is the required energy identity. Conversely this identity and the Euler equation give the two equations for I. Uniqueness follows directly from its positive square root. The assertion does not extend through a turning point where numerator and denominator both vanish, or to a constant equilibrium path. □

This construction is covariant under changes of configuration coordinates. For gauge independence one must use the reduced metric or keep the scalar potential in the full covariant tangent. Arbitrary time-dependent gauge changes followed by discarding the scalar potential do not preserve the kinetic energy. The numerical construction below uses the specified temporal-gauge representative throughout. Nor does a Jacobi metric on configuration space itself define a Lorentzian spacetime metric.

There is also a useful approximation statement. Let a fixed C^3 regular path remain in a compact nonturning region, with G, V smooth there. Its polygonal interpolants at parameter spacing at most δ have duration error $O(\delta^2)$. To see this, write the duration integrand as $F(q,v) = \sqrt{G_q(v,v)/(2(E-V(q)))}$. On each short segment, the configuration error is $O(h^2)$, the velocity error is $O(h)$, and the latter has zero integral. Expand F to first order: freezing its velocity derivative at the midpoint cancels that integral, and the derivative's variation and quadratic remainder

both integrate to $O(h^3)$. Summing gives the assertion. Smoothness is used away from zero velocity and $E - V = 0$. This bound compares exact curve integrals; floating-point quadrature and errors in the supplied samples require separate control.

The executable consumer reads only the ordered decoded configuration coordinates from a bounded self-reading software episode. Local patch states, port responses, retained records and feedback restoration belong to that episode's independent provenance check. The clock consumer uses neither the recorded velocities, action timestamps nor repair counts. It retains the same complete scalar-basis derivative in its kinetic form and supplies the initial energy $E = 1969/160 + (493/120)\sqrt{5}$. Piecewise linear interpolation gives a duration on each configuration segment. A separate implementation integrates unrestricted tetrahedral fields instead of the producer's one-dimensional symmetric formulas; a nonlinear reparameterization of each segment checks duration invariance. The producer, verifier and data are `code/electromagnetism/whitney_ephemeris_clock.py`, `code/electromagnetism/verify_whitney_ephemeris_clock.py` and `code/electromagnetism/runtime/whitney_ephemeris_clock_receipt.json`.

The clock is internal to the supplied autonomous model: its rate is reconstructed from change and the action-energy convention. Selecting that action, energy and physical preparation from observer histories, assigning physical units, and comparing this duration with laboratory clocks remain additional tasks. It neither constructs a quantum clock observable nor identifies an arbitrary event ordering with proper time.

Spatial consistency of the same coupled action. The dressed scalar construction admits an action and first-variation estimate on the original solid under spatial refinement. The estimate uses the same edge integrals, Whitney potentials and potential-dependent scalar basis, rather than replacing them by a different finite-element family. Whitney interpolation and its commuting exterior derivative are standard [45]; the additional point here is to control the nonlinear dressing and its full derivative in this particular action.

Let $\mathcal{T}_0$ be the twenty-tetrahedron cone, $I = [0, T]$, and let $\mathcal{T}_\delta$ be conforming refinements respecting every macro-face, with maximum diameter $\delta \le 1$ and a common bound on diameter divided by inradius. Time remains continuous. Give $U = (A, \phi, \psi)$ the norm

$$\|U\|_X = \max_{K \in \mathcal{T}_0} \big(\|U\|_{L^\infty(I; W^{2,\infty}(K))} + \|\partial_t U\|_{L^\infty(I; W^{2,\infty}(K))} \big).$$

Use continuous representatives on each closed macro-cell and require $\phi, \psi \in H^1(\Omega)$ and $A \in H(\mathrm{curl}; \Omega)$ at each time, with the same trace conditions on their time derivatives. The componentwise interpretation applies to complex ψ. Let $a_{ij} = \int_{v_i}^{v_j} A \cdot dx$, take nodal samples of ϕ, ψ, and write $A_\delta = I_1 A$, $\phi_\delta = I_0 \phi$, $\Psi_\delta = \sum_i \lambda_i e^{ie\theta_i} \psi(v_i)$, where $\theta_i = \int_{v_i}^{x} I_1 A \cdot dx = \sum_j a_{ij} \lambda_j$. These paths integrate the interpolated potential, not the original smooth potential. Continuous tangential traces agree on shared faces, hence on their edges; connectivity of the cells incident to an edge makes its oriented integral single valued. Scalar face traces agree as well. No global continuity of the normal component of A is required. Define $S_\delta(U)$ by exact space and time integration of (10) with these fields. Define $S(U)$ by the same continuum scalar-electrodynamics density with A, ϕ, ψ, $E = -\partial_t A - \nabla \phi$ and $B = \mathrm{curl}\, A$.

Theorem A.12 (Smooth-window action consistency)**.** *Fix finite T, R, e, m^2, g, the macro-mesh and a shape-regularity bound. There is a constant C_R, independent of the refinement, such that for every $\|U\|_X \le R$,*

$$|S_\delta(U) - S(U)| + \sup_{\substack{V \in X \\ \|V\|_X \le 1}} |DS_\delta(U)[V] - DS(U)[V]| \le C_R \delta. \tag{16}$$

Variations obey the same trace requirements. The derivative is the full real derivative with respect to all three sampled fields, including the dependence of the scalar basis on the edge potential. This estimate is a consistency statement on bounded smooth test fields; it does not assert convergence of nonlinear trajectories or of arbitrary finite-energy data.

Proof. All following bounds are uniform on each small tetrahedron of diameter h, with constants depending only on the displayed data. Shape regularity gives $|\nabla\lambda_i| \le C/h$. The usual elementary Taylor estimates for nodal interpolation give $\|I_0 f - f\|_\infty \le Ch^2\|f\|_{W^{2,\infty}}$ and $\|\nabla(I_0 f - f)\|_\infty \le Ch\|f\|_{W^{2,\infty}}$. Whitney edge interpolation reproduces constant vector fields, is locally bounded in the sup norm, and commutes with the exterior derivative by Stokes's theorem. Consequently

$$\|I_1 A - A\|_\infty + \|\operatorname{curl}(I_1 A - A)\|_\infty \le Ch\|A\|_{W^{2,\infty}}.$$

For clarity, the curl estimate also follows without a general projection theorem: subtract the affine Taylor polynomial of A. Both its curl and the interpolated curl are constant; their fluxes through every face agree by Stokes's theorem and exact sampled edge circulations. The face normals span $\mathbb{R}^3$, so the curls agree. The remaining edge integrals are $O(h^3)$, while curls of edge basis functions are $O(h^{-2})$. The same estimates apply to $\partial_t A$, and to a variation $V = (B, \eta, \zeta)$, by linearity. Thus the electric and magnetic fields and their variations have $O(h)$ errors.

For the scalar field, antisymmetry of a_{ij} gives the exact cancellation

$$\sum_i \lambda_i\theta_i = \sum_{ij} \lambda_i a_{ij}\lambda_j = 0.$$

The same identity holds after differentiating in space, time or a potential direction. Put $r_i = \psi(v_i) - \psi(x)$ and $q(s) = e^{ies} - 1 - ies$. The exact error decomposition is

$$\Psi_\delta - I_0\psi = \psi(x)\sum_i \lambda_i q(\theta_i) + \sum_i \lambda_i(e^{ie\theta_i} - 1)r_i.$$

Here $\theta_i, \partial_t\theta_i, r_i, \partial_t r_i = O(h)$, their spatial gradients are bounded, and $q(\theta_i) = O(h^2)$, $q'(\theta_i) = O(h)$. Differentiating the displayed identity gives

$$\|\Psi_\delta - \psi\|_\infty + \|\partial_t(\Psi_\delta - \psi)\|_\infty \le C_R h^2, \qquad \|\nabla(\Psi_\delta - \psi)\|_\infty \le C_R h.$$

In particular the time estimate retains the phase velocity, and does not hold by simply deleting it from the definition.

To check that differentiation in the fields preserves these estimates, write $b_{ij} = \int_{v_i}^{v_j} B \cdot dx$ and $\beta_i = \sum_j b_{ij}\lambda_j$. The full derivative is

$$Z_\delta := D\Psi_\delta(U)[V] = \sum_i \lambda_i e^{ie\theta_i}(\zeta(v_i) + ie\beta_i\psi(v_i)).$$

The first term has the scalar interpolation estimates just proved. For the second, use $\sum_i \lambda_i\beta_i = 0$ to replace $e^{ie\theta_i}\psi(v_i)$ by $e^{ie\theta_i}\psi(v_i) - \psi(x) = O(h)$. Since $\beta_i, \partial_t\beta_i = O(h)$ and their spatial gradients are bounded, the resulting sum is $O(h^2)$, its time derivative is $O(h^2)$, and its spatial gradient is $O(h)$. These estimates are proportional to $\|V\|_X$. In particular $Z_\delta - \zeta$ has those same three bounds. Its time derivative includes all four terms

$$\partial_t Z_\delta = \sum_i \lambda_i e^{ie\theta_i}\left[\dot\zeta_i + ie\dot\beta_i\psi_i + ie\beta_i\dot\psi_i + ie\dot\theta_i(\zeta_i + ie\beta_i\psi_i)\right].$$

No second time derivative is used.

The covariant quantities and their variations are

$$Q_\delta = \partial_t \Psi_\delta + ie\phi_\delta \Psi_\delta, \qquad P_\delta = \nabla \Psi_\delta - ieA_\delta \Psi_\delta,$$
$$\delta Q_\delta = \partial_t Z_\delta + ie(I_0\eta)\Psi_\delta + ie\phi_\delta Z_\delta, \qquad \delta P_\delta = \nabla Z_\delta - ie(I_1 B)\Psi_\delta - ieA_\delta Z_\delta.$$

They are uniformly bounded and differ from their continuum counterparts by at most $C_R h$, with the variation bounds proportional to $\|V\|_X$. Finally the action density is $\frac{1}{2}|E|^2 - \frac{1}{2}|B|^2 + |Q|^2 - |P|^2 - m^2|\psi|^2 - g|\psi|^4/2$. Its first variation is the sum of the corresponding bilinear pairings, with matter potential contribution $-2(m^2 + g|\psi|^2)\,\mathrm{Re}(\overline{\psi}\,\zeta)$. On the bounded range just obtained both expressions are Lipschitz in their arguments. Integrating their $C_R h$ differences over $I \times \Omega$, and using $h \le \delta$, proves the result. The estimates hold almost everywhere for the stated Sobolev time regularity; all spatial operations use the Lipschitz representatives on each macro-cell. □

Two controls delimit the argument. On the tetrahedron with vertices

$$0, \qquad (h,0,0), \qquad (0,h,0), \qquad (0,0,h),$$

let $A = \nabla(x_1^2)$. The weighted original-potential path sum is $x_1^2 - hx_1$, whereas the weighted Whitney-potential path sum is exactly zero. Interchanging these two path definitions therefore changes the interpolant. For a smooth bump b supported near an interior vertex with $b(0) = 1$, the fields $b_\epsilon(x) = b(x/\epsilon)$ satisfy $\|b_\epsilon\|_{H^1}^2 = \epsilon^3\|b\|_{L^2}^2 + \epsilon\|\nabla b\|_{L^2}^2 \to 0$, while the nodal value stays one. Thus this nodal construction cannot inherit an estimate for arbitrary three-dimensional H^1 data by point sampling.

`Lean/Screen/WhitneySpatialConsistency.lean` checks the finite antisymmetric cancellation and error-decomposition algebra; the spatial estimate itself is the analytic proof above. The executable refinement probe in `code/electromagnetism/whitney_spatial_consistency.py` uses manufactured nonzero-curvature fields on refinements of the same cone, with independent quadrature and directional finite-difference checks. These numerical checks test the implementation, not the uniform theorem or a physical observer history. The controlled comparison retains a supplied geometry, time coordinate and scalar-electrodynamics action.

A controlled continuum trajectory in the same matter action. The charged action contains a nonlinear real-matter sector for which a spatial solution estimate can be proved. The electric and magnetic fields vanish in this sector; the scalar quartic interaction remains active. The construction first specifies a uniformly regular refinement of the actual solid, then compares its trajectories with a supplied smooth continuum solution. It does not assume that consistency on smooth test functions controls arbitrary finite-element errors.

Proposition A.13 (A uniform conforming cone refinement)**.** *There is a nested conforming refinement $\mathcal{T}_n$, $n = 2^k$, of the twenty-tetrahedron cone with $20n^3$ cells. It respects every macro-face and has*

$$\frac{1}{2n} < \operatorname{diam} K < \frac{6}{n}, \qquad r_K > \frac{1}{12n}, \qquad \frac{\operatorname{diam} K}{r_K} < 72 \quad (K \in \mathcal{T}_n).$$

Here r_K is the inradius. The constants are independent of n.

Proof. Give the macro-vertices one global order. On an ordered tetrahedron (v_0, v_1, v_2, v_3), use cumulative barycentric coordinates

$$x = v_0 + By, \qquad B = (v_1 - v_0,\ v_2 - v_1,\ v_3 - v_2), \qquad 0 \le y_3 \le y_2 \le y_1 \le 1.$$

Intersect this ordered simplex with the Kuhn triangulation of the $1/n$ cubical lattice. In a cube based at b/n, its tetrahedra have vertices

$$\frac{b}{n}, \quad \frac{b+e_{\pi(1)}}{n}, \quad \frac{b+e_{\pi(1)}+e_{\pi(2)}}{n}, \quad \frac{b+(1,1,1)}{n}, \qquad \pi \in S_3.$$

Retain those contained in the ordered simplex. The hyperplanes $y_i = j/n$ and $y_i - y_l = j/n$ describe this triangulation. They include the ordered-simplex boundary and subdivide its interior into disjoint open tetrahedra. Their volume is $1/(6n^3)$, giving n^3 cells per macro-tetrahedron. The hyperplanes at scale n occur at scale $2n$, so these refinements are nested. On a macro-face a cumulative coordinate becomes an endpoint or two adjacent coordinates agree. The induced arrangement is precisely the lower-dimensional construction in the inherited vertex order. Thus both incident macro-cells give the same face subdivision. This is the standard edgewise construction [41]; the coordinate description makes its compatibility explicit here.

Every cell is a translate of $BP\widehat{K}/n$, where P is a coordinate permutation and $\widehat{K}$ is one fixed Kuhn tetrahedron. Its diameter is $\sqrt{3}$ and inradius is $1/(2+2\sqrt{2})$. The centre is the first vertex of every ordered macro-cell. With $\varphi = (1+\sqrt{5})/2$, its matrix satisfies $\|B\|_F^2 = \varphi + 10 < 12$ and $|\det B| = 3+\sqrt{5} > 5$. Hence $\sigma_{\max}(B) < \sqrt{12}$ and $\sigma_{\min}(B) > 5/12$: use $|\det B| \le \sigma_{\min}(B)\sigma_{\max}(B)^2$. Linear maps bound diameters above and below by these singular values, and map an inscribed ball to an ellipsoid containing a ball of radius $\sigma_{\min}(B) r_{\widehat{K}}$. The displayed estimates follow. □

Lemma A.14 (The real sector is invariant under the full action)**.** *On any conforming tetrahedral refinement, set $a = \phi = 0$ and take real scalar coefficients and real velocities. Solutions of*

$$(\ddot{u}_n, v_n) + (\nabla u_n, \nabla v_n) + m^2(u_n, v_n) + g(u_n^3, v_n) = 0 \quad \text{for all } v_n \in V_n \tag{17}$$

solve all equations of the original charged action, for any real e. Here V_n is the continuous piecewise-affine real space on $\mathcal{T}_n$, including its boundary degrees of freedom. All Gauss constraints and electromagnetic fields vanish exactly.

Proof. At $a = 0$ the dressed interpolation is exactly $u_n = W_0 u$. Its variation in any real edge direction is $ie\sum_i \lambda_i\beta_i u_i$, which is purely imaginary. Its time and spatial derivatives are also purely imaginary, including the terms containing the edge-direction velocity and the real scalar velocity. The additional term $-ieB_n u_n$ in the covariant spatial variation is imaginary. Every scalar kinetic, gradient and potential first variation pairs these terms with real fields and takes the real part, giving zero. Variations in imaginary scalar coefficients and in the scalar potential vanish for the same reason. Maxwell terms vanish because both field strengths are zero. Real scalar variations give the displayed wave equation after cancelling the common factor two. This argument tests all variations of the full action; it does not drop the dressing derivative. Uniqueness of the finite classical equations then makes this sector invariant. Its local charge and current vanish, even when $g > 0$ gives nonlinear scalar motion. □

Let Ω be the fixed solid and let $a_1(v,w) = (\nabla v, \nabla w) + (v,w)$. Define the H^1 Ritz projection by

$$a_1(R_n v, v_n) = a_1(v, v_n) \qquad (v_n \in V_n). \tag{18}$$

The added mass term controls the constant mode; no boundary value is fixed by this definition. The uniform mesh gives

$$\|R_n v - v\|_{H^1} \le C n^{-1} \|v\|_{H^2}, \qquad \|R_n v\|_{H^1} \le \|v\|_{H^1}. \tag{19}$$

Indeed, orthogonal projection is a contraction in a_1, and its error is no larger than the piecewise-affine interpolation error. The latter follows by scaling the fixed reference tetrahedra and the usual H^2-to-H^1 affine interpolation estimate. In three dimensions H^2 has continuous representatives, so its nodal values are defined. No L^2 duality estimate or elliptic regularity assertion is needed.

Theorem A.15 (Conditional nonlinear matter trajectory convergence)**.** *Fix $T < \infty$, $m^2 > 0$ and $g \geq 0$. Suppose a real continuum solution satisfies*

$$u_{tt} - \Delta u + m^2 u + g u^3 = 0, \qquad \partial_\nu u = 0 \text{ on } \partial\Omega, \qquad u \in C^2([0,T]; H^2(\Omega)). \tag{20}$$

The equation and boundary condition may equivalently be stated weakly against every $H^1(\Omega)$ test function. Let u_n solve the finite equation with $u_n(0) = R_n u(0)$ and $\dot{u}_n(0) = R_n u_t(0)$. Then

$$\sup_{0\leq t\leq T} \left(\|u_n(t) - u(t)\|_{H^1} + \|\dot{u}_n(t) - u_t(t)\|_{L^2} \right) \leq C_T n^{-1}. \tag{21}$$

The constant depends on the fixed domain, couplings, time window and displayed reference norms, but not on the refinement. These are trajectories of the full charged action in its invariant real sector. The theorem assumes the indicated smooth continuum reference exists.

Proof. The finite equation conserves

$$E_n = \tfrac{1}{2}\|\dot{u}_n\|_2^2 + \tfrac{1}{2}\|\nabla u_n\|_2^2 + \tfrac{m^2}{2}\|u_n\|_2^2 + \tfrac{g}{4}\|u_n\|_4^4.$$

Its initial value is uniformly bounded: use Ritz contraction and the fixed-domain embeddings $H^1 \hookrightarrow L^4, L^6$. Positive m^2 therefore bounds $\|u_n\|_{H^1}$ and $\|\dot{u}_n\|_2$ uniformly on the whole interval. For each finite mesh the positive mass matrix and this bound also give global classical existence of the polynomial ordinary differential equation.

Put $\eta = R_n u - u$, $\theta = u_n - R_n u$. The time-independent bounded projection commutes with both time derivatives. Equation (19) bounds η, η_t, η_{tt} in H^1 by C/n. Subtracting the full continuum weak equation gives, for every $v_n \in V_n$,

$$(\ddot{\theta}, v_n) + a_1(\theta, v_n) = -(\eta_{tt}, v_n) - (m^2 - 1)(\theta + \eta, v_n) - g(u_n^3 - u^3, v_n).$$

The Ritz identity cancelled the spatial derivative of η; there is no inverse estimate on $\dot{\theta}$. The cubic factorization and Hölder's inequality imply

$$\|v^3 - w^3\|_2 \leq C(\|v\|_{H^1}^2 + \|w\|_{H^1}^2)\|v - w\|_{H^1}.$$

Apply this with $v = u_n, w = u$, using the uniform bounds above. Testing with $\dot{\theta}$ and writing $\mathcal{E} = \frac{1}{2}\|\dot{\theta}\|_2^2 + \frac{1}{2}\|\theta\|_{H^1}^2$ yields $\dot{\mathcal{E}} \leq C\mathcal{E} + Cn^{-2}$. The chosen initialization gives $\mathcal{E}(0) = 0$. Gronwall's inequality and the projection estimates prove the claim. For other uniformly energy-bounded initializations, the same proof adds $C_T(\|u_n(0) - R_n u(0)\|_{H^1} + \|\dot{u}_n(0) - R_n u_t(0)\|_2)$ to the right-hand side. □

The smooth-reference class has nonzero localized examples. Choose nonzero C_c^∞ real data in a ball strictly inside the cone and first solve on $\mathbb{R}^3$. The standard local semilinear Klein–Gordon theorem and finite propagation speed [42] give a smooth solution on a positive interval. At the regularity needed here, local existence also follows directly from the Klein–Gordon Duhamel map in $CH^4 \cap C^1 H^3$, since the cubic is locally Lipschitz from H^4 to H^3; the equation gives $u_{tt} \in CH^2$. Take the time window shorter than both this existence interval and the distance of the initial support to the cone boundary. Finite propagation then leaves an open zero-field collar at that boundary,

so restriction to Ω satisfies the stated homogeneous Neumann condition. This uses standard local continuum PDE input, not a source-derived physical state.

This result supplies an actual nonlinear spatial trajectory comparison, with zero electromagnetic current. Electrically active complex matter requires a further stability argument for its field-dependent kinetic metric and constraints. The present theorem supplies neither that argument nor physical geometry, a calibrated clock or a physical matter species.

`code/electromagnetism/whitney_real_continuum.py` constructs the ordered refinements, checks a quadratic Ritz comparator with all boundary variations free, and integrates a compact real pulse under the autonomous finite quartic-wave equation. Its exact polynomial cell integrals are independently reconstructed by the verifier. Finite mesh and numerical time-step comparisons test the implementation; the uniform geometric and continuum error statements are the analytic proofs above, not numerical error certificates or Lean formalizations.

Complex matter in a fixed magnetic background. The dressed scalar action also admits a controlled complex continuum limit with a nonzero magnetic field. Here the electromagnetic potential is prescribed and held fixed in the variation. This is a matter equation in an external field, rather than an invariant sector of the complete matter–Maxwell equations.

On the fixed solid Ω, take

$$A(x) = c + \tfrac{1}{2} B \times x, \qquad \phi = 0, \qquad D_A = \nabla - ieA,$$

where $c, B \in \mathbb{R}^3$ and $e \in \mathbb{R}$ are fixed. Use the conforming uniform meshes $\mathcal{T}_n$, $n = 2^k$, of Proposition A.13. For every oriented edge set $a_{ij} = \int_{v_i}^{v_j} A \cdot dx$. On a tetrahedron define

$$b_i^A(x) = \lambda_i(x) e^{ie\theta_i(x)}, \qquad \theta_i(x) = \sum_j a_{ij} \lambda_j(x), \qquad J_n^A u = \sum_i b_i^A u(v_i).$$

Let V_n^A be the complex space obtained from these functions with one coefficient at each global vertex, including boundary vertices.

Lemma A.16 (Conforming magnetic approximation)**.** *The Whitney interpolant of A equals A on every cell, and $\theta_i = \int_{v_i}^{x} A \cdot dx$. The dressed space V_n^A is a subspace of $H^1(\Omega; \mathbb{C})$. There is a constant independent of n such that*

$$\|J_n^A u - u\|_{H^1} \le C n^{-1} \|u\|_{H^2} \qquad (u \in H^2(\Omega; \mathbb{C})). \tag{22}$$

Proof. Write $A(x) = c + Lx$, where $L^{\mathsf{T}} = -L$. The local Whitney vector space is exactly $\{d + Kx : K^{\mathsf{T}} = -K\}$. To check this directly, $\lambda_i \nabla \lambda_j - \lambda_j \nabla \lambda_i$ has skew linear part, and its six oriented edge integrals give the identity matrix. Thus the six-dimensional space is unisolvent for those integrals, and reproduces A. This is the lowest-order edge space in the usual Whitney construction [45]. Skewness gives $\int_{v_i}^{x} A \cdot dx = A(v_i) \cdot (x - v_i)$, an affine function of x, whose vertex values are a_{ij}. On a shared face, the barycentric functions for vertices off that face vanish. The remaining phases use the same straight paths within the face, and their traces agree. The resulting piecewise smooth functions are continuous and therefore belong to H^1. Their vertex values are the coefficients, so the global basis is linearly independent.

For the approximation bound, fix a cell K of diameter h and a vertex x_0. Put $A_0 = A(x_0)$, $\chi(x) = A_0 \cdot x$, and $w = e^{-ie\chi} u$. Then on that cell

$$J_n^A u = e^{ie\chi(x)} \sum_i \lambda_i(x) e^{ie\delta_i(x)} w(v_i), \quad \delta_i = \int_{v_i}^{x} (A - A_0) \cdot dx.$$

Uniform shape regularity gives $\|\delta_i\|_\infty \le Ch^2$, $\|\nabla\delta_i\|_\infty \le Ch$, and $\|\nabla\lambda_i\| \le C/h$. Consequently, with $r = \sum_i \lambda_i(e^{ie\delta_i}-1)w(v_i)$,

$$\|r\|_{L^2(K)} \le Ch^2|K|^{1/2}\Big(\sum_i |w(v_i)|^2\Big)^{1/2}, \quad \|\nabla r\|_{L^2(K)} \le Ch|K|^{1/2}\Big(\sum_i |w(v_i)|^2\Big)^{1/2}.$$

The constants may absorb the fixed upper bound on cell diameters. Point evaluation on the reference tetrahedron is bounded on H^2 in three dimensions. Scaling therefore bounds the last weighted nodal norm by $C(\|w\|_{L^2(K)} + h\|\nabla w\|_{L^2(K)} + h^2\|D^2 w\|_{L^2(K)})$. The ordinary affine interpolation bound for $I_0 w - w$, together with the estimate for r, gives $\|J_n^A u - u\|_{H^1(K)} \le Ch\|u\|_{H^2(K)}$. Multiplication by $e^{\pm ie\chi}$ has uniformly bounded H^2 operator norm, since A_0 remains bounded on the fixed solid. Sum the squared local estimates and use $h < 6/n$. No point evaluation on arbitrary H^1 data is used. □

For $m^2 > 0$, use the Hermitian form (linear in its first entry)

$$a_A(v,w) = \int_\Omega D_A v \cdot \overline{D_A w} + m^2 v\overline{w}\, dx.$$

Its norm is equivalent to H^1, with constants depending only on $m^2, e, \|A\|_\infty$. Indeed, $\|\nabla v\|_2 \le \|D_A v\|_2 + |e|\|A\|_\infty\|v\|_2$, and the reverse inequality follows by the same triangle inequality. Define the complex magnetic Ritz projection by

$$a_A(R_n^A u, v_n) = a_A(u, v_n) \quad (v_n \in V_n^A).$$

Orthogonal projection, norm equivalence and (22) imply

$$\|R_n^A u\|_{H^1} \le C\|u\|_{H^1}, \qquad \|R_n^A u - u\|_{H^1} \le Cn^{-1}\|u\|_{H^2}. \tag{23}$$

Theorem A.17 (Complex magnetic matter trajectory convergence)**.** *Fix the preceding background, $m^2 > 0$, $g \ge 0$, and a finite time interval $[0,T]$. Suppose a complex reference $u \in C^2([0,T]; H^2(\Omega))$ satisfies*

$$(u_{tt}, v) + a_A(u,v) + g(|u|^2 u, v) = 0 \quad (v \in H^1(\Omega;\mathbb{C})). \tag{24}$$

Here $(v,w) = \int v\overline{w}$. In strong notation this is $u_{tt} + D_A^* D_A u + m^2 u + g|u|^2 u = 0$, with magnetic Neumann condition $\nu \cdot D_A u = 0$. Let $u_n \in V_n^A$ satisfy the same equation against all $v_n \in V_n^A$, with $u_n(0) = R_n^A u(0)$, $\dot u_n(0) = R_n^A u_t(0)$. Then the finite solution exists for every time and

$$\sup_{0\le t\le T} (\|u_n - u\|_{H^1} + \|\dot u_n - u_t\|_{L^2}) \le C_T n^{-1}. \tag{25}$$

The constant is independent of refinement. These are precisely the scalar Euler equations of the original dressed action with A held fixed, $\phi = 0$, and exact spatial integration.

Proof. The fixed basis has a positive definite mass matrix and no phase velocity. Taking the real part of the equation tested with $\dot u_n$ shows conservation of

$$E_n = \tfrac{1}{2}\|\dot u_n\|_2^2 + \tfrac{1}{2}a_A(u_n,u_n) + \tfrac{g}{4}\|u_n\|_4^4.$$

The initial energies are uniformly bounded by Ritz stability and the fixed-domain embedding $H^1 \hookrightarrow L^4$. Coercivity yields uniform H^1 field and L^2 velocity bounds. At each fixed mesh these also prevent finite-time escape of the finite polynomial ordinary differential equation, proving global existence.

Put $\eta = R_n^A u - u$, $\theta = u_n - R_n^A u$. The projection is time independent and bounded; it commutes with both time derivatives. Thus η, η_t, η_{tt} are $O(n^{-1})$ in H^1, uniformly in time. Ritz orthogonality gives the exact error equation

$$(\theta_{tt}, v_n) + a_A(\theta, v_n) = -(\eta_{tt}, v_n) - g(|u_n|^2 u_n - |u|^2 u, v_n).$$

For complex numbers the inequality $\big||z|^2 z - |w|^2 w\big| \le 2(|z|^2 + |w|^2)|z - w|$, followed by Hölder and $H^1 \hookrightarrow L^6$, yields

$$\||v|^2 v - |w|^2 w\|_2 \le C(\|v\|_{H^1}^2 + \|w\|_{H^1}^2)\|v - w\|_{H^1}.$$

All norms in its coefficient are uniformly bounded by the energy and the reference hypothesis. Take $v_n = \dot\theta$, take real parts, and set $\mathcal{E} = \frac{1}{2}\|\dot\theta\|_2^2 + \frac{1}{2}a_A(\theta, \theta)$. Coercivity and Young's inequality give $\dot{\mathcal{E}} \le C\mathcal{E} + Cn^{-2}$. The initial error energy is zero. Gronwall and the projection estimates prove the bound. Uniformly energy-bounded alternative initializations add their initial H^1 field and L^2 velocity distances from the Ritz data, multiplied by C_T. This proof uses no inverse inequality or mesh-dependent stability constant. □

The reference hypothesis admits localized complex examples. Extend the background smoothly with bounded derivatives outside a neighborhood of the solid, without changing it there. In the magnetic wave equation the additional connection and potential terms have at most one spatial derivative. The usual local wave Duhamel contraction in $CH^4 \cap C^1H^3$, with forcing in CH^3, therefore applies to smooth compactly supported complex data. The cubic is locally Lipschitz in this space. The equation gives $u_{tt} \in CH^2$. The standard local wave energy argument gives finite propagation speed one, since the principal part is unchanged [42]. For a short interval before the support reaches the boundary, restriction to the solid has a zero-field boundary collar and obeys the magnetic Neumann condition. For example, $u(0) = f$, $u_t(0) = i\omega f$ with a nonzero real compact bump and $e\omega \neq 0$ has $\mathrm{Im}(\overline{u}u_t) = \omega f^2 \neq 0$. The matter charge need not vanish.

Time independence makes the magnetic mass matrix fixed, and exact Whitney reproduction removes a potential-consistency defect. Those two facts are the essential restrictions of this result. In particular, such charged data generally violate the Gauss equation for $E = 0$; the prescribed field is not varied, and its sources and backreaction are not derived. The theorem does not establish the self-consistent charged continuum or select a physical background, preparation or laboratory clock.

`code/electromagnetism/whitney_magnetic_continuum.py` evaluates nonzero-curvature complex Ritz errors on the same refinements and evolves a finite nonlinear complex pulse. The independent verifier reconstructs the mesh, phases and covariant derivatives using a different quadrature and checks an independent time integration. Gauge changes, shared-face traces and omitted phase derivatives supply falsifying controls. These are finite numerical comparisons with declared tolerances; (25) is an analytic paper proof, not an interval simulation certificate or a Lean formalization. The three pointwise norm estimates in `Lean/Screen/MagneticMatterStability.lean` are Lean-checked; they include the stronger cubic constant 3/2. The Sobolev, Ritz and continuum trajectory arguments above remain analytic proofs.

A Hilbert realization of the coupled finite action. The charged action also admits an interacting quantum Hamiltonian once a quantization prescription is supplied. The kinetic energy depends on the matter field and its gauge-dependent interpolation. Its reduction therefore uses that kinetic metric, rather than the Maxwell mass matrix alone. The construction retains the fixed cone, its unwrapped real edge coefficients, the autonomous scalar action with $m^2, g \ge 0$, and $e \neq 0$.

Write $y = (a, \psi) \in \mathcal{Y} = \mathbb{R}^{42} \times \mathbb{C}^{13}$, regarded as a real manifold. Let $W(a)\psi = \Psi_a$, $S(a,\psi)b = \partial_a(W(a)\psi)[b]$, and define

$$G_y(v,v) = v_a^{\mathsf{T}} M v_a + 2\|W(a)v_\psi + S(a,\psi)v_a\|^2_{L^2(\Omega)}, \tag{26}$$

$$V(y) = \frac{1}{2}a^{\mathsf{T}} K a + \|D_x\Psi_a\|^2_{L^2} + m^2\|\Psi_a\|^2_{L^2} + \frac{g}{2}\|\Psi_a\|^4_{L^4}. \tag{27}$$

The factor two in G follows from the normalization of the scalar kinetic term. Nodal interpolation and $M > 0$ imply that G is a smooth positive metric, including at $\psi = 0$. The potential is smooth and nonnegative.

Theorem A.18 (Gauge reduction and a neutral interacting Hilbert space)**.** *The mean-zero real gauge group has a global smooth quotient*

$$\mathcal{Q} = \ker(D^{\mathsf{T}} M) \times \mathbb{C}^{13} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}.$$

Eliminating the twelve mean-zero scalar-potential coefficients from (10) *defines a smooth positive metric γ on $\mathcal{Q}$. In fixed Euclidean-orthonormal slice coordinates there are constants $c_*, C_* > 0$ such that*

$$\frac{c_*|w|^2}{1+|\psi|^2} \le \gamma_y(w,w) \le C_*(1+|\psi|^2)|w|^2. \tag{28}$$

This metric is complete. With $J_y = (0, i\psi)$ and c the remaining constant scalar potential, the reduced action is

$$L_{\mathrm{red}}(y, \dot{y}, c) = \frac{1}{2}\gamma_y(\dot{y} + ecJ_y, \dot{y} + ecJ_y) - V(y). \tag{29}$$

Its Hamiltonian at $c = 0$ is $H_{\mathrm{cl}}(y,p) = \frac{1}{2}\gamma_y^{-1}(p,p) + V(y)$, with the remaining Gauss constraint $p(J_y) = 0$.

Supply $\hbar > 0$, the Riemannian volume $d\mu = d\mathrm{vol}_\gamma$, and Laplace–Beltrami quantization. Then $\mathcal{H} = L^2(\mathcal{Q}, d\mu)$ has a nonnegative self-adjoint Hamiltonian $\widehat{H}$: the operator $-\hbar^2\Delta_\gamma/2 + V$ on $C_c^\infty(\mathcal{Q})$ is essentially self-adjoint, and its unique self-adjoint extension is its Friedrichs realization. The closed, nonzero subspace

$$\mathcal{H}_0 = \{f \in \mathcal{H} : f(a, e^{i\alpha}\psi) = f(a,\psi) \text{ in } L^2 \text{ for every } \alpha \in \mathbb{R}\} \tag{30}$$

reduces $\widehat{H}$. Its restriction $\widehat{H}_0$ is self-adjoint on $\mathcal{D}(\widehat{H}) \cap \mathcal{H}_0$ and generates the strongly continuous unitary evolution $\exp(-it\widehat{H}_0/\hbar)$. Thus a single interacting finite action supplies both a classical constrained Hamiltonian and, under the declared prescription, a compatible neutral quantum state space.

Proof. Let $\mathfrak{g}_0 = \{\xi \in \mathbb{R}^{13} : \sum_i \xi_i = 0\}$. Its action is $(a,\psi) \mapsto (a + D\xi, e^{ie\xi}\psi)$. Because the incidence matrix has kernel the constants, $D^{\mathsf{T}}MD$ is positive on $\mathfrak{g}_0$. Every orbit therefore meets $\mathcal{Q}$ exactly once: solve $D^{\mathsf{T}}MD\xi = -D^{\mathsf{T}}Ma$ in $\mathfrak{g}_0$. This solution depends smoothly on a. Together with the inverse gauge action it gives a global product description $\mathcal{Y} \cong \mathcal{Q} \times \mathfrak{g}_0$, so no singular quotient is needed for these twelve gauge directions.

For a gauge velocity define $R_y\xi = (D\xi, ie\xi \odot \psi)$. Differentiating the exact interpolation gauge identity gives

$$S(a,\psi)D\xi + W(a)(ie\xi \odot \psi) = ie(W_0\xi)\Psi_a. \tag{31}$$

Consequently the full Lagrangian is exactly $L = G_y(\dot{y} + R_y\phi, \dot{y} + R_y\phi)/2 - V(y)$. This identity retains the derivative of the dressing; discarding it would change both the scalar-potential equation

and the reduced metric. Time-independent gauge transformations are isometries of G and preserve V, by the same pointwise unit-phase covariance.

Choose any basis of $\mathfrak{g}_0$. At $y \in \mathcal{Q}$ let R denote its vertical matrix and put $I_y = R^{\mathsf{T}} G_y R$. The matrix I_y is positive: the Maxwell contribution gives $\xi^{\mathsf{T}} D^{\mathsf{T}} M D \xi > 0$ for nonzero $\xi \in \mathfrak{g}_0$. For $w \in T_y\mathcal{Q}$, set

$$\begin{aligned} \eta_*(w) &= -I_y^{-1} R^{\mathsf{T}} G_y w, \\ \gamma_y(w,w) &= G_y(w,w) - (R^{\mathsf{T}} G_y w)^{\mathsf{T}} I_y^{-1} (R^{\mathsf{T}} G_y w). \end{aligned}$$

Completing the square yields the exact identity

$$G_y(w + R\eta, w + R\eta) = \gamma_y(w,w) + (\eta - \eta_*)^{\mathsf{T}} I_y (\eta - \eta_*).$$

Thus γ is independent of the chosen vertical basis and smooth. If $\gamma_y(w,w) = 0$, positivity of G gives $w = -R\eta_*$. Since w is tangent to the Coulomb slice, $D^{\mathsf{T}} M D \eta_* = 0$, so $\eta_* = 0$ and $w = 0$.

The stronger bound uses the actual dressed basis. On a tetrahedron T, choose a node maximizing $|z_i|$. Where $\lambda_i \geq 3/4$, unit-modulus phases and the triangle inequality give $|W(a)z| \geq (2\lambda_i - 1)|z_i| \geq |z_i|/2$. This region has volume $|T|/64$, so

$$\|W(a)z\|^2_{L^2(T)} \geq \frac{|T|}{1024} |z_T|^2.$$

Summing over tetrahedra gives $\|W(a)z\|^2 \geq c_W |z|^2$, uniformly in all unwrapped a, with $c_W = \min_T |T|/1024 > 0$. The fixed path coefficients also give $\|W(a)\| \leq C$ and $\|S(a,\psi)\| \leq C|\psi|$. Let T_c and B be orthonormal bases of the Coulomb and mean-zero spaces, and write $w = (T_c b, z)$. For every η, the vertical identity and $T_c^{\mathsf{T}} M D = 0$ give

$$\begin{aligned} E_\eta &:= G(w + R_y(B\eta), w + R_y(B\eta)) \\ &= b^{\mathsf{T}} T_c^{\mathsf{T}} M T_c b + \eta^{\mathsf{T}} B^{\mathsf{T}} D^{\mathsf{T}} M D B \eta + 2\|F_\eta\|^2, \\ F_\eta &= Wz + S T_c b + ie(W_0 B\eta)\Psi_a. \end{aligned}$$

Both displayed Maxwell matrices are positive. The remaining two terms in $F_\eta - Wz$ have norm at most $C|\psi|(|b| + |\eta|)$. Consequently $|b|^2 + |z|^2 \leq C(1 + |\psi|^2) E_\eta$ for every η. Taking the minimum proves the lower bound in (28); taking $\eta = 0$ proves the upper bound. Constants concern this fixed mesh. In global coordinates q, the lower bound implies $\gamma \geq c_*(1 + |q|^2)^{-1} I$. An escaping curve therefore has length at least $\sqrt{c_*}$ times its radial arsinh $|q|$ variation and cannot have finite length. Smooth positivity on compact sets then proves completeness.

Write $\phi = \eta + c\mathbf{1}$, with $\eta \in \mathfrak{g}_0$. The twelve η equations are precisely the preceding minimization, with $w = \dot{y} + ecJ_y$. Substitution proves (29). Elimination is variationally valid because the eliminated derivative $\partial L/\partial \eta$ vanishes identically at its unique solution. The Legendre transform gives

$$p = \gamma_y(\dot{y} + ecJ_y, \,\cdot\,), \qquad H(y,p,c) = \frac{1}{2}\gamma_y^{-1}(p,p) + V(y) - ec\, p(J_y).$$

Variation of c, with $e \neq 0$, gives the remaining constraint. It is the constant component of the original thirteen Gauss equations, not an extra neutrality assumption added after quantization.

For the analytic construction use the Hermitian form, initially on $C_c^\infty(\mathcal{Q})$,

$$q(f,k) = \frac{\hbar^2}{2} \int_{\mathcal{Q}} \gamma^{-1}(d\overline{f}, dk)\, d\mu + \int_{\mathcal{Q}} V \overline{f} k \, d\mu. \tag{32}$$

Its domain is dense because μ has a smooth positive density in the global Euclidean coordinates. The form is nonnegative and closable. Indeed, if $f_n \to 0$ in L^2 and f_n is form-Cauchy, its differentials and $\sqrt{V} f_n$ have respective L^2 limits. Integration against compactly supported smooth vector fields, using the metric divergence, shows that the first limit is zero. Local boundedness of V shows that the second limit is zero on every compact set and hence globally. Therefore $q(f_n, f_n) \to 0$, proving closability.

Let $\overline{q}$ be its closure. The closed-form representation theorem gives a unique nonnegative self-adjoint operator for this particular closed form; this is the Friedrichs realization of the displayed differential operator. Explicitly, with the inner product linear in its second argument,

$$\mathcal{D}(\widehat{H}) = \{ f \in \mathcal{D}(\overline{q}) : \exists b \in \mathcal{H} \ \forall k \in \mathcal{D}(\overline{q}),\ \overline{q}(k, f) = \langle k, b \rangle \}, \qquad \widehat{H} f = b.$$

The representation theorem and its symmetry version are standard functional analysis [44]; the reduction and coefficient hypotheses needed here have been verified above. This argument identifies the Friedrichs realization. Completeness strengthens this identification: apply the essential-self-adjointness theorem of Shubin [43, Theorem 1.1], with zero magnetic one-form and scalar potential $2V/\hbar^2$. The manifold and measure are smooth, the metric is complete, and this potential is smooth and nonnegative. Thus the minimal operator has exactly one self-adjoint extension.

The residual circle $T_\alpha(a, \psi) = (a, e^{i\alpha}\psi)$ preserves the slice, its vertical spaces, G, and V. It therefore preserves γ, μ, and q. The pullbacks $\mathcal{U}_\alpha f = f \circ T_{-\alpha}$ are unitary and strongly continuous, first on compactly supported smooth functions by dominated convergence and then on $\mathcal{H}$ by density. They preserve the closed form and its domain. Uniqueness in the displayed operator-domain characterization gives $\mathcal{U}_\alpha \widehat{H} = \widehat{H} \mathcal{U}_\alpha$ with domain preservation. Their bounded group average $P_0 = (2\pi)^{-1} \int_0^{2\pi} \mathcal{U}_\alpha \, d\alpha$ is the orthogonal projection onto $\mathcal{H}_0$ and commutes with the resolvent of $\widehat{H}$. Hence this subspace reduces the operator, and its restriction is self-adjoint. Smooth compactly supported functions of a and $\sum_i |\psi_i|^2$ give nonzero invariant vectors; the subspace is infinite-dimensional. The space $P_0 C_c^\infty(\mathcal{Q})$ is an operator core for the restriction: averaging preserves smoothness, its support lies in the compact circle-saturation of the original support, and averaging is contractive in the graph norm because it commutes with $\widehat{H}$. Apply P_0 to the compactly supported smooth graph approximants supplied by essential self-adjointness. It is also a form core. Spectral calculus supplies the stated strongly continuous unitary group. For initial states in $\mathcal{D}(\widehat{H}_0)$ it solves $i\hbar \partial_t f = \widehat{H}_0 f$ in the strong sense; for every Hilbert state it supplies norm-preserving evolution. □

With the convention $\mathcal{U}_{es} = \exp(-is\widehat{Q}/\hbar)$, the self-adjoint charge generator acts on compactly supported smooth functions as

$$\widehat{Q} = -i\hbar e \sum_i \left(u_i \frac{\partial}{\partial v_i} - v_i \frac{\partial}{\partial u_i} \right), \qquad \psi_i = u_i + i v_i.$$

Thus $\mathcal{H}_0 = \ker \widehat{Q}$ expresses the quantum constant Gauss constraint. It allows matter amplitudes and neutral correlations; it does not freeze the scalar field at zero. The point $\psi = 0$, where the circle has a stabilizer, remains an ordinary point of $\mathcal{Q}$. Taking circle-invariant Hilbert vectors avoids imposing a smooth structure on a singular full orbit space.

Proposition A.19 (A normalized neutral initial state)**.** *In the orthonormal* 56*-dimensional slice coordinates write* $d\mu = \rho(q)\, dq$*, with* $\rho = \sqrt{\det \gamma}$*. For any supplied* $\sigma > 0$*, the function*

$$f_\sigma(q) = \rho(q)^{-1/2} (2\pi\sigma^2)^{-14} \exp\left(-\frac{|q|^2}{4\sigma^2} \right) \tag{33}$$

is a normalized vector in $\mathcal{D}(\widehat{H}_0)$. Its initial multiplication observables satisfy

$$\mathbb{E}\|\Psi_a\|_{L^2}^2 = \frac{4}{5}\sigma^2|\Omega|, \qquad \mathbb{E}\|\Psi_a\|_{L^4}^4 = \frac{48}{35}\sigma^4|\Omega|, \tag{34}$$

$$\mathbb{E}\frac{a^{\mathsf{T}}Ka}{2} = \frac{\sigma^2}{2}\operatorname{tr}(T_c^{\mathsf{T}}KT_c). \tag{35}$$

Proof. The measure $|f_\sigma|^2 d\mu$ is exactly $N(0,\sigma^2 I_{56})$. The residual circle acts orthogonally and preserves ρ, so f_σ is normalized and neutral.

For the operator-domain condition, every derivative of W in a is uniformly bounded, and the corresponding derivatives of S are linear in ψ. The vertical Gram matrix has the fixed positive lower bound $B^{\mathsf{T}}D^{\mathsf{T}}MDB$. Differentiating its inverse and the Schur formula shows polynomial growth of all metric derivatives. The global metric bound gives polynomial growth of γ^{-1} and its derivatives; $\partial_j \log\rho = \frac{1}{2}\operatorname{tr}(\gamma^{-1}\partial_j\gamma)$ gives the same control for derivatives of $\log\rho$. The potential and its derivatives grow polynomially, with $V(q) \le C(1+|q|)^4$. Thus f_σ and the differential expression $(-\hbar^2\Delta_\gamma/2 + V)f_\sigma$ lie in $L^2(d\mu)$: after multiplication by $\sqrt{\rho}$, each differentiated term is bounded by a polynomial times a Gaussian. Integration by parts against C_c^∞ puts the state in the adjoint domain; essential self-adjointness identifies that domain with $\mathcal{D}(\widehat{H})$.

Conditional on a, independent circular Gaussian nodal coefficients give pointwise second and fourth moments $2\sigma^2\sum_i \lambda_i^2$ and $8\sigma^4(\sum_i \lambda_i^2)^2$. The simplex averages of $\sum_i \lambda_i^2$ and $(\sum_i \lambda_i^2)^2$ are $2/5$ and $6/35$, respectively. This proves the matter identities; the magnetic identity follows from the transverse Gaussian covariance. □

This is a selected initial state, without a ground-state, physical-preparation, or computed-time-history assertion.

The metric coefficients and their Schur reduction are approximated by positive element quadrature in `code/electromagnetism/whitney_interacting_quantum.py`; independent tests check the actual cone, gauge covariance, scalar normalization, Gauss elimination, and the nondegenerate zero-matter limit. These finite checks compare exact simplex moments at $a = 0$ and quadrature orders four, five and six at a fixed nonzero configuration. The latter comparison measures quadrature sensitivity, not a certified integration error. The tests do not prove closability or self-adjointness; those are the analytic arguments above. The quantum prescription chooses a measure, operator ordering and $\hbar$; the extension is unique for that operator. The code evaluates Gaussian half-densities and exact moment formulae; its curved-measure amplitudes use the approximate metric density. No equivalence with quantization before reduction, unique quantization, quantum trajectory computation, continuum interacting field theory, physical-state selection, or Born-rule selection follows. The established attachment is to the same finite charged action and its full kinetic metric.

One field readout in the classical and quantum descriptions. The reconstruction of the charged action also fixes a class of observables on its neutral Hilbert space. This identification uses the same dressed field, volume and magnetic reconstruction in both descriptions; it does not identify a classical trajectory with a quantum state.

Use the configuration space $\mathcal{Q}$, metric volume $d\mu$ and neutral Hilbert space $\mathcal{H}_0$ of Theorem A.18. For real $f \in L^\infty(\Omega)$ and a real square-integrable two-form b, define the configuration functions

$$\begin{aligned} I_f(a,\psi) &= \int_\Omega f|\Psi_a|^2\,dx, \\ J_f(a,\psi) &= \int_\Omega f|\Psi_a|^4\,dx, \\ B_b(a,\psi) &= \int_\Omega \langle b, d(W_1 a)\rangle\,dx. \end{aligned} \tag{36}$$

The form pairing and volume use the supplied cone metric. These functions are finite and real on every configuration and are invariant under the nodal gauge transformations. In particular, they are invariant under the residual circle on $\mathcal{Q}$. Spatial smearings specify which field information is read; they do not supply a physical detector model.

Theorem A.20 (A common reconstructed observable algebra)**.** *Let $\mathcal{O} = (O_1, \dots, O_k) : \mathcal{Q} \to \mathbb{R}^k$ be any finite list of functions from (36). Each multiplication operator*

$$\widehat{O}_j F = O_j F, \qquad \mathcal{D}(\widehat{O}_j) = \{F \in \mathcal{H}_0 : O_j F \in L^2(\mathcal{Q}, d\mu)\}$$

is self-adjoint on $\mathcal{H}_0$. They strongly commute, with joint spectral projection-valued measure

$$\mathsf{P}(B)F = \mathbf{1}_{\mathcal{O}^{-1}(B)} F, \qquad B \subseteq \mathbb{R}^k \text{ Borel.} \tag{37}$$

For every normalized $F \in \mathcal{H}_0$, its spectral probability law is

$$\nu_F(B) = \langle F, \mathsf{P}(B)F \rangle = \int_{\mathcal{O}^{-1}(B)} |F(q)|^2 \, d\mu(q). \tag{38}$$

For every bounded real Borel detector function $d : \mathbb{R}^k \to \mathbb{R}$, the operator $d(\widehat{\mathcal{O}})$ is multiplication by $d \circ \mathcal{O}$. Thus the classical readout $d(\mathcal{O}(q))$ and its quantum spectral law use exactly the same configuration function. No ground-state assumption is required.

Proof. The dressed interpolation is smooth in its finite coefficient variables, with unit-modulus phases, and the cone has finite volume. On compact coefficient sets its scalar fields and their coefficient derivatives are uniformly bounded. Dominated convergence proves continuity of I_f, J_f; B_b is a continuous linear function of a. Gauge covariance of Ψ_a and $d^2 = 0$ prove their invariance.

For any one such real function O, truncation by $\mathbf{1}_{\{|O| \le n\}}$ approximates every neutral Hilbert vector and lies in the multiplication domain. This proves density. The operator is symmetric. Multiplication by $(O \pm i)^{-1}$ is bounded and preserves neutrality; for every $F \in \mathcal{H}_0$ it gives a domain vector mapped to F by $\widehat{O} \pm i$. Both resolvents are therefore everywhere defined, proving self-adjointness. The same argument on $\mathcal{H}$ gives its full space realization. Invariance of O makes every spectral projection commute with the residual-circle action and its neutral projection, so $\mathcal{H}_0$ is a reducing subspace.

The indicator multipliers in (37) are orthogonal projections. Disjoint countable unions give strong countable additivity by dominated convergence, and $\mathsf{P}(\mathbb{R}^k) = I_{\mathcal{H}_0}$. Its coordinate spectral integrals have precisely the stated multiplication domains. This proves joint spectrality and strong commutation. Integrating a bounded Borel function proves the detector formula and (38). □

For any initial normalized neutral state, the established unitary group gives $F_t = e^{-it\widehat{H}_0/\hbar} F$, and the same law defines ν_{F_t} at every model time. This defines time-dependent probabilities without claiming that those probabilities have been numerically computed. For the Gaussian half-density in Proposition A.19, all polynomial coefficient moments are finite. In particular, the two exact scalar moments in (34) are the spectral expectations of $\widehat{I}_1$ and $\widehat{J}_1$ from this construction.

The charged field Ψ_a itself is not circle invariant; its multiplication operator does not in general preserve $\mathcal{H}_0$. Consequently the theorem concerns neutral field readouts, rather than asserting that every gauge-fixed coefficient is a physical observable. It also supplies no momentum-ordering prescription for a local quantum current or kinetic-energy density. The spectral probability rule is that of the declared Hilbert realization; its identification with experimental outcomes remains a physical measurement assumption.

Proposition A.21 (Continuum convergence of neutral detector readouts)**.** *Under the hypotheses and initialization of Theorem A.15, fix finitely many real bounded spatial smearings. The scalar readouts*

$$I_f(u_n) = \int_\Omega f u_n^2 \, dx, \qquad J_f(u_n) = \int_\Omega f u_n^4 \, dx$$

converge to the corresponding continuum readouts uniformly on $[0,T]$ *at rate* $O(n^{-1})$*. Every fixed Lipschitz function of this finite readout vector has the same rate. In particular, bounded Lipschitz detector responses have a controlled classical continuum limit.*

Proof. The trajectory theorem bounds u_n, u uniformly in H^1, and their difference there is $O(n^{-1})$. Hence

$$|I_f(u_n) - I_f(u)| \le \|f\|_\infty (\|u_n\|_2 + \|u\|_2)\|u_n - u\|_2.$$

Factorization of the fourth powers and Hölder's inequality give

$$|J_f(u_n) - J_f(u)| \le C\|f\|_\infty (\|u_n\|_4^3 + \|u\|_4^3)\|u_n - u\|_4.$$

Use the fixed-domain embedding $H^1 \hookrightarrow L^4$ and then the Lipschitz inequality for the detector function. □

The continuum corollary concerns the full action's invariant real sector, where electromagnetic fields and current vanish. It proves no convergence of the quantum states or spectral measures across refinements. No quantum–classical state correspondence, calibrated detector, physical spacetime identification or source-selected action follows from sharing the observable functions. The common input is the stated cone action and its reconstruction and quantization rules.

Comparing quantum readouts at fixed energy and resolution. The common neutral observable algebra permits a precise comparison with a declared effective field model. A fixed ultraviolet regulator is allowed; the statement below does not require an interacting limit at arbitrarily high energy. It separates a generator estimate, a state map, and a detector estimate. None of those comparison estimates follows from a classical invariant sector alone.

Let $H_{\mathrm{e}} \ge 0$ and H_h be self-adjoint operators on separable complex Hilbert spaces $\mathcal{H}_{\mathrm{e}}$ and $\mathcal{H}_h$. Write $U_j(t) = \exp(-itH_j/\hbar)$, with $\hbar > 0$, and let $J : \mathcal{H}_{\mathrm{e}} \to \mathcal{H}_h$ be a linear isometry. Fix $E > 0$ and the *fixed spectral projection* $P = \mathbf{1}_{[0,E]}(H_{\mathrm{e}})$. Thus P commutes with $U_{\mathrm{e}}(t)$ and $H_{\mathrm{e}}P$ is bounded. A neutral comparison uses neutral Hilbert spaces and effects throughout.

Theorem A.22 (Energy-window effect comparison)**.** *Suppose*

$$J \operatorname{Ran} P \subset \mathcal{D}(H_h), \qquad \|(H_h J - J H_{\mathrm{e}})P\| \le \epsilon_H. \tag{39}$$

The residual is a bounded map into the full *target Hilbert space; its component orthogonal to* $J \operatorname{Ran} P$ *is included. Let* $0 \le F_h \le I$ *and* $0 \le F_{\mathrm{e}} \le I$ *be bounded effects with*

$$\|P(J^* F_h J - F_{\mathrm{e}})P\| \le \epsilon_D. \tag{40}$$

Let ρ *and* ρ_h *be density operators, and put* $D(\alpha,\beta) = \frac{1}{2}\|\alpha - \beta\|_1$*. Assume*

$$D(\rho_h, J\rho J^*) \le \epsilon_0, \qquad \eta_\rho := \operatorname{Tr}((I-P)\rho) \le \eta < 1, \qquad \epsilon_0, \epsilon_H, \epsilon_D, \eta \ge 0.$$

For every real t*, the probabilities*

$$p_h(t) = \operatorname{Tr}\big(F_h U_h(t) \rho_h U_h(t)^*\big), \qquad p_{\mathrm{e}}(t) = \operatorname{Tr}\big(F_{\mathrm{e}} U_{\mathrm{e}}(t) \rho U_{\mathrm{e}}(t)^*\big)$$

obey

$$|p_h(t) - p_{\mathrm{e}}(t)| \le \min\left\{1,\ \epsilon_0 + 2\sqrt{\eta} + \frac{|t|\epsilon_H}{\hbar} + \epsilon_D\right\}. \tag{41}$$

In particular, an exactly energy-supported preparation has no tail term. The same bound is uniform over any declared detector class satisfying (40) *with one constant.*

Proof. Put $\mathcal{K} = \operatorname{Ran} P$ and $A = H_{\mathrm{e}}|_{\mathcal{K}}$. The spectral theorem gives $\|A\| \le E$. For $J_E = J|_{\mathcal{K}}$, the residual $R = H_h J_E - J_E A$ is bounded by hypothesis. Consequently $H_h J_E = J_E A + R$ is bounded from $\mathcal{K}$ to $\mathcal{H}_h$; in particular J_E is continuous into the operator domain equipped with its graph norm. For $x \in \mathcal{K}$ and fixed $t \ge 0$, the path

$$s \longmapsto U_h(t-s) J_E e^{-isA/\hbar} x$$

is strongly differentiable on $[0,t]$, with derivative

$$\frac{i}{\hbar} U_h(t-s) R e^{-isA/\hbar} x.$$

Indeed, the bounded A orbit is differentiable; its image lies in $\mathcal{D}(H_h)$, and its Hamiltonian image is continuous by the displayed bounded expression for $H_h J_E$. The usual difference quotient for the product therefore uses the strong derivative of U_h on its operator domain. This verifies the domain step for an unbounded H_h; it does not assume operator-norm differentiability of U_h. Integration and unitarity give

$$\|(U_h(t)J - JU_{\mathrm{e}}(t))P\| \le |t|\epsilon_H/\hbar. \tag{42}$$

The same argument on the reversed interval proves the result for $t < 0$.

Let $p = \operatorname{Tr}(P\rho) = 1 - \eta_\rho > 0$ and $\rho_E = P\rho P/p$. A purification Ψ of ρ has normalized projected purification $\Psi_E = (P \otimes I)\Psi/\sqrt{p}$ with overlap $\sqrt{p}$. For unit vectors x, y, diagonalization on their two-dimensional span gives

$$D(|x\rangle\langle x|, |y\rangle\langle y|) = \sqrt{1 - |\langle x, y\rangle|^2} \le \|x - y\|.$$

Contractivity under partial trace proves the normalized conditioning bound

$$D(\rho, \rho_E) \le \sqrt{\eta_\rho} \le \sqrt{\eta}. \tag{43}$$

This argument retains coherences across the energy cut; no commutation of ρ with P is assumed.

Apply (42) to a purification of ρ_E. Tensoring a bounded Hilbert-space operator with the identity preserves its norm. The two evolved purifications therefore differ by at most $|t|\epsilon_H/\hbar$, and their reduced states have at most this trace distance. Initial preparation contributes ϵ_0 by unitary invariance. Replacing ρ by ρ_E on each side contributes at most $2\sqrt{\eta}$ by (43), isometric embedding, and unitary invariance.

Finally $U_{\mathrm{e}}(t)\rho_E U_{\mathrm{e}}(t)^*$ is supported in P. Its expectations of $J^* F_h J$ and F_{e} differ by at most ϵ_D. For any two density operators and any effect $0 \le F \le I$, $|\operatorname{Tr} F(\alpha - \beta)| \le D(\alpha, \beta)$: the positive and negative parts of the trace-zero self-adjoint difference each have trace $\|\alpha - \beta\|_1/2$. These facts and the triangle inequality give the second bound in (41); both probabilities lie in $[0,1]$, giving the first. □

If a preparation has the form-energy bound $\operatorname{Tr}(H_{\mathrm{e}}^{1/2} \rho H_{\mathrm{e}}^{1/2}) \le \overline{E}$, then $\eta_\rho \le \overline{E}/E$. This follows by integrating $\mathbf{1}_{(E,\infty)}(\lambda) \le \lambda/E$ against its spectral probability measure. A finite sth spectral moment similarly gives $\eta_\rho \le M_s/E^s$ for $s > 0$. Such tail bounds require their own certificates; a Gaussian label or a spatial bandwidth alone supplies none.

Why compression is insufficient. Take $\mathcal{H}_{\mathrm{e}} = \mathbb{C}$, $H_{\mathrm{e}} = 0$, $\mathcal{H}_h = \mathbb{C}^2$, $Jz = (z, 0)$, and $H_h = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$. The compressed residual $J^* H_h J - H_{\mathrm{e}}$ vanishes. The full residual $H_h J$ has norm one. With $F_{\mathrm{e}} = I$, $F_h = JJ^*$ and the exactly mapped initial pure state, the probabilities are 1 and $\cos^2(t/\hbar)$. They differ by one at $t = \pi\hbar/2$. Thus a compressed Galerkin matrix can pass an exact algebraic comparison while its omitted coupling changes the observed probability completely. Lean verifies the interaction-picture norm estimate, the constant-one bounded-effect inequality and this explicit two-level control. The spectral-domain and mixed-state trace-class steps above are analytic proofs.

For the charged Whitney Hamiltonian, the self-adjoint neutral operator and bounded neutral detector functions are available from Theorems A.18 and A.20. Applying this comparison additionally requires an effective Hamiltonian, an isometry respecting the intended preparations, and certified residual, detector and tail bounds on that same pair of models. The full curved kinetic operator enters the residual. Neither its coefficients nor couplings to omitted modes may be discarded. The theorem supplies no numerical comparison accuracy, quantum refinement limit, experimental identification or source-selected action.

The spacetime derivation paper constructs a complementary massive scalar comparison on prepared golden source addresses. A declared tensor action has a full oscillator Fock realization and compact, separated preparation and detector smearings. Spatial refinement with symmetric local time steps shrinking faster than the minimum coordinate gap gives the same free-field detector limit. At $q = 233$ and step 2^{-17}, reading the nearest update within dimensionless time $[0.95, 1]$ gives probability error below 0.050078 and response above 0.137746 throughout the window. The bound includes full frequency leakage, both localization tails, time stepping and the original-vacuum covariance change. These are mathematical update trajectories, not an executed tensor history. The action, boundary, protected addresses, quantization and model time are declared. The charged Whitney interaction, native observer realization and physical clock attachment retain their own hypotheses.

A $q = 5$ classical execution composes the declared scalar action with local averaging probes, retained records, restoration feedback and routed readout. Subsequent field updates consume the actual restoration writers. Across $[15\tau, 17\tau]$, where $\tau^2 = 1/245$, the paired separated response exceeds 0.01576 while the total comparison error is below 0.01364. The bound includes both preparations, feedback error propagated into later states, finite-action time stepping, detector error, causal holding and clock uncertainty. All auxiliary events are retained. Source addresses, action, exact routing and classical control operations remain supplied, and the clock is a model parameter. This finite comparison establishes no physical source selection or quantum outcome law.

A trial quantum history with a global norm bound. The complete reduced metric also permits an explicit, error-controlled trial history in the full interacting Hilbert space. The bound below is deliberately conservative: its time interval is extremely short in the supplied dimensionless units. It does not furnish an ordinary-physics benchmark, a physical clock, or an exact computed Hamiltonian trajectory.

Retain the same fixed cone and set $e = 1/4$, $m^2 = 1/2$, $g = 1/4$, $\hbar = 1$, and $\sigma = 1/2$. Let H be the neutral Hamiltonian of Theorem A.18 and $f = f_\sigma$ the normalized state of Proposition A.19. All configuration coordinates remain present: $q = (a, \psi) \in \mathcal{Q} \cong \mathbb{R}^{30} \times \mathbb{C}^{13}$, with Euclidean-orthonormal coordinates on the Coulomb subspace. Write $X = |a|^2$, $Y = |\psi|^2$, and $\rho = \sqrt{\det \gamma}$.

Theorem A.23 (Controlled potential-phase trial)**.** *The normalized neutral trial path*

$$v(t, q) = e^{-itV(q)/\hbar} f(q)$$

belongs to $\mathcal{D}(H)$ and is differentiable in the Hilbert norm. If $u(t) = e^{-itH/\hbar}f$, then

$$\|u(t) - v(t)\|^2_{L^2(d\mathrm{vol}_\gamma)} \le \frac{(\overline{E} + \overline{K})|t| + \overline{B}|t|^3/6}{\hbar}, \tag{44}$$

where the following rational constants are admissible:

$$\overline{K} = 135829605095437, \qquad \overline{E} = \frac{2173273682146345}{16}, \qquad \overline{B} = \frac{24441816695556087}{4}.$$

In particular, $\|u(t) - v(t)\| \le 1/10$ for every $|t| \le 2^{-55}$. The trial configuration probability measure is exactly $N(0, \sigma^2 I_{56})$ at every time; its phase changes, while its configuration density stays fixed.

Proof. The argument separates a form comparison from explicit coefficient bounds. Define

$$K_0 = \frac{\hbar^2}{2}\int |df|^2_{\gamma^{-1}}\, d\mu, \qquad E_0 = K_0 + \int V|f|^2\, d\mu, \qquad B_0 = \int |dV|^2_{\gamma^{-1}}|f|^2\, d\mu.$$

The polynomial derivative bounds used for the initial state also apply to $e^{-itV/\hbar}f$ on bounded time intervals. Thus $v(t)$, its formal Hamiltonian image, and $Vv(t)$ lie in $L^2(d\mu)$. The adjoint-domain characterization and essential self-adjointness put $v(t)$ in $\mathcal{D}(H)$; differentiation under its Gaussian envelope gives $i\hbar\dot{v} = Vv$. Gauge invariance of V gives neutrality, and the phase has unit modulus.

Let $\mathfrak{t}$ denote the kinetic quadratic form. Since f is real and positive, $\mathfrak{t}(v(s), v(s)) = K_0 + s^2 B_0/2$. Positivity of V and conservation of the exact Hamiltonian energy give $\mathfrak{t}(u(s), u(s)) \le E_0$. Differentiating the overlap and using the form Cauchy inequality yields

$$\left|\frac{d}{ds}\|u(s) - v(s)\|^2\right| \le \frac{2}{\hbar}\sqrt{E_0}\sqrt{K_0 + s^2 B_0/2} \le \frac{E_0 + K_0 + s^2 B_0/2}{\hbar}.$$

Integration proves (44) for any valid upper bounds on these three constants. This comparison uses the full kinetic form and its Riemannian measure.

The canonical geometry has volume at most 18, tetrahedron volumes at least 5/6, squared edge lengths at most 4, and $|\nabla\lambda_i| \le 2$. Every node belongs to at least five tetrahedra and every edge to at least two. These inequalities are checked in $\mathbb{Q}(\sqrt{5})$ from the canonical vertices and incidence lists. They imply the useful uniform bounds

$$M \ge \frac{1}{81}I, \qquad W(a)^* W(a) \ge \frac{1}{128}I. \tag{45}$$

For the first, an affine vector field F on a tetrahedron satisfies $\int_T |F|^2 \ge |T| \sum_i |F(v_i)|^2/20$, while its six edge integrals obey $\sum_{ij} |a_{ij}|^2 \le 6 \sum_i |F(v_i)|^2$. Summing gives $M \ge 1/72$, which implies the displayed bound. For the second, choose a largest local nodal coefficient $|z_i|$. On $\lambda_i \ge 1/2$, $|Wz| \ge (2\lambda_i - 1)|z_i|$, independently of the phases, and

$$\frac{1}{|T|}\int_{\lambda_i \ge 1/2} (2\lambda_i - 1)^2\, dx = 3\int_{1/2}^1 (2s-1)^2(1-s)^2\, ds = \frac{1}{80}.$$

Thus the summed lower bound is 5/384, which exceeds 1/128.

The radial edges give $|D\xi|^2 \ge |\xi|^2$ for mean-zero ξ: their contribution is $\sum_{i=1}^{12}(\xi_i - \xi_0)^2 = |\xi|^2 + 13\xi_0^2$. The mean-zero vertical Gram matrix I consequently satisfies $I \ge 1/81$. For a slice velocity (b, z) and mean-zero gauge velocity η, let $E_\eta = G(w + R\eta, w + R\eta)$. Maxwell orthogonality

gives $|b|^2+|\eta|^2 \le 81E_\eta$. The vertical interpolation identity and $\|W\| \le \sqrt{18}$, $\|S\| \le \sqrt{18}|\psi|/4$ give $\|Wz\|^2 \le (1+729Y/2)E_\eta$. Applying (45) and minimizing over η proves

$$\gamma^{-1} \le A(Y)I, \qquad A(Y) = 209 + 46656Y. \tag{46}$$

The density derivative can be bounded without differentiating a matrix inverse twice. The combined slice and vertical tangent matrix has constant determinant: after permuting columns its edge block is the fixed matrix $[T_c, DB]$ and its scalar diagonal block is the identity. Schur factorization therefore gives $\log\rho = \text{constant} + \frac{1}{2}\log\det G - \frac{1}{2}\log\det I$. Put $J = (S, W)$. For any full velocity $w = (w_a, w_\psi)$, (45) implies

$$|w_a| \le 9\sqrt{G(w,w)}, \qquad |w_\psi| \le (8+108|\psi|)\sqrt{G(w,w)}.$$

Each real phase-path row has Euclidean norm at most one. Differentiating J and using $2\|Jw\|^2 \le G(w,w)$ yields, for a configuration direction h,

$$|dG[h](w,w)| \le 3(17+441|\psi|/4)|h|G(w,w).$$

For $\chi = W_0 B$, the identity $I = I_0 + 2e^2\int \chi^{\mathsf{T}}\chi|\Psi_a|^2$ similarly gives $|dI[h](\eta,\eta)| \le 27(1+|\psi|/4)|h|I(\eta,\eta)$. Taking traces in dimensions 68 and 12 proves

$$|\nabla\log\rho| \le 1896 + 11286\sqrt{Y}. \tag{47}$$

All gradients here are in the orthonormal slice coordinates; bounds for unit full-configuration directions restrict to those coordinates.

For completeness, the remaining potential bounds follow directly from the same geometry. Whitney curl has squared operator bound 384 on each cell. The pointwise estimates

$$|\Psi_a| \le |\psi|, \qquad |D_x\Psi_a| \le (4+9|a|/4)|\psi|,$$
$$|d(D_x\Psi_a)[b,z]| \le (4+9|a|/4)|z| + (13/4+9|a|/16)|\psi||b|$$

retain the derivative of the gauge-dependent dressing. The phase-gradient bound uses $|\nabla p_i\cdot a| \le 4|a|$ and $|\mathbf{A}(a)| \le 5|a|$. Squaring and integrating gives

$$V \le V_*(X,Y) := 3456X + 585Y + \frac{729}{4}XY + \frac{9}{4}Y^2,$$
$$|\nabla V|^2 \le P(X,Y) := 4\cdot 6912^2 X + 2313^2 Y^2 + 4\cdot 1152^2 X^2Y^2 + \frac{81}{4}Y^4 + 3\cdot 1170^2 Y + 3\cdot 1152^2 X^2 Y + 243Y^3.$$

For example, the vector and scalar gradient norms are bounded by $6912\sqrt{X}+(2313/2)Y+1152XY+(9/4)Y^2$ and $1170\sqrt{Y}+1152X\sqrt{Y}+9Y^{3/2}$; Cauchy's inequality for four and three summands gives the displayed P.

Under $|f|^2 d\mu$, the independent variables X, Y have laws $\sigma^2\chi^2_{30}$ and $\sigma^2\chi^2_{26}$. The Gaussian integration-by-parts recurrence gives exactly

$$\mathbb{E}[(\sigma^2\chi^2_d)^k] = \sigma^{2k}\prod_{j=0}^{k-1}(d+2j).$$

Since $df = -f(d\log\rho + q\cdot dq/\sigma^2)/2$, admissible constants are the finite polynomial expectations

$$\overline{K} = \frac{1}{4}\mathbb{E}\Big[A(Y)\{2\cdot 1896^2 + 2\cdot 11286^2 Y + 16(X+Y)\}\Big],$$
$$\overline{E} = \overline{K} + \mathbb{E}[V_*] = \overline{K} + \frac{619353}{16}, \qquad \overline{B} = \mathbb{E}[A(Y)P(X,Y)].$$

Exact rational evaluation gives the stated values. The right side of (44) is increasing in $|t|$; at 2^{-55} it is less than $0.007541 < 1/100$, whereas the preceding dyadic time 2^{-54} fails that target. This proves coverage of the entire stated interval, including negative times. □

The packet `code/electromagnetism/runtime/whitney_quantum_history_receipt.json` records five trial times and four configuration probes. Potential values and the relative phase angles at those probes are exact elements of $\mathbb{Q}(\sqrt{5})$; an independent verifier reconstructs their simplex integrals and all envelope constants. The sampled phases illustrate the trial function; the norm bound concerns the full Hilbert space. Global polynomial integration includes the entire Gaussian tail, with no Monte Carlo, quadrature, or floating-point allowance used to certify the error. A separate numerical evaluator retains the full metric density and marks its approximate coefficients explicitly. The certificate inherits the supplied action, geometry, quantum prescription, initial state and time units. Its tiny conservative horizon is not physical state preparation, authenticated observer activity, spatial convergence, or an interacting continuum quantum field theory.

Charged phase-space data and neutral Gaussian packets. The charged trajectory can supply packet parameters only after its positions and velocities are expressed in the Coulomb quotient. This is a preparation construction in the full interacting Hilbert space of Theorem A.18. A family of prepared packets at classical sample times does not constitute quantum propagation.

Let B have orthonormal columns spanning the mean-zero gauge space, let T_c span $\ker(D^{\mathsf{T}}M)$ orthonormally, and put $L_0 = B^{\mathsf{T}}D^{\mathsf{T}}MDB$. For a differentiable full configuration $y(t) = (a(t), \psi(t))$, define

$$\begin{aligned} \xi &= -BL_0^{-1}B^{\mathsf{T}}D^{\mathsf{T}}Ma, & \dot\xi &= -BL_0^{-1}B^{\mathsf{T}}D^{\mathsf{T}}M\dot a,\\ a_C &= a + D\xi, & \psi_C &= e^{ie\xi}\psi,\\ \dot a_C &= \dot a + D\dot\xi, & \dot\psi_C &= e^{ie\xi}(\dot\psi + ie\dot\xi\odot\psi). \end{aligned} \tag{48}$$

Products and exponentials on nodal coefficients are componentwise. The quotient coordinates are $q = (T_c^{\mathsf{T}}a_C, \operatorname{Re}\psi_C, \operatorname{Im}\psi_C) \in \mathbb{R}^{56}$, with velocity $w = \dot q$. The unit phase generator is $J(a,u,v) = (0,-v,u)$; its parameter is the phase angle, without the charge factor e.

Proposition A.24 (Coulomb phase-space preparation)**.** *For a temporal-gauge trajectory satisfying all thirteen Gauss equations, the transformed scalar potential is $\phi_C = -\dot\xi$, its constant component vanishes, and the Schur minimizer is exactly $-\dot\xi$. The reduced cotangent is $p = \gamma_q w$ and satisfies $p(Jq) = 0$. For the recorded symmetric sector, with radial coefficient α,*

$$\xi = \frac{\alpha}{13}(12,-1,\dots,-1), \qquad a_C = 0. \tag{49}$$

At its exact initial datum, with $\mathcal{V} = |\Omega| = 10 + 10\sqrt{5}/3$,

$$q_a = p_a = 0, \quad \psi_C = \mathbf{1}, \quad p_{\operatorname{Re}\psi} = 0, \quad p_{\operatorname{Im}\psi} = \mathcal{V}(3/10, -1/40, \dots, -1/40). \tag{50}$$

Proof. The gauge law for the temporal covariant derivative is $\phi \mapsto \phi - \dot{\xi}$; differentiating the nodal gauge law gives (48). The transformed velocity plus vertical scalar-potential term is the image of the original velocity under the time-independent gauge differential. Gauge covariance therefore preserves every scalar-potential equation. Positivity of its mean-zero Hessian makes $-\dot{\xi}$ the unique minimizer. Since $\sum \xi_i = 0$, the constant potential remains zero. The reduced Legendre formula and constant Gauss equation give the cotangent and its zero moment map. Without Gauss, the displayed rechart remains valid but the equality of its transformed potential and Schur minimizer need not hold.

The radial incidence equation is $\xi_j - \xi_0 = -\alpha$, and its mean-zero solution gives (49). Initially the edge dressing derivative vanishes because all nodal coefficients equal one. Write A_0 for the scalar nodal mass matrix. The transformed covariant velocity is the original $i(3, -1, \dots, -1)$, so the scalar cotangent is $2A_0 i(3, -1, \dots, -1)$. Simplex integration gives $(A_0)_{00} = \mathcal{V}/10$, $(A_0)_{0j} = \mathcal{V}/80$, and the boundary row sum $\mathcal{V}/16$. These identities prove (50). Maxwell orthogonality $T_c^{\mathsf{T}} M D = 0$ gives the zero edge cotangent. Its constant charge is $\mathcal{V}(3/10 - 12/40) = 0$. □

Supply a width $\sigma > 0$, momentum p, center q, and $\hbar > 0$. In Lebesgue measure on all 56 quotient coordinates, let the normalized seed be

$$g_{q,p,\sigma}(x) = (2\pi\sigma^2)^{-14} \exp\left(-\frac{|x-q|^2}{4\sigma^2} + \frac{i}{\hbar} p \cdot (x-q) \right). \tag{51}$$

Its position covariance is $\sigma^2 I_{56}$; no transverse direction has been removed. The unitary half-density map is $f \mapsto \sqrt{\rho} f$, where $\rho = \sqrt{\det \gamma}$. The circle acts orthogonally on the scalar coordinates and fixes the thirty edge coordinates, so its average P_0 has the same pullback expression in both measures.

Proposition A.25 (A nonvanishing neutral projection)**.** *Write $q_s, p_s \in \mathbb{R}^{26}$ for the scalar parts and set*

$$A = \frac{|q_s|^2}{4\sigma^2} + \frac{\sigma^2 |p_s|^2}{\hbar^2}, \qquad B_* = \frac{p(Jq)}{\hbar}. \tag{52}$$

Then $A \geq |B_*|$ and

$$n^2 := \|P_0 g_{q,p,\sigma}\|^2 = e^{-A} I_0\left(\sqrt{A^2 - B_*^2} \right) > 0. \tag{53}$$

Consequently $F = \rho^{-1/2} P_0 g / n$ is a normalized vector in $\mathcal{D}(\widehat{H}_0)$. At the exact initial datum and $\hbar = 1$, each width $\sigma \in \{1/4, 1/2, 1\}$ satisfies

$$B_* = 0, \quad A = \frac{13}{4\sigma^2} + \frac{39\sigma^2 \mathcal{V}^2}{400} < 64, \qquad n^2 > \frac{1}{64}. \tag{54}$$

Proof. The Cauchy and arithmetic–geometric mean inequalities give $|B_*| \leq |p_s||q_s|/\hbar \leq A$. Completing the Gaussian square in $\langle g, \mathcal{U}_\theta g \rangle$ gives

$$\exp\left[-A(1 - \cos\theta) - i B_* \sin\theta \right].$$

Its circle average is n^2. Expanding $\exp[(A - B_*)e^{i\theta}/2 + (A + B_*)e^{-i\theta}/2]$ and retaining equal powers gives $e^{-A} \sum_{k \geq 0} ((A^2 - B_*^2)/4)^k / (k!)^2$, proving (53). This also defines I_0 and proves positivity, including the equality case $A = |B_*|$. For $B_* = 0$ it is the usual real circle-integral representation.

Polynomial coefficient and density-derivative bounds from Proposition A.19 apply to every translated, modulated Gaussian. Thus $\rho^{-1/2} g$ belongs to $\mathcal{D}(\widehat{H})$. The bounded circle projection preserves that domain and commutes with the Hamiltonian. Division by $n > 0$ proves the domain claim. The same argument gives finite moments of every polynomial multiplication observable.

Initially $|q_s|^2 = 13$ and $|p_s|^2 = 39\mathcal{V}^2/400$. Since $\mathcal{V} < 18$, direct rational substitution at the three stated widths gives $A < 64$. On $|\theta| \le 1/8$, $A(1-\cos\theta) \le A\theta^2/2 < 1/2$. Hence the integral is larger than $e^{-1/2}/(8\pi) > 1/64$, using $e^{-1/2} > 1/2$ and $\pi < 4$. □

Projection changes the state: a nonzero center is a parameter for an orbit, not the scalar one-point expectation. In the Lebesgue half-density representation, circle invariance gives $\langle x_s\rangle = \langle -i\hbar\nabla_s\rangle = 0$. For $B_* = 0$, direct Gaussian integration gives the useful invariant nodal-radius observable

$$\langle |x_s|^2\rangle_F = 26\sigma^2 + \frac{|q_s|^2}{2}(1+r) - \frac{2\sigma^4|p_s|^2}{\hbar^2}(1-r), \qquad r = \frac{I_1(A)}{I_0(A)}. \tag{55}$$

Indeed, inserting $|x_s|^2$ in the overlap adds $26\sigma^2 + |q_s|^2(1+\cos\theta)/2 - 2\sigma^4|p_s|^2(1-\cos\theta)/\hbar^2$; the imaginary term is proportional to B_*. The cosine average is I_1/I_0. For arbitrary B_*, the same calculation gives $26\sigma^2 + |q_s|^2/2 - 2\sigma^4|p_s|^2/\hbar^2 + 2\sigma^2 z I_1(z)/I_0(z)$, where $z = \sqrt{A^2 - B_*^2}$; its last term is zero when $z = 0$. The thirty edge coordinates retain their seed covariance, independently of this scalar projection. This nodal observable is not a supplied laboratory measurement operator.

The residual required for propagation. The half-density change of measure does not flatten the Hamiltonian. Writing $a^{ij} = (\gamma^{-1})^{ij}$ and $\ell = \log\rho$, its Lebesgue expression is

$$\widetilde{H} = -\frac{\hbar^2}{2}\partial_i(a^{ij}\partial_j) + V + U_\rho, \qquad U_\rho = \frac{\hbar^2}{4}\partial_i(a^{ij}\partial_j\ell) + \frac{\hbar^2}{8}(\partial_i\ell)a^{ij}(\partial_j\ell). \tag{56}$$

This follows by differentiating $\rho^{-1/2}g$; the two first-order cross terms cancel by symmetry of a^{ij}. For any smooth seed path in the strong domain, let $r_g = i\hbar\dot{g} - \widetilde{H}g$. Where $n(t) = \|P_0 g(t)\| > 0$, the normalized projection has residual norm at most $2\|r_g(t)\|/n(t)$: differentiating the projected norm and using self-adjointness gives $|\dot{n}| \le \|P_0 r_g\|/\hbar$. Duhamel's formula therefore gives

$$\left\| e^{-it\widehat{H}_0/\hbar}F(0) - F(t)\right\| \le \frac{2}{\hbar}\int_0^{|t|}\frac{\|r_g(s)\|}{n(s)}\,ds \tag{57}$$

for forward time, and with the reversed path for negative time. The initial bound on n alone does not bound this integral. Continuous center and covariance defects, the full variable coefficients and U_ρ, their numerical errors, and all Gaussian tails must enter an actual residual certificate. The prepared samples supply none of those propagation estimates. In particular, no error bound on a $1/40$ interval at $\hbar = 1$ follows from the preparation or from a semiclassical limit.

The executable packet at `code/electromagnetism/runtime/whitney_quantum_packet_receipt.json` recharts two adjacent classical samples and checks all mean-zero multipliers and reduced cotangents. Its independent checker uses exact-degree simplex moments at $a_C = 0$, including the complete 68-coordinate kinetic metric and all 56 reduced directions. Matrix operations and the second classical sample are numerical, while the displayed initial cotangent and projection lower bound are analytic identities. The packet specifies three full-dimensional seeds and their neutral projections. It does not select a physical preparation, evolve their covariances, identify them with a computed quantum history, or establish interacting continuum field theory.

B State-and-Law Habitat

This appendix records the compact convex state-and-law setting in which a closure map of the observer layer would act.

The OPH inputs used here are:

1. the patch net $P \mapsto A(P)$ of von Neumann algebras together with isotony and overlap restriction maps;

2. a compatible local state family, plus the separately declared global-extension interface used by this appendix to represent it on the inductive-limit algebra;

3. the finite premise-A3 information-projection branch and the separate optimizer-pushforward interface, which together supply a common finite family of gauge-invariant local constraint observables across cutoffs;

4. on the fixed-cutoff branch, the derived finite type-I presentations and compact boundary gluing groups used elsewhere in the OPH package.

The law slot is encoded by the finite expectation-value coordinates of the retained A1-generated constraint family used by premise A3. That choice furnishes a compact convex state-and-law habitat for the framework.

Theorem (Internal state-and-law habitat theorem). Fix finitely many screen patches $P_1, \dots, P_N$ in the standard OPH setup. For each i, let

$$M_i := A(P_i), \qquad M_{ij} := A(P_i \cap P_j),$$

where M_i is the patch von Neumann algebra and $M_{ij} \subset M_i, M_j$ is the overlap algebra. Let

$$S_i := S(M_i) \subset M_i^*$$

be the full state space of M_i, endowed with the weak* topology $\sigma(M_i^*, M_i)$.

Choose a finite family of self-adjoint gauge-invariant local constraint observables

$$\mathcal{O} = \{O_1, \dots, O_m\}$$

from the A1-generated premise-A3 constraint grammar, with each O_a supported in one of the chosen patches; write $i(a)$ for an index such that $O_a \in M_{i(a)}$.

Define the overlap-consistent state sector

$$X_{\mathrm{ov}} := \left\{ (\omega_1, \dots, \omega_N) \in \prod_{i=1}^{N} S_i : \omega_i|_{M_{ij}} = \omega_j|_{M_{ij}} \text{ for all } i, j \right\}.$$

Define the OPH law-coordinate map

$$c : X_{\mathrm{ov}} \to \mathbb{R}^m, \qquad c(\omega_1, \dots, \omega_N) := \big(\omega_{i(1)}(O_1), \dots, \omega_{i(m)}(O_m)\big).$$

Let

$$X_{\mathcal{O}} := \left\{ (\omega_1, \dots, \omega_N, \ell) \in \left(\prod_{i=1}^{N} S_i \right) \times \mathbb{R}^m : (\omega_1, \dots, \omega_N) \in X_{\mathrm{ov}},\ \ell = c(\omega_1, \dots, \omega_N) \right\}.$$

Then:

1. the ambient product

$$E := \left(\prod_{i=1}^{N} M_i^* \right) \times \mathbb{R}^m$$

is a Banach space for any product norm;

2. $X_{\mathcal{O}}$ is nonempty and convex;

3. $X_{\mathcal{O}}$ is norm-closed in E;

4. $X_{\mathcal{O}}$ is compact in the product topology

$$\tau := \left(\prod_{i=1}^{N} \sigma(M_i^*, M_i) \right) \times \text{Euclidean topology on } \mathbb{R}^m;$$

Proof. Each M_i is a von Neumann algebra in the OPH patch net, hence a Banach space, so each dual M_i^* is Banach. A finite product of Banach spaces is Banach, hence so is E.

For each i, the state space

$$S_i = \{\omega \in M_i^* : \omega \geq 0,\ \omega(1) = 1\}$$

is convex and weak* compact by Banach–Alaoglu, because it is a weak* closed subset of the dual unit ball. For every pair (i, j), restriction along $M_{ij} \subset M_i$ and $M_{ij} \subset M_j$ defines affine weak* continuous maps

$$r_{ij} : S_i \to S(M_{ij}), \qquad r_{ji} : S_j \to S(M_{ij}).$$

Hence X_{ov} is an intersection of affine equalizer sets, so it is convex and τ-closed inside $\prod_i S_i$. Since $\prod_i S_i$ is compact and convex in the product weak* topology, X_{ov} is compact and convex as well.

Nonemptiness comes from the OPH global state on the inductive-limit algebra: its restrictions

$$(\omega|_{M_1}, \ldots, \omega|_{M_N})$$

belong to X_{ov} because the restrictions agree on every overlap by construction.

Each coordinate $\omega_{i(a)} \mapsto \omega_{i(a)}(O_a)$ is affine and weak* continuous, since evaluation at a fixed algebra element is weak* continuous. Thus c is affine and τ-continuous. The graph map

$$G : X_{\text{ov}} \to E, \qquad G(x) := (x, c(x)),$$

is affine and continuous, so its image $X_{\mathcal{O}} = G(X_{\text{ov}})$ is convex and τ-compact.

To see norm-closedness in E, let $(x_n, c(x_n)) \to (x, \ell)$ in norm. Norm convergence implies weak* convergence on each coordinate, so $x \in X_{\text{ov}}$ because X_{ov} is weak* closed. Since c is weak* continuous, $c(x_n) \to c(x)$, while norm convergence in $\mathbb{R}^m$ also gives $c(x_n) \to \ell$. Hence $\ell = c(x)$, so $(x, \ell) \in X_{\mathcal{O}}$.

Finally, $X_{\mathcal{O}}$ is a compact convex subset of the locally convex topological vector space E equipped with τ. □

The additional inputs needed for a strange-loop closure theorem are:

1. a closure map T built from internal operations such as overlap repair, premise-A3 information reprojection, an explicit collar-recovery interface, and record-sector coarse-graining;

2. a proof that the stronger "observer-supporting" or "selected-world" criteria carve out a nonempty T-invariant observer-supporting subset of $X_{\mathcal{O}}$;

3. any uniqueness, contraction, or Lyapunov-type stability estimate for that closure map.

The habitat theorem supplies only the ambient Banach/compact-convex setting. It does not define a canonical internal self-map, it does not identify any nonempty invariant observer-supporting subset, and it does not prove uniqueness or stability. In particular, nonemptiness and compact-convexity of the ambient habitat do not by themselves imply the existence of a nonempty invariant observer-supporting sector. This does not erase the exact finite packet branch proved on the consensus surface: there the quotient normal-form map pushes forward to a continuous affine idempotent self-map of the finite packet simplex, with fixed points exactly the packets supported on quotient normal forms. Any further strange-loop reading of the ambient habitat as a closure story for existence is interpretive only and is not part of the recovered theorem package.

C Observer Registry and Clock-Naturality Certificate

For finite exported observer systems, a theorem-faithful evidence bundle separates executor metadata from observer semantics. It contains:

(i) a semantic history DAG or labeled poset $\mathsf{H}_O = (E_O, \preceq_O, \ell_O)$, with event keys computed from semantic payload, observer token, visible footprint, and semantic parents;

(ii) a provenance envelope containing worker ID, repair iteration, retry count, queue position, timestamp, and packet latency, none of which may enter the semantic event key unless declared as physical input;

(iii) local observer registry groupoids with disjoint patch-observer, cap-observer, and future-observer namespaces, explicit birth-event classes, continuation arrows, split/merge lineage arrows, and overlap functors preserving those data;

(iv) a global registry descent certificate: cocycle agreement, trivial monodromy, no duplicate global observer identifiers, no mixed patch/cap/future namespace collisions, and no reuse of local anchor-patch identifiers as local observer indices;

(v) observer-algebra extraction data

$$\mathsf{ObsAlg}(\widehat{q}_{\mathrm{nf}}) = (\mathcal{A}_O, \omega_O, \mathcal{R}_O, \ldots)$$

whose outputs are state-preserving isomorphic for history-equivalent or implementation-equivalent terminal augmented normal forms;

(vi) a support-cap chart map J_O natural with those observer-algebra isomorphisms; and

(vii) a clock instrument Θ_O with affine reparameterization and residual bound for state, calibration, and record-readout errors.

Theorem C.1 (Finite observer-clock certificate soundness)**.** *On a finite exported model, suppose the augmented repair relation on $\widehat{Q} = \{(q, [\mathsf{H}])\}$ terminates, terminal states are exactly the consistent physical states with completed semantic records, every augmented critical pair has a history-coherent join, implementation steps are linearizable to semantic commits or stutters, registry descent passes the groupoid cocycle and monodromy checks, and the observer-algebra/chart/clock maps satisfy the state-preserving and affine-residual certificates above. Then worker count, queue order, retry delivery, scheduler counters, and packet latency that is not declared physical input do not change the observer-readable history class, registry class, or clock law beyond the declared affine clock residual.*

Proof. Termination plus the augmented local diamond gives confluence by Newman's lemma for terminal history classes rather than only physical quotient states. Linearizability says every implementation execution projects to the same semantic relation up to stutters, so executor metadata cannot create new semantic events. Registry descent identifies the same global observer groupoid across local presentations. State-preserving observer-algebra extraction and natural support-cap charts transport the modular flow along the same observer identity, while the clock instrument certificate bounds the operational readout after the declared affine reparameterization. □

Remark C.2 (Falsifiers)**.** *The certificate fails if semantic event identifiers depend on worker IDs, timestamps, queue positions, repair iteration counters, or retries; if duplicate retry delivery creates two visible record commits; if two namespaces collide in the global registry; if registry cocycles or lineage arrows fail on overlaps; if histories from two executors are not labeled-poset isomorphic; or if clock residuals exceed the declared bound. These are public evidence-bundle failures that exceed field-name mismatches.*

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- Absolute irreducibility of real representations and Schur's lemma over an algebraically closed field [147, 182, 248].
- Additivity of relative modular Hamiltonians and interaction energy in statistical mechanics [226, 123].
- Ancilla dilations and conditional-expectation obstructions in finite-dimensional operator algebra [230, 259].
- AQFT, quantum information, operational observer frameworks [172, 225, 132].
- Axiomatic finite Lorentz chart and causal-order interfaces [273, 197].
- Burnside's theorem on the span of a group representation [128, 199].
- Busch-Gleason theorems deriving the Born rule from effect-valuation additivity [167, 130].
- Canonical oscillator quantization, bosonic Fock space and finite-element normal modes [123, 119].
- Christiansen-Halvorsen simplicial gauge actions and gauge-dependent finite elements [135, 48].
- CHSH classical bounds for diagonal states and free-operation closures in quantum resource theories [138, 133].
- Classical linear observation and feedback with retained records [254].
- compact-group averaging [126].
- Conditional expectations, Kraus representations, projector dephasing semigroups, Banach-algebra exponentials, and GKSL generators [262, 196, 112, 245, 209, 168].
- Coulomb reduction [215, 179].
- Courant conditions and subluminal motion on lattices [142, 143].
- Deterministic replay and transparency-log verification in software engineering [203, 202].

- Directed systems, finite matrix algebras, state restriction, natural transformations, and algebraic generators [121, 213, 150].
- Discrete geodesics and direct-sum variational principles [38].
- Discrete variational mechanics with constrained step alphabets and impulse-momentum balances [38].
- Discrete variational principles for lattice charges and the electromotive balance [38, 193].
- Duhamel estimates and quantum trace-distance bounds [231, 225].
- edgewise_Kuhn_simplex_refinement [198, 41, 163].
- Einstein's covariance argument for the inertia of energy [157, 158].
- Energy norms for leapfrog schemes and the complex structure of linear Hamiltonian flows [174, 218].
- energy_Gronwall_estimate [170, 159].
- Equivariant lattice field theory under finite symmetry groups [219, 214].
- exact_Gaussian_chi_squared_moments [188].
- Fibred records and product-type packaging of dynamical data [234, 185].
- Finite AQFT nets, isotony and algebraic locality, presheaf descent, conditional expectations, and tensor-split no-signalling [171, 212, 262, 258].
- Finite commutative C-star algebras, projective measurements, sharp-state no-cloning, and mixed-state no-broadcasting [150, 264, 270, 152, 117].
- Finite frame functions, informationally complete qubit measurements, Gleason and Busch representation theorems, and operator-algebra generation [167, 129, 237, 130, 128].
- Finite Gibbs information projection, discrete variational calculus, conditional resampling, Legendre transforms, and inverse-problem identifiability [37, 38, 165, 243, 260].
- Finite matrix tensor products, partial traces and matrix-unit generation [225, 150].
- Finite reversible Green–Kubo theory, Poisson resolvents, graph diffusion, Fick and Fourier laws [36, 36, 194, 169, 161, 162].
- Finite stochastic maps, automorphisms of finite function algebras, locally constant public flows, inner automorphisms of full matrix algebras, unitary exponentials, and the von Neumann equation [230, 122, 264].
- Finite zero-error channel coding and regular representations of A5 [252, 251].
- Finite-state Markov-chain quotient selection, stationary H-theorems, communicating classes, and lumping/intertwining [191, 244, 144].
- finite_dimensional_ODE_numerical_integration_and_refinement_controls [173, 174].
- Fixed-point methods, zero-error capacity, and standard RG/display conventions [261, 252].
- Frequency standards and counting processes as clocks [240, 148].
- Galois-conjugate three-dimensional character data over Q(sqrt 5) [186, 141].
- Gaussian overlap [139, 228].
- Gibbons-Hawking de Sitter entropy and static-patch relations [166].

- Gleason-type uniqueness of quantum states and operational axiomatics of instruments [167, 130, 151].
- Icosahedral coordinates and active finite tomography are classical linear constructions [145].
- Inertia of internal energy in relativistic composite bodies [157].
- instrument-channel compatibility with projective measurements [211, 176].
- Inverse problems of the calculus of variations and Lagrangian nonuniqueness [177, 153, 146, 178].
- Isometric identification of quotient inner-product spaces with rest fibers of Lorentz frames [221, 229].
- Jacobi_Maupertuis_lapse_elimination [184, 113, 116].
- Jaynes maximum-entropy thermodynamics, Lindblad-Uhlmann relative-entropy monotonicity, and stochastic-thermodynamics H-theorems [187, 208, 263, 250].
- Joint checkpoint channels, zero-error capacity, deflationary maps, and estimator identifiability [252, 149, 204].
- Kac's lemma and return-time analysis of finite Markov chains [189, 206].
- Kernel quotients, finite abstract rewriting, Church-Rosser normal-form uniqueness, and quotient-compatible dynamics [137, 224, 115].
- kinetic_form_Cauchy_Schwarz_and_unitary_energy_conservation [239, 190].
- Lattice minimal coupling of point charges and the gauge-invariance/current-conservation equivalence [268, 219].
- Lattice paths of point charges and their continuum worldlines [268, 219].
- Leapfrog integrators as time-h maps of continuous flows and the elliptic normal form of symplectic maps [174, 205].
- Legendre reduction [113, 216].
- local_semilinear_Klein_Gordon_existence_and_finite_propagation [249, 257].
- magnetic Galerkin approximation [154, 124].
- Maxwell-Klein-Gordon discretization [134].
- Model-theoretic simultaneous satisfaction of axiom packages on one structure [180].
- Normal modes of leapfrog Maxwell schemes and the dodecahedral graph spectrum [272, 142, 127].
- Null Lagrangians and total-divergence freedom in the discrete calculus of variations [227, 181].
- Order data, affine clock synchronization, proper time on inertial worldlines, and chart-covariant Lorentz geometry [273, 156, 221].
- Pauli matrices, Hermitian two-by-two matrix models of Minkowski space, celestial spheres, hyperboloid frames, and Lorentz rest spaces [232, 238, 221].
- Proper-time clocks on piecewise-linear worldlines and sub-sampled Markov chains [156, 206].
- Proper-time versus coordinate-time clocks on piecewise-linear worldlines and the time-dilation factor [156, 241].
- Reparametrization-invariant point-particle actions and the einbein gauge [125, 236].

- Ritz_projection_and_Sobolev_embedding [266, 109].
- Schema validators and reproducibility checkers [233, 246, 247].
- Self-consistency equations, finite quotient probabilities, Poisson limits, monoid actions, and standard cosmological posterior comparison [160, 192, 110].
- semilinear wave energy estimates [159, 253].
- Separable states satisfy local-hidden-variable bounds (Werner 1989) and the CHSH classical bound for product-diagonal states [265, 138].
- SI 2019 definitional constants and unit-conversion algebra [118, 256].
- Special-relativistic four-momentum kinematics [131, 241].
- Spectra of graph Laplacians and Hodge decompositions on polyhedral complexes [136, 207, 155].
- staggered integrators with time-step calibration [272, 174].
- Symmetry-adapted normal modes of polyhedral graphs and the icosahedral group's representations [269, 164].
- Third-law formulations on cofinal regulator nets and density-of-states bounds [111, 271].
- Third-law formulations via spectral gaps and density-of-states bounds [217, 111].
- Typed multi-sorted structures and their extensions [180, 234].
- Typed operational semantics, causal orders, modular parameters, worldline coordinates, and affine clock gauges [235, 200, 140].
- Typed scientific workflows and commuting-diagram dependency systems [210, 255, 242].
- validated interval Taylor integration [220, 223].
- Variation of coupled Maxwell-matter actions and bound polarization charge [183, 201].
- variational integrators and classical record-assisted inversion [38, 174].
- Whitney edge-element Maxwell actions [120, 222].
- Whitney finite elements [267, 120].
- Whitney finite elements and commuting cochain interpolation [267, 114].
- Whitney finite elements, gauge reduction, scalar electrodynamics and standard operator/ODE theory [120, 135, 239, 175].
- Zero-error channel codes, finite record gluing, monotone finite maps, and finite-size transfer methods [252, 195, 261].

References

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). https://doi.org/10.1103/PhysRevD.88.124026.

[8] J. Myrheim, *Statistical geometry*, CERN preprint TH-2538 (1978).

[9] D. A. Meyer, *The Dimension of Causal Sets*, PhD thesis, Massachusetts Institute of Technology (1988).

[10] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). https://doi.org/10.1103/PhysRevD.67.024034.

[11] D. M. T. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010). https://doi.org/10.1103/PhysRevLett.104.181301.

[12] F. Dowker and L. Glaser, *Causal set d'Alembertians for various dimensions*, Class. Quantum Grav. **30**, 195016 (2013). https://doi.org/10.1088/0264-9381/30/19/195016.

[13] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[14] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[15] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[16] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/observable_normal_forms.pdf.

[17] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at GitHub PDF.

[18] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*, 2026. Available at GitHub PDF.

[19] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/reality_as_consensus_protocol.pdf.

[20] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/screen_microphysics_and_observer_synchronization.pdf.

[21] B. Müller, M. Poneder, A. Osika, K. Xue, and D. A. Friedman, *Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[22] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/cosmology/oph_dark_matter_paper.pdf.

[23] B. Müller and J. Hill, *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/yang_mills_gap_clay_problem.pdf.

[24] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Archived manuscript. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/archive/observer_patch_holography_as_string_vacuum_selector.pdf.

[25] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, D. A. Friedman, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/flagship/from_observer_consensus_to_standard_physics.pdf.

[26] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/executable_observer_construction.pdf.

[27] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf.

[28] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/closure_loop. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8. Finite local-domain revision

and data: https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506, https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/closure_loop. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence.

[29] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography.

[30] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31.

[31] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," 2025, arXiv:2503.14738. https://arxiv.org/abs/2503.14738. Official chains and checksum manifest: https://data.desi.lbl.gov/public/papers/y3/bao-cosmo-params/README.html.

[32] M. Filzinger et al., "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons," *Physical Review Letters* **130**, 253001 (2023). https://doi.org/10.1103/PhysRevLett.130.253001.

[33] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/.

[34] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917.

[35] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931), https://doi.org/10.1103/PhysRev.37.405.

[36] M. S. Green, "Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II," *Journal of Chemical Physics* **22**, 398–413 (1954), https://doi.org/10.1063/1.1740082; R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957), https://doi.org/10.1143/JPSJ.12.570.

[37] I. Csiszár, "I-Divergence Geometry of Probability Distributions and Minimization Problems," *The Annals of Probability* **3**, 146–158 (1975), https://doi.org/10.1214/aop/1176996454.

[38] J. E. Marsden and M. West, "Discrete Mechanics and Variational Integrators," *Acta Numerica* **10**, 357–514 (2001), https://doi.org/10.1017/S096249290100006X.

[39] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[40] S. B. Gryb, "Jacobi's principle and the disappearance of time," *Physical Review D* **81**, 044035 (2010), https://arxiv.org/abs/0804.2900.

[41] H. Edelsbrunner and D. R. Grayson, "Edgewise subdivision of a simplex," *Discrete & Computational Geometry* **24**, 707–719 (2000), https://doi.org/10.1007/s004540010063.

[42] T. Tao, "Non-linear wave equations," author lecture notes, https://www.math.ucla.edu/~tao/Dispersive/wave.html.

[43] M. Shubin, "Essential self-adjointness for semi-bounded magnetic Schrödinger operators on non-compact manifolds," *Journal of Functional Analysis* **186**, 92–116 (2001), https://arxiv.org/abs/math/0007019.

[44] A. Ibort, F. Lledó, and J. M. Pérez-Pardo, "On self-adjoint extensions and symmetries in quantum mechanics," arXiv:1402.5537 (2014), https://arxiv.org/abs/1402.5537.

[45] D. N. Arnold, R. S. Falk, and R. Winther, "Finite element exterior calculus: from Hodge theory to numerical stability," *Bulletin of the American Mathematical Society* **47**, 281–354 (2010), https://doi.org/10.1090/S0273-0979-10-01278-4.

[46] B. He and F. L. Teixeira, "Geometric finite element discretization of Maxwell equations in primal and dual spaces," arXiv:physics/0503013 (2005), https://arxiv.org/abs/physics/0503013.

[47] E. Nelson, "Analytic vectors," *Annals of Mathematics* **70**, 572–615 (1959), https://doi.org/10.2307/1970331.

[48] S. H. Christiansen and T. G. Halvorsen, "A simplicial gauge theory," *Journal of Mathematical Physics* **53**, 033501 (2012), https://doi.org/10.1063/1.3692167.

[49] M. H. A. Newman, "On theories with a combinatorial definition of 'equivalence'," *Ann. of Math. (2)* **43** (1942), 223–243.

[50] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28** (1972), 251–257.

[51] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **16** (1975), 985–1007.

[52] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17** (1976), 303–321.

[53] R. Brunetti, D. Guido, and R. Longo, "Modular Structure and Duality in Conformal Quantum Field Theory," *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.

[54] H.-W. Wiesbrock, "Half-Sided modular inclusions of von-Neumann-Algebras," *Commun. Math. Phys.* **157** (1993), 83–92.

[55] H. Araki and L. Zsidó, "Extension of the structure theorem of Borchers and its application to half-sided modular inclusions," *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.

[56] D. Guido, R. Longo, and H.-W. Wiesbrock, "Extensions of Conformal Nets and Superselection Structures," *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.

[57] H.-W. Wiesbrock, "Modular Intersections of von-Neumann-Algebras in Quantum Field Theory," *Commun. Math. Phys.* **193** (1998), 269–285.

[58] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14** (1976), 870–892.

[59] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.

[60] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.

[61] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.

[62] Y. Chen, D. Stanford, H. Tang, and Z. Yang, "Negative shocks versus static patch holography," arXiv:2607.14042 [hep-th] (2026). https://arxiv.org/abs/2607.14042

[63] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.

[64] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, "Proof of the quantum null energy condition," *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.

[65] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, "Quantum focusing conjecture," *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.

[66] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, "A general proof of the quantum null energy condition," *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.

[67] E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14** (1973), 1938–1941.

[68] E. H. Wichmann, "Density matrices arising from incomplete measurements," *J. Math. Phys.* **4** (1963), 884–896.

[69] F. Hiai, M. Ohya, and M. Tsukada, "Sufficiency, KMS condition and relative entropy in von Neumann algebras," *Pacific J. Math.* **96** (1981), 99–109.

[70] M. Takesaki, "Conditional expectations in von Neumann algebras," *J. Funct. Anal.* **9** (1972), 306–321.

[71] D. Petz, "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra," *Commun. Math. Phys.* **105** (1986), 123–131.

[72] D. Petz, "Sufficiency of channels over von Neumann algebras," *Quart. J. Math.* **39** (1988), 97–108.

[73] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.

[74] P. Hayden, R. Jozsa, D. Petz, and A. Winter, "Structure of states which satisfy strong subadditivity of quantum entropy with equality," *Commun. Math. Phys.* **246** (2004), 359–374.

[75] S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Invent. Math.* **98** (1989), 157–218.

[76] S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Commun. Math. Phys.* **131** (1990), 51–107.

[77] T. Tannaka, "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen," *Tohoku Math. J.* **45** (1938), 1–12.

[78] M. G. Krein, "A principle of duality for a bicompact group and a square block algebra," *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.

[79] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32** (1974), 438–441.

[80] S. L. Glashow, J. Iliopoulos, and L. Maiani, "Weak interactions with lepton-hadron symmetry," *Phys. Rev. D* **2** (1970), 1285–1292.

[81] E. Witten, "An SU(2) anomaly," *Phys. Lett. B* **117** (1982), 324–328.

[82] S. Dimopoulos, S. Raby, and F. Wilczek, "Supersymmetry and the scale of unification," *Phys. Rev. D* **24** (1981), 1681–1683.

[83] U. Amaldi, W. de Boer, and H. Fürstenau, "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP," *Phys. Lett. B* **260** (1991), 447–455.

[84] D. J. Gross and W. Taylor, "Two-dimensional QCD is a string theory," *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.

[85] F. Peter and H. Weyl, "Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe," *Math. Ann.* **97** (1927), 737–755.

[86] A. W. Knapp, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.

[87] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, "Topological phases from higher gauge symmetry in 3+1D," *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.

[88] S. Chandrasekharan and U.-J. Wiese, "Quantum link models: A discrete approach to gauge theories," *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.

[89] W. Donnelly and A. C. Wall, "Entanglement entropy of electromagnetic edge modes," *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.

[90] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence," *JHEP* **06** (2015), 149, arXiv:1503.06237.

[91] M. A. Levin and X.-G. Wen, "String-net condensation: A physical mechanism for topological phases," *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.

[92] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, "Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections," *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.

[93] Clay Mathematics Institute, "Yang–Mills & the Mass Gap." Available at https://www.claymath.org/millennium/yang-mills-the-maths-gap/.

[94] A. Jaffe and E. Witten, "Quantum Yang–Mills Theory," official Clay Mathematics Institute problem description. Available at https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.

[95] L. Wu, "Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition," *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.

[96] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," *Communications in Mathematical Physics* **31** (1973), 83–112.

[97] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

[98] Particle Data Group, "Review of Particle Physics," *Phys. Rev. D* **110** (2024), 030001. Available at https://pdg.lbl.gov/2024/download/db2024.pdf.

[99] Particle Data Group, "Electroweak Model and Constraints on New Physics," Available at https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf.

[100] A. Strominger and C. Vafa, *Microscopic Origin of the Bekenstein–Hawking Entropy*, Phys. Lett. B **379**, 99–104 (1996). https://arxiv.org/abs/hep-th/9601029.

[101] J. M. Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*, Adv. Theor. Math. Phys. **2**, 231–252 (1998). https://arxiv.org/abs/hep-th/9711200.

[102] A. Ashtekar and J. Lewandowski, *Background Independent Quantum Gravity: A Status Report*, Class. Quantum Grav. **21**, R53–R152 (2004). https://arxiv.org/abs/gr-qc/0404018.

[103] C. Rovelli and L. Smolin, *Discreteness of area and volume in quantum gravity*, Nucl. Phys. B **442**, 593–622 (1995). https://arxiv.org/abs/gr-qc/9411005.

[104] S. Wolfram, *A Class of Models with the Potential to Represent Fundamental Physics*, section 8.12, Basic Concepts of Quantum Mechanics (2020). https://www.wolframphysics.org/technical-introduction/potential-relation-to-physics/basic-concepts-of-quantum-mechanics/. See also *Observer Theory* (2023), https://wolframinstitute.org/output/observer-theory.

[105] J. Gorard, M. Namuduri and X. D. Arsiwalla, *ZX-Calculus and Extended Hypergraph Rewriting Systems I: A Multiway Approach to Categorical Quantum Information Theory* (2020). https://arxiv.org/abs/2010.02752.

[106] E. R. Weinstein, *Geometric Unity*, draft of April 1, 2021, sections 2.5–3. https://geometricunity.nyc3.digitaloceanspaces.com/Geometric_Unity-Draft-April-1st-2021.pdf.

[107] R. Mosseri and R. Dandoloff, *Geometry of entangled states, Bloch spheres and Hopf fibrations*, J. Phys. A **34**, 10243–10252 (2001). https://arxiv.org/abs/quant-ph/0108137.

[108] W. Donnelly and L. Freidel, *Local subsystems in gauge theory and gravity*, JHEP **09**, 102 (2016). https://arxiv.org/abs/1601.04744.

[109] Adams, R. A., and Fournier, J. J. F. (2003). *Sobolev Spaces*. Academic Press. ISBN 978-0-12-044143-3.

[110] Aghanim, N., and others (2020). "Planck 2018 results. VI. Cosmological parameters." *Astronomy & Astrophysics* **641**, A6. https://doi.org/10.1051/0004-6361/201833910

[111] Aizenman, M., and Lieb, E. H. (1981). "The third law of thermodynamics and the degeneracy of the ground state for lattice systems." *Journal of Statistical Physics* **24**, 279–297. https://doi.org/10.1007/BF01007649

[112] Alicki, R., and Lendi, K. (2007). *Quantum Dynamical Semigroups and Applications*. Springer. https://doi.org/10.1007/3-540-70861-8

[113] Arnold, V. I. (1989). *Mathematical Methods of Classical Mechanics*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4757-2063-1

[114] Arnold, D. N., Falk, R. S., and Winther, R. (2006). "Finite element exterior calculus, homological techniques, and applications." *Acta Numerica* **15**, 1–155. https://doi.org/10.1017/S0962492906210018

[115] Baader, F., and Nipkow, T. (1998). *Term Rewriting and All That*. Cambridge University Press. https://doi.org/10.1017/CBO9781139172752

[116] Barbour, J. B., and Bertotti, B. (1982). "Mach's principle and the structure of dynamical theories." *Proceedings of the Royal Society of London. A. Mathematical and Physical Sciences* **382**, 295–306. https://doi.org/10.1098/rspa.1982.0102

[117] Barnum, H., Caves, C. M., Fuchs, C. A., Jozsa, R., and Schumacher, B. (1996). "Noncommuting mixed states cannot be broadcast." *Physical Review Letters* **76**, 2818–2821. https://doi.org/10.1103/PhysRevLett.76.2818

[118] Bureau International des Poids et Mesures (2019). "The International System of Units (SI)." https://www.bipm.org/en/publications/si-brochure

[119] Boffi, D. (2010). "Finite element approximation of eigenvalue problems." *Acta Numerica* **19**, 1–120. https://doi.org/10.1017/S0962492910000012

[120] Bossavit, A. (1988). "Whitney forms: a class of finite elements for three-dimensional computations in electromagnetism." *IEE Proceedings A: Physical Science, Measurement and Instrumentation, Management and Education, Reviews* **135**, 493–500. https://doi.org/10.1049/ip-a-1.1988.0077

[121] Bratteli, O. (1972). "Inductive limits of finite dimensional C*-algebras." *Transactions of the American Mathematical Society* **171**, 195–234. https://doi.org/10.1090/S0002-9947-1972-0312282-2

[122] Bratteli, O., and Robinson, D. W. (1987). *Operator Algebras and Quantum Statistical Mechanics 1: C*- and W*-Algebras, Symmetry Groups, Decomposition of States*. Springer. https://doi.org/10.1007/978-3-662-02520-8

[123] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[124] Brenner, S. C., and Scott, L. R. (2008). *The Mathematical Theory of Finite Element Methods*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-75934-0

[125] Brink, L., Deser, S., Zumino, B., Di Vecchia, P., and Howe, P. (1976). "Local supersymmetry for spinning particles." *Physics Letters B* **64**, 435–438. https://doi.org/10.1016/0370-2693(76)90115-5

[126] Bröcker, T., and tom Dieck, T. (1985). *Representations of Compact Lie Groups*. Springer. https://doi.org/10.1007/978-3-662-12918-0

[127] Brouwer, A. E., and Haemers, W. H. (2012). *Spectra of Graphs*. Springer, New York (Universitext). https://doi.org/10.1007/978-1-4614-1939-6

[128] Burnside, W. (1905). "On the condition of reducibility of any group of linear substitutions." *Proceedings of the London Mathematical Society* **s2-3**, 430–434. https://doi.org/10.1112/plms/s2-3.1.430

[129] Busch, P. (1991). "Informationally complete sets of physical quantities." *International Journal of Theoretical Physics* **30**, 1217–1227. https://doi.org/10.1007/BF00671008

[130] Busch, P. (2003). "Quantum states and generalized observables: a simple proof of Gleason's theorem." *Physical Review Letters* **91**, 120403. https://doi.org/10.1103/PhysRevLett.91.120403

[131] Byckling, E., and Kajantie, K. (1973). *Particle Kinematics*. Wiley. ISBN 0-471-12885-6.

[132] Chiribella, G., D'Ariano, G. M., and Perinotti, P. (2011). "Informational derivation of quantum theory." *Physical Review A* **84**, 012311. https://doi.org/10.1103/PhysRevA.84.012311

[133] Chitambar, E., and Gour, G. (2019). "Quantum resource theories." *Reviews of Modern Physics* **91**, 025001. https://doi.org/10.1103/RevModPhys.91.025001

[134] Christiansen, S. H., and Halvorsen, T. G. (2011). "Discretizing the Maxwell-Klein-Gordon equation by the lattice gauge theory formalism." *IMA Journal of Numerical Analysis* **31**, 1–24. https://doi.org/10.1093/imanum/drp019

[135] Christiansen, S. H., and Halvorsen, T. G. (2011). "A gauge invariant discretization on simplicial grids of the Schrödinger eigenvalue problem in an electromagnetic field." *SIAM Journal on Numerical Analysis* **49**, 331–345. https://doi.org/10.1137/090757502

[136] Chung, F. R. K. (1997). *Spectral Graph Theory*. American Mathematical Society. https://doi.org/10.1090/cbms/092

[137] Church, A., and Rosser, J. B. (1936). "Some properties of conversion." *Transactions of the American Mathematical Society* **39**, 472–482. https://doi.org/10.1090/S0002-9947-1936-1501858-0

[138] Clauser, J. F., Horne, M. A., Shimony, A., and Holt, R. A. (1969). "Proposed experiment to test local hidden-variable theories." *Physical Review Letters* **23**, 880–884. https://doi.org/10.1103/PhysRevLett.23.880

[139] Combescure, M., and Robert, D. (2012). *Coherent States and Applications in Mathematical Physics*. Springer. https://doi.org/10.1007/978-94-007-0196-0

[140] Connes, A., and Rovelli, C. (1994). "Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories." *Classical and Quantum Gravity* **11**, 2899–2917. https://doi.org/10.1088/0264-9381/11/12/007

[141] Conway, J. H., Curtis, R. T., Norton, S. P., Parker, R. A., and Wilson, R. A. (1985). *Atlas of Finite Groups: Maximal Subgroups and Ordinary Characters for Simple Groups*. Clarendon Press, Oxford. ISBN 0-19-853199-0.

[142] Courant, R., Friedrichs, K., and Lewy, H. (1928). "Über die partiellen Differenzengleichungen der mathematischen Physik." *Mathematische Annalen* **100**, 32–74. https://doi.org/10.1007/BF01448839

[143] Courant, R., Friedrichs, K., and Lewy, H. (1967). "On the partial difference equations of mathematical physics." *IBM Journal of Research and Development* **11**, 215–234. https://doi.org/10.1147/rd.112.0215

[144] Cover, T. M., and Thomas, J. A. (2006). *Elements of Information Theory*. Wiley. https://doi.org/10.1002/047174882X

[145] Coxeter, H. S. M. (1973). *Regular Polytopes*. Dover. ISBN 0-486-61480-8.

[146] Currie, D. G., and Saletan, E. J. (1966). "q-Equivalent Particle Hamiltonians. I. The Classical One-Dimensional Case." *Journal of Mathematical Physics* **7**, 967–974. https://doi.org/10.1063/1.1705010

[147] Curtis, C. W., and Reiner, I. (1962). *Representation Theory of Finite Groups and Associative Algebras*. Interscience. https://doi.org/10.1090/chel/356

[148] Daley, D. J., and Vere-Jones, D. (2003). *An Introduction to the Theory of Point Processes, Volume I: Elementary Theory and Methods*. Springer, New York (Probability and its Applications). https://doi.org/10.1007/b97277

[149] Davey, B. A., and Priestley, H. A. (2002). *Introduction to Lattices and Order*. Cambridge University Press. https://doi.org/10.1017/CBO9780511809088

[150] Davidson, K. R. (1996). *C*-Algebras by Example*. American Mathematical Society. https://doi.org/10.1090/fim/006

[151] Davies, E. B., and Lewis, J. T. (1970). "An operational approach to quantum probability." *Communications in Mathematical Physics* **17**, 239–260. https://doi.org/10.1007/BF01647093

[152] Dieks, D. (1982). "Communication by EPR devices." *Physics Letters A* **92**, 271–272. https://doi.org/10.1016/0375-9601(82)90084-6

[153] Douglas, J. (1941). "Solution of the inverse problem of the calculus of variations." *Transactions of the American Mathematical Society* **50**, 71–128. https://doi.org/10.1090/S0002-9947-1941-0004740-5

[154] Du, Q., Gunzburger, M. D., and Peterson, J. S. (1992). "Analysis and Approximation of the Ginzburg-Landau Model of Superconductivity." *SIAM Review* **34**, 54–81. https://doi.org/10.1137/1034003

[155] Eckmann, B. (1944). "Harmonische Funktionen und Randwertaufgaben in einem Komplex." *Commentarii Mathematici Helvetici* **17**, 240–255. https://doi.org/10.1007/BF02566245

[156] Einstein, A. (1905). "Zur Elektrodynamik bewegter Körper." *Annalen der Physik* **322**, 891–921. https://doi.org/10.1002/andp.19053221004

[157] Einstein, A. (1905). "Ist die Trägheit eines Körpers von seinem Energieinhalt abhängig?." *Annalen der Physik* **323**, 639–641. https://doi.org/10.1002/andp.19053231314

[158] Einstein, A. (1907). "Über die vom Relativitätsprinzip geforderte Trägheit der Energie." *Annalen der Physik* **328**, 371–384. https://doi.org/10.1002/andp.19073280713

[159] Evans, L. C. (2010). *Partial Differential Equations*. American Mathematical Society. https://doi.org/10.1090/gsm/019

[160] Feller, W. (1968). *An Introduction to Probability Theory and Its Applications, Volume 1*. Wiley. ISBN 0-471-25708-7.

[161] Fick, A. (1855). "Ueber Diffusion." *Annalen der Physik* **170**, 59–86. https://doi.org/10.1002/andp.18551700105

[162] Fourier, J. B. J. (1822). *Théorie analytique de la chaleur*. Firmin Didot, Paris; reissued by Cambridge University Press (Cambridge Library Collection), 2009. https://doi.org/10.1017/CBO9780511693229

[163] Freudenthal, H. (1942). "Simplizialzerlegungen von beschränkter Flachheit." *Annals of Mathematics* **43**, 580. https://doi.org/10.2307/1968813

[164] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[165] Geman, S., and Geman, D. (1984). "Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images." *IEEE Transactions on Pattern Analysis and Machine Intelligence* **PAMI-6**, 721–741. https://doi.org/10.1109/TPAMI.1984.4767596

[166] Gibbons, G. W., and Hawking, S. W. (1977). "Cosmological event horizons, thermodynamics, and particle creation." *Physical Review D* **15**, 2738–2751. https://doi.org/10.1103/PhysRevD.15.2738

[167] Gleason, A. M. (1957). "Measures on the closed subspaces of a Hilbert space." *Journal of Mathematics and Mechanics* **6**, 885–893. https://doi.org/10.1512/iumj.1957.6.56050

[168] Gorini, V., Kossakowski, A., and Sudarshan, E. C. G. (1976). "Completely positive dynamical semigroups of N-level systems." *Journal of Mathematical Physics* **17**, 821–825. https://doi.org/10.1063/1.522979

[169] Grady, L. J., and Polimeni, J. R. (2010). *Discrete Calculus*. Springer. https://doi.org/10.1007/978-1-84996-290-2

[170] Gronwall, T. H. (1919). "Note on the derivatives with respect to a parameter of the solutions of a system of differential equations." *Annals of Mathematics* **20**, 292. https://doi.org/10.2307/1967124

[171] Haag, R., and Kastler, D. (1964). "An algebraic approach to quantum field theory." *Journal of Mathematical Physics* **5**, 848–861. https://doi.org/10.1063/1.1704187

[172] Haag, R. (1996). *Local Quantum Physics*. Springer. https://doi.org/10.1007/978-3-642-61458-3

[173] Hairer, E., Nørsett, S. P., and Wanner, G. (1993). *Solving Ordinary Differential Equations I: Nonstiff Problems*. Springer, Berlin (Springer Series in Computational Mathematics). https://doi.org/10.1007/978-3-540-78862-1

[174] Hairer, E., Lubich, C., and Wanner, G. (2006). *Geometric Numerical Integration: Structure-Preserving Algorithms for Ordinary Differential Equations*. Springer. https://doi.org/10.1007/3-540-30666-8

[175] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[176] Heinosaari, T., and Ziman, M. (2011). *The Mathematical Language of Quantum Theory: From Uncertainty to Entanglement*. Cambridge University Press. https://doi.org/10.1017/CBO9781139031103

[177] von Helmholtz, H. (1887). "Ueber die physikalische Bedeutung des Princips der kleinsten Wirkung." *Journal für die reine und angewandte Mathematik* **100**, 137–166. https://doi.org/10.1515/crll.1887.100.137

[178] Henneaux, M. (1982). "Equations of motion, commutation relations and ambiguities in the Lagrangian formalism." *Annals of Physics* **140**, 45–64. https://doi.org/10.1016/0003-4916(82)90334-7

[179] Henneaux, M., and Teitelboim, C. (1992). *Quantization of Gauge Systems*. Princeton University Press. https://doi.org/10.1515/9780691213866

[180] Hodges, W. (1993). *Model Theory*. Cambridge University Press. https://doi.org/10.1017/CBO9780511551574

[181] Hydon, P. E., and Mansfield, E. L. (2004). "A Variational Complex for Difference Equations." *Foundations of Computational Mathematics* **4**, 187–217. https://doi.org/10.1007/s10208-002-0071-9

[182] Isaacs, I. M. (1976). *Character Theory of Finite Groups*. Academic Press. https://doi.org/10.1090/chel/359

[183] Jackson, J. D. (1999). *Classical Electrodynamics*. Wiley, New York. ISBN 0-471-30932-X.

[184] Jacobi, C. G. J. (1884). *Vorlesungen über Dynamik*. G. Reimer, Berlin. https://doi.org/10.5962/bhl.title.18726

[185] Jacobs, B. (1999). *Categorical Logic and Type Theory*. Elsevier. https://doi.org/10.1016/S0049-237X(98)X8028-6

[186] James, G., and Liebeck, M. (2001). *Representations and Characters of Groups*. Cambridge University Press. https://doi.org/10.1017/CBO9780511814532

[187] Jaynes, E. T. (1957). "Information Theory and Statistical Mechanics." *Physical Review* **106**, 620–630. https://doi.org/10.1103/PhysRev.106.620

[188] Johnson, N. L., Kotz, S., and Balakrishnan, N. (1994). *Continuous Univariate Distributions, Volume 1*. Wiley. ISBN 0-471-58495-9.

[189] Kac, M. (1947). "On the notion of recurrence in discrete stochastic processes." *Bulletin of the American Mathematical Society* **53**, 1002–1010. https://doi.org/10.1090/S0002-9904-1947-08927-8

[190] Kato, T. (1995). *Perturbation Theory for Linear Operators*. Springer, Berlin (Classics in Mathematics). https://doi.org/10.1007/978-3-642-66282-9

[191] Kemeny, J. G., and Snell, J. L. (1976). *Finite Markov Chains*. Springer-Verlag, New York (Undergraduate Texts in Mathematics). ISBN 0-387-90192-2.

[192] Kilp, M., Knauer, U., and Mikhalev, A. V. (2000). *Monoids, Acts and Categories*. De Gruyter. https://doi.org/10.1515/9783110812909

[193] Kogut, J. B. (1979). "An introduction to lattice gauge theory and spin systems." *Reviews of Modern Physics* **51**, 659–713. https://doi.org/10.1103/RevModPhys.51.659

[194] Komorowski, T., Landim, C., and Olla, S. (2012). *Fluctuations in Markov Processes: Time Symmetry and Martingale Approximation*. Springer, Berlin (Grundlehren der mathematischen Wissenschaften). https://doi.org/10.1007/978-3-642-29880-6

[195] Körner, J., and Orlitsky, A. (1998). "Zero-error information theory." *IEEE Transactions on Information Theory* **44**, 2207–2229. https://doi.org/10.1109/18.720537

[196] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Springer. https://doi.org/10.1007/3-540-12732-1

[197] Kronheimer, E. H., and Penrose, R. (1967). "On the structure of causal spaces." *Proceedings of the Cambridge Philosophical Society* **63**, 481–501. https://doi.org/10.1017/S030500410004144X

[198] Kuhn, H. W. (1960). "Some combinatorial lemmas in topology." *IBM Journal of Research and Development* **4**, 518–524. https://doi.org/10.1147/rd.45.0518

[199] Lam, T. Y. (1998). "A theorem of Burnside on matrix rings." *American Mathematical Monthly* **105**, 651–653. https://doi.org/10.1080/00029890.1998.12004939

[200] Lamport, L. (1978). "Time, clocks, and the ordering of events in a distributed system." *Communications of the ACM* **21**, 558–565. https://doi.org/10.1145/359545.359563

[201] Landau, L. D., and Lifshitz, E. M. (1975). *The Classical Theory of Fields*. Pergamon Press. ISBN 978-0-08-025072-4.

[202] Laurie, B., Langley, A., and Kasper, E. (2013). "Certificate Transparency." https://doi.org/10.17487/RFC6962

[203] LeBlanc, T. J., and Mellor-Crummey, J. M. (1987). "Debugging parallel programs with Instant Replay." *IEEE Transactions on Computers* **C-36**, 471–482. https://doi.org/10.1109/TC.1987.1676929

[204] Lehmann, E. L., and Casella, G. (1998). *Theory of Point Estimation*. Springer, New York (Springer Texts in Statistics). https://doi.org/10.1007/b98854

[205] Leimkuhler, B., and Reich, S. (2004). *Simulating Hamiltonian Dynamics*. Cambridge University Press. https://doi.org/10.1017/CBO9780511614118

[206] Levin, D. A., and Peres, Y. (2017). *Markov Chains and Mixing Times*. American Mathematical Society. https://doi.org/10.1090/mbk/107

[207] Lim, L. H. (2020). "Hodge Laplacians on graphs." *SIAM Review* **62**, 685–715. https://doi.org/10.1137/18M1223101

[208] Lindblad, G. (1975). "Completely positive maps and entropy inequalities." *Communications in Mathematical Physics* **40**, 147–151. https://doi.org/10.1007/BF01609396

[209] Lindblad, G. (1976). "On the generators of quantum dynamical semigroups." *Communications in Mathematical Physics* **48**, 119–130. https://doi.org/10.1007/BF01608499

[210] Ludäscher, B., Altintas, I., Berkley, C., Higgins, D., Jaeger, E., Jones, M., et al. (2006). "Scientific workflow management and the Kepler system." *Concurrency and Computation: Practice and Experience* **18**, 1039–1065. https://doi.org/10.1002/cpe.994

[211] Lüders, G. (1950). "Über die Zustandsänderung durch den Meßprozeß." *Annalen der Physik* **443**, 322–328. https://doi.org/10.1002/andp.19504430510

[212] Mac Lane, S., and Moerdijk, I. (1992). *Sheaves in Geometry and Logic: A First Introduction to Topos Theory*. Springer. https://doi.org/10.1007/978-1-4612-0927-0

[213] Mac Lane, S. (1998). *Categories for the Working Mathematician*. Springer. https://doi.org/10.1007/978-1-4757-4721-8

[214] Mandula, J. E., Zweig, G., and Govaerts, J. (1983). "Representations of the rotation reflection symmetry group of the four-dimensional cubic lattice." *Nuclear Physics B* **228**, 91–108. https://doi.org/10.1016/0550-3213(83)90399-1

[215] Marsden, J., and Weinstein, A. (1974). "Reduction of symplectic manifolds with symmetry." *Reports on Mathematical Physics* **5**, 121–130. https://doi.org/10.1016/0034-4877(74)90021-4

[216] Marsden, J. E., and Ratiu, T. S. (1999). *Introduction to Mechanics and Symmetry: A Basic Exposition of Classical Mechanical Systems*. Springer, New York (Texts in Applied Mathematics). https://doi.org/10.1007/978-0-387-21792-5

[217] Masanes, L., and Oppenheim, J. (2017). "A general derivation and quantification of the third law of thermodynamics." *Nature Communications* **8**, 14538. https://doi.org/10.1038/ncomms14538

[218] McDuff, D., and Salamon, D. (2017). *Introduction to Symplectic Topology*. Oxford University Press. https://doi.org/10.1093/oso/9780198794899.001.0001

[219] Montvay, I., and Münster, G. (1994). *Quantum Fields on a Lattice*. Cambridge University Press. https://doi.org/10.1017/CBO9780511470783

[220] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[221] Naber, G. L. (2012). *The Geometry of Minkowski Spacetime: An Introduction to the Mathematics of the Special Theory of Relativity*. Springer. https://doi.org/10.1007/978-1-4419-7838-7

[222] Nédélec, J. C. (1980). "Mixed finite elements in $\mathbf{R}^3$." *Numerische Mathematik* **35**, 315–341. https://doi.org/10.1007/BF01396415

[223] Nedialkov, N. S., Jackson, K. R., and Corliss, G. F. (1999). "Validated solutions of initial value problems for ordinary differential equations." *Applied Mathematics and Computation* **105**, 21–68. https://doi.org/10.1016/S0096-3003(98)10083-8

[224] Newman, M. H. A. (1942). "On theories with a combinatorial definition of "equivalence"." *Annals of Mathematics* **43**, 223. https://doi.org/10.2307/1968867

[225] Nielsen, M. A., and Chuang, I. L. (2010). *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press. https://doi.org/10.1017/CBO9780511976667

[226] Ohya, M., and Petz, D. (1993). *Quantum Entropy and Its Use*. Springer. https://doi.org/10.1007/978-3-642-57997-4

[227] Olver, P. J. (1993). *Applications of Lie Groups to Differential Equations*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-4350-2

[228] Olver, F. W. J., Lozier, D. W., Boisvert, R. F., and Clark, C. W. (2010). *NIST Handbook of Mathematical Functions*. Cambridge University Press. ISBN 978-0-521-19225-5.

[229] O'Neill, B. (1983). *Semi-Riemannian Geometry with Applications to Relativity*. Academic Press, New York. ISBN 0-12-526740-1.

[230] Paulsen, V. (2002). *Completely Bounded Maps and Operator Algebras*. Cambridge University Press. https://doi.org/10.1017/CBO9780511546631

[231] Pazy, A. (1983). *Semigroups of Linear Operators and Applications to Partial Differential Equations*. Springer. https://doi.org/10.1007/978-1-4612-5561-1

[232] Penrose, R., and Rindler, W. (1984). *Spinors and Space-Time, Volume 1*. Cambridge University Press. https://doi.org/10.1017/CBO9780511564048

[233] Pezoa, F., Reutter, J. L., Suarez, F., Ugarte, M., and Vrgoč, D. (2016). "Foundations of JSON Schema." In *Proceedings of the 25th International Conference on World Wide Web (WWW '16)*, 263–273. International World Wide Web Conferences Steering Committee. https://doi.org/10.1145/2872427.2883029

[234] Pierce, B. C. (2002). *Types and Programming Languages*. MIT Press. ISBN 0-262-16209-1.

[235] Plotkin, G. D. (2004). "A structural approach to operational semantics." *The Journal of Logic and Algebraic Programming* **60-61**, 17–139. https://doi.org/10.1016/j.jlap.2004.05.001

[236] Polchinski, J. (1998). *String Theory. Volume 1*. Cambridge University Press. https://doi.org/10.1017/CBO9780511816079

[237] Prugovečki, E. (1977). "Information-theoretical aspects of quantum measurement." *International Journal of Theoretical Physics* **16**, 321–331. https://doi.org/10.1007/BF01807146

[238] Ratcliffe, J. G. (2019). *Foundations of Hyperbolic Manifolds*. Springer International Publishing. https://doi.org/10.1007/978-3-030-31597-9

[239] Reed, M., and Simon, B. (1980). *Methods of Modern Mathematical Physics I: Functional Analysis*. Academic Press. ISBN 0-12-585050-6.

[240] Riehle, F. (2004). *Frequency Standards: Basics and Applications*. Wiley-VCH, Weinheim. https://doi.org/10.1002/3527605991

[241] Rindler, W. (2006). *Relativity: Special, General, and Cosmological*. Oxford University Press. https://doi.org/10.1093/oso/9780198567318.001.0001

[242] Ringer, T., Palmskog, K., Sergey, I., Gligoric, M., and Tatlock, Z. (2019). "QED at Large: A Survey of Engineering of Formally Verified Software." *Foundations and Trends in Programming Languages* **5**, 102–281. https://doi.org/10.1561/2500000045

[243] Rockafellar, R. T. (1970). *Convex Analysis*. Princeton University Press. https://doi.org/10.1515/9781400873173

[244] Rogers, L. C. G., and Pitman, J. W. (1981). "Markov Functions." *The Annals of Probability* **9**, 573–582. https://doi.org/10.1214/aop/1176994363

[245] Rudin, W. (1991). *Functional Analysis*. McGraw-Hill. ISBN 0-07-054236-8.

[246] Rundgren, A., Jordan, B., and Erdtman, S. (2020). "JSON Canonicalization Scheme (JCS)." https://doi.org/10.17487/RFC8785

[247] Sandve, G. K., Nekrutenko, A., Taylor, J., and Hovig, E. (2013). "Ten simple rules for reproducible computational research." *PLoS Computational Biology* **9**, e1003285. https://doi.org/10.1371/journal.pcbi.1003285

[248] Schur, I. (1905). "Neue Begründung der Theorie der Gruppencharaktere." *Sitzungsberichte der Königlich Preußischen Akademie der Wissenschaften zu Berlin*, 406–432. https://zbmath.org/?q=an:36.0194.01

[249] Segal, I. E. (1963). "The global Cauchy problem for a relativistic scalar field with power interaction." *Bulletin de la Société Mathématique de France* **91**, 129–135. https://doi.org/10.24033/bsmf.1593

[250] Seifert, U. (2012). "Stochastic thermodynamics, fluctuation theorems and molecular machines." *Reports on Progress in Physics* **75**, 126001. https://doi.org/10.1088/0034-4885/75/12/126001

[251] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[252] Shannon, C. (1956). "The zero error capacity of a noisy channel." *IRE Transactions on Information Theory* **2**, 8–19. https://doi.org/10.1109/TIT.1956.1056798

[253] Shatah, J., and Struwe, M. (2000). *Geometric Wave Equations*. American Mathematical Society. https://doi.org/10.1090/cln/002

[254] Sontag, E. D. (1998). *Mathematical Control Theory*. Springer. https://doi.org/10.1007/978-1-4612-0577-7

[255] Spivak, D. I. (2014). *Category Theory for the Sciences*. The MIT Press. https://doi.org/10.7551/mitpress/9995.001.0001

[256] Stock, M., Davis, R., de Mirandés, E., and Milton, M. J. T. (2019). "The revision of the SI—the result of three decades of progress in metrology." *Metrologia* **56**, 022001. https://doi.org/10.1088/1681-7575/ab0013

[257] Strauss, W. A. (1989). *Nonlinear Wave Equations*. American Mathematical Society (CBMS Regional Conference Series in Mathematics 73). https://doi.org/10.1090/cbms/073

[258] Summers, S. J. (1990). "On the independence of local algebras in quantum field theory." *Reviews in Mathematical Physics* **2**, 201–247. https://doi.org/10.1142/S0129055X90000090

[259] Takesaki, M. (1972). "Conditional expectations in von Neumann algebras." *Journal of Functional Analysis* **9**, 306–321. https://doi.org/10.1016/0022-1236(72)90004-3

[260] Tarantola, A. (2005). *Inverse Problem Theory and Methods for Model Parameter Estimation*. SIAM. https://doi.org/10.1137/1.9780898717921

[261] Tarski, A. (1955). "A lattice-theoretical fixpoint theorem and its applications." *Pacific Journal of Mathematics* **5**, 285–309. https://doi.org/10.2140/pjm.1955.5.285

[262] Tomiyama, J. (1957). "On the projection of norm one in W*-algebras." *Proceedings of the Japan Academy* **33**, 608–612. https://doi.org/10.3792/pja/1195524885

[263] Uhlmann, A. (1977). "Relative entropy and the Wigner-Yanase-Dyson-Lieb concavity in an interpolation theory." *Communications in Mathematical Physics* **54**, 21–32. https://doi.org/10.1007/BF01609834

[264] von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Springer. https://doi.org/10.1007/978-3-642-61409-5

[265] Werner, R. F. (1989). "Quantum states with Einstein-Podolsky-Rosen correlations admitting a hidden-variable model." *Physical Review A* **40**, 4277–4281. https://doi.org/10.1103/PhysRevA.40.4277

[266] Wheeler, M. F. (1973). "A priori L2 error estimates for Galerkin approximations to parabolic partial differential equations." *SIAM Journal on Numerical Analysis* **10**, 723–759. https://doi.org/10.1137/0710062

[267] Whitney, H. (1957). *Geometric Integration Theory*. Princeton University Press. https://doi.org/10.1515/9781400877577

[268] Wilson, K. G. (1974). "Confinement of quarks." *Physical Review D* **10**, 2445–2459. https://doi.org/10.1103/PhysRevD.10.2445

[269] Wilson, J. E. B., Decius, J. C., and Cross, P. C. (1980). *Molecular Vibrations: The Theory of Infrared and Raman Vibrational Spectra*. Dover Publications. ISBN 0-486-63941-X.

[270] Wootters, W. K., and Zurek, W. H. (1982). "A single quantum cannot be cloned." *Nature* **299**, 802–803. https://doi.org/10.1038/299802a0

[271] Wreszinski, W. F., and Abdalla, E. (2009). "A Precise Formulation of the Third Law of Thermodynamics." *Journal of Statistical Physics* **134**, 781–792. https://doi.org/10.1007/s10955-009-9693-5

[272] Yee, K. S. (1966). "Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media." *IEEE Transactions on Antennas and Propagation* **14**, 302–307. https://doi.org/10.1109/TAP.1966.1138693

[273] Zeeman, E. C. (1964). "Causality implies the Lorentz group." *Journal of Mathematical Physics* **5**, 490–493. https://doi.org/10.1063/1.1704140

Statements and Declarations

Funding declaration: The authors declare that no external grants or dedicated third-party research funding were received for the preparation of this manuscript.

Author affiliations declaration: Bernhard Mueller is affiliated with Pragma Research Inc. Alexander Osika is affiliated with EtherWorks. Daniel Ari Friedman is affiliated with the Active Inference Institute and FractiAI. No institutional affiliations were declared for Mario Poneder, Kai Xue, Ben Cassie, or Peter Nguyen. Correspondence email addresses are provided in the correspondence declaration.

Consent to Participate declaration: not applicable.

Consent to Publish declaration: not applicable.

Author Contribution declaration: Bernhard Mueller developed the OPH concept, mathematical formulation, synthesis, main manuscript text, and build package. Mario Poneder contributed the exact conditional port-current construction, verifier, proof packet, receipt, and tests, as well as proof and validation work on the hierarchy certificates, the edge-entropy area-law route to the Einstein/Newton coupling, coupling normalization, particle and neutrino certificates, and manuscript/book consistency. Alexander Osika contributed to the hardware path, experimental prototype framing, hardware evidence boundary, and manuscript review. Kai Xue contributed consistency checks, simulator review, visualization review, and prototype engineering review. Ben Cassie

contributed hardware-evidence review, implementation-claim review, experimental-prototype framing, and manuscript review. Peter Nguyen contributed proof review, branch-boundary analysis, Yang–Mills and compact-gauge analysis, finite packet-quotient closure, fine-structure certification, and manuscript consistency. Daniel Ari Friedman contributed independent red-team audits of the papers, the claim and premise ledgers, the Lean library and the custody records, and the programme-obstruction audit. All authors reviewed and approved the manuscript.

Correspondence declaration: Correspondence should be addressed to Bernhard Mueller, bernhard@floatingpragma.ai, and Alexander Osika, alex@snrgystudios.com.

Data Availability declaration: Source manuscripts, public PDFs, simulation code, and implementation notes are available through the OPH repository and its linked public artifacts. The exact-federation archive publishes the level-six configuration and seeds, primitive seam arrays, initial and terminal loads, per-schedule descent ledgers, quotient hashes, per-carrier response kernels and a standalone exact verifier, with the level-five to level-ten settlement receipts, and the closure-loop archive publishes the carrier event logs, recovered specifications and producer-free verifier [28]. Hardware and prototype evidence supports no claim of this manuscript; the finite receipts, verifier code and configurations behind the stated implementation claims are in the repository.

Code Availability declaration: The code is in the OPH repository, and the executable observer with its protocol records is in its model directory [29]. The archived large run pins the simulator revision that produced it and supplies byte-hash and simulator-independent normal-form verification for the curated evidence subset.

Ethics declaration: not applicable.

Competing Interests declaration: The authors participate in OPH-related research and implementation programs, including prototype hardware and simulator development. The authors declare no external competing financial interests for this manuscript.

Use of AI-assisted tools declaration: AI-assisted tools were used for language editing, drafting assistance, formatting support, and manuscript preparation. The authors reviewed and are responsible for all scientific content, claims, references, and declarations.

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.